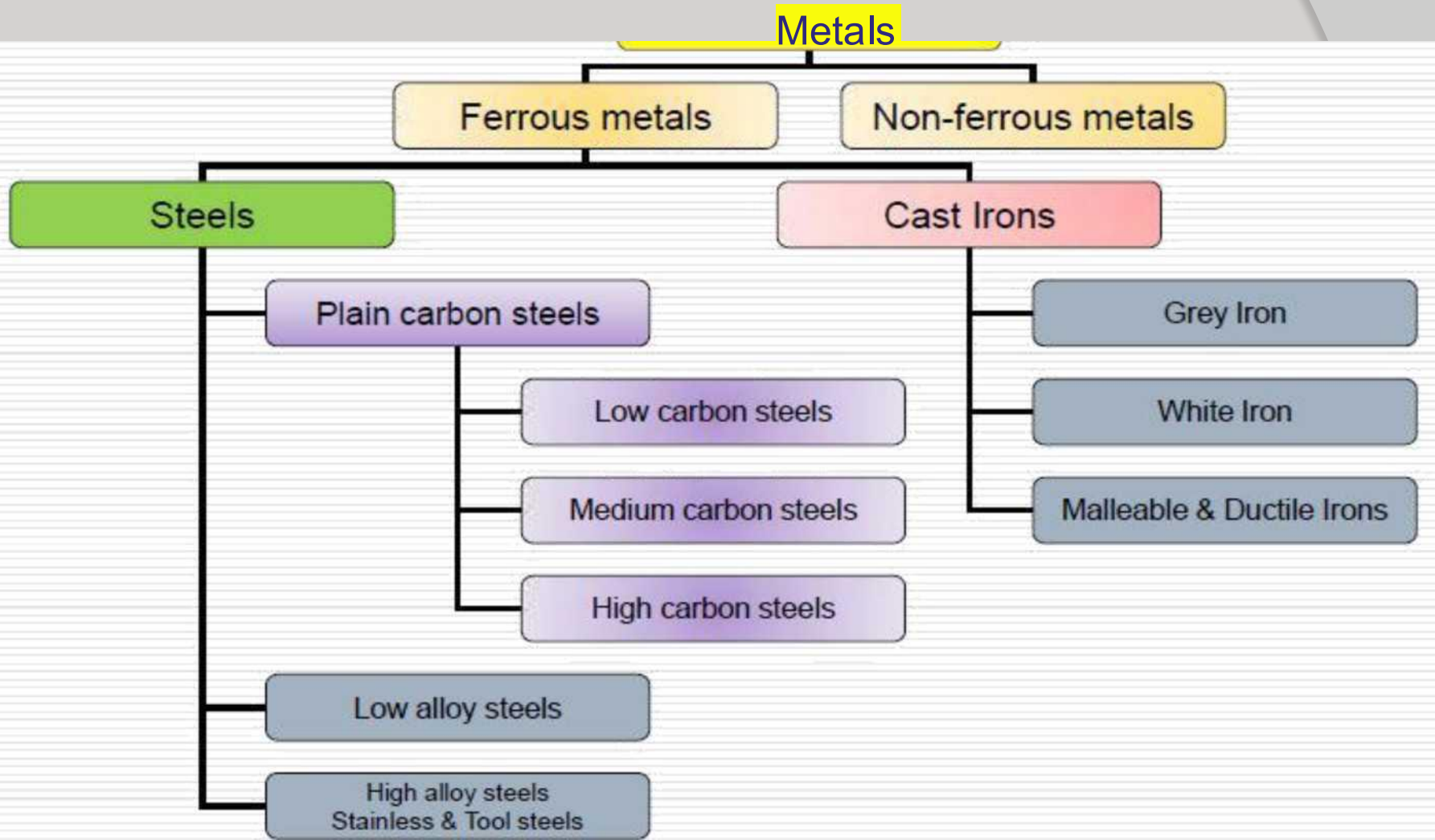


Material Science & Engineering

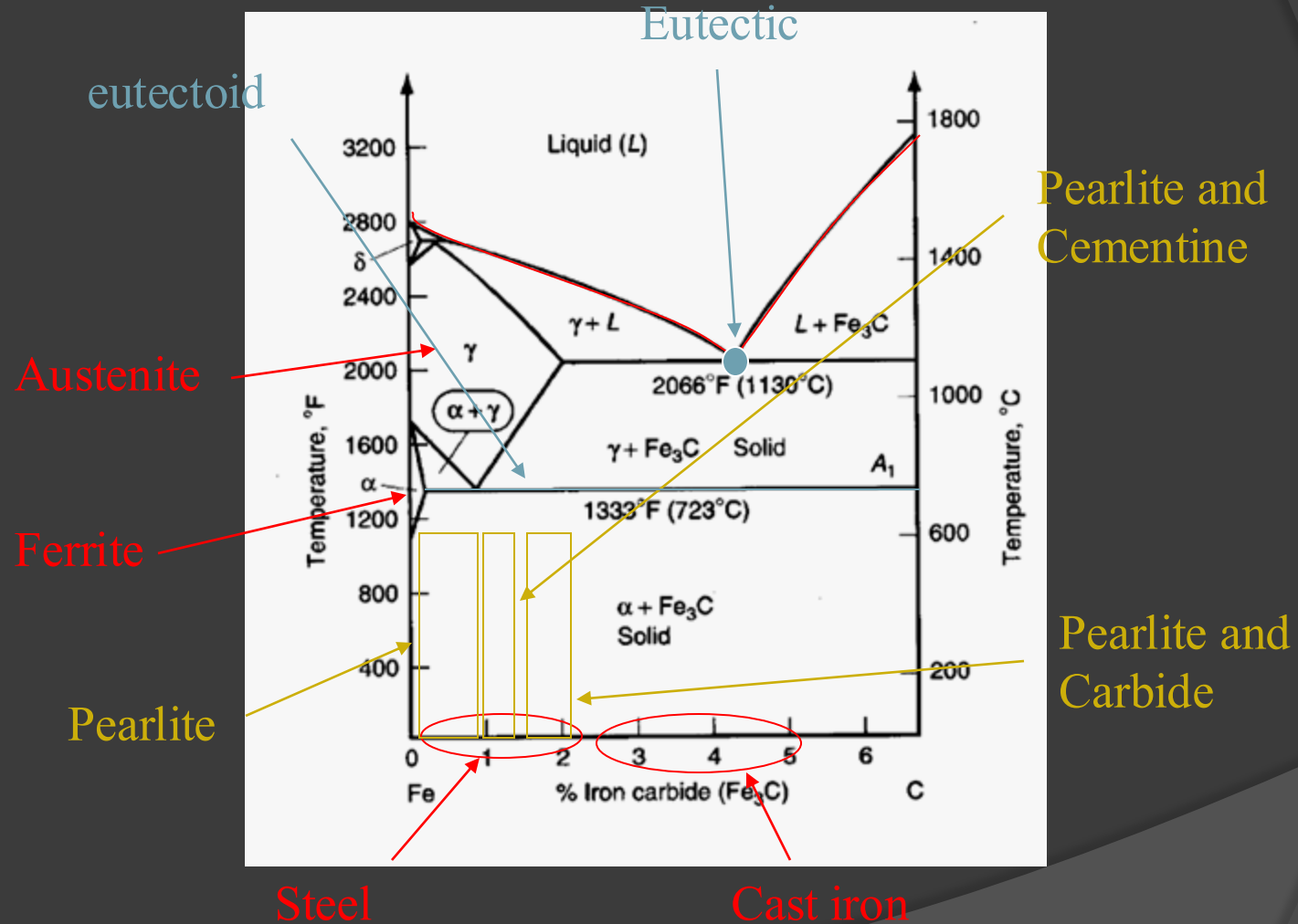
Outline

- Introduction
- Cooling curve for pure iron
- Definition of structures
- Iron-Carbon equilibrium phase diagram – Sketch
- **The Iron-Iron Carbide Diagram - Explanation**
- The Austenite to ferrite / cementite transformation
- Nucleation & growth of pearlite
- Effect of C %age on the microstructure of steel
- Relationship b/w C %age & mechanical properties of steel

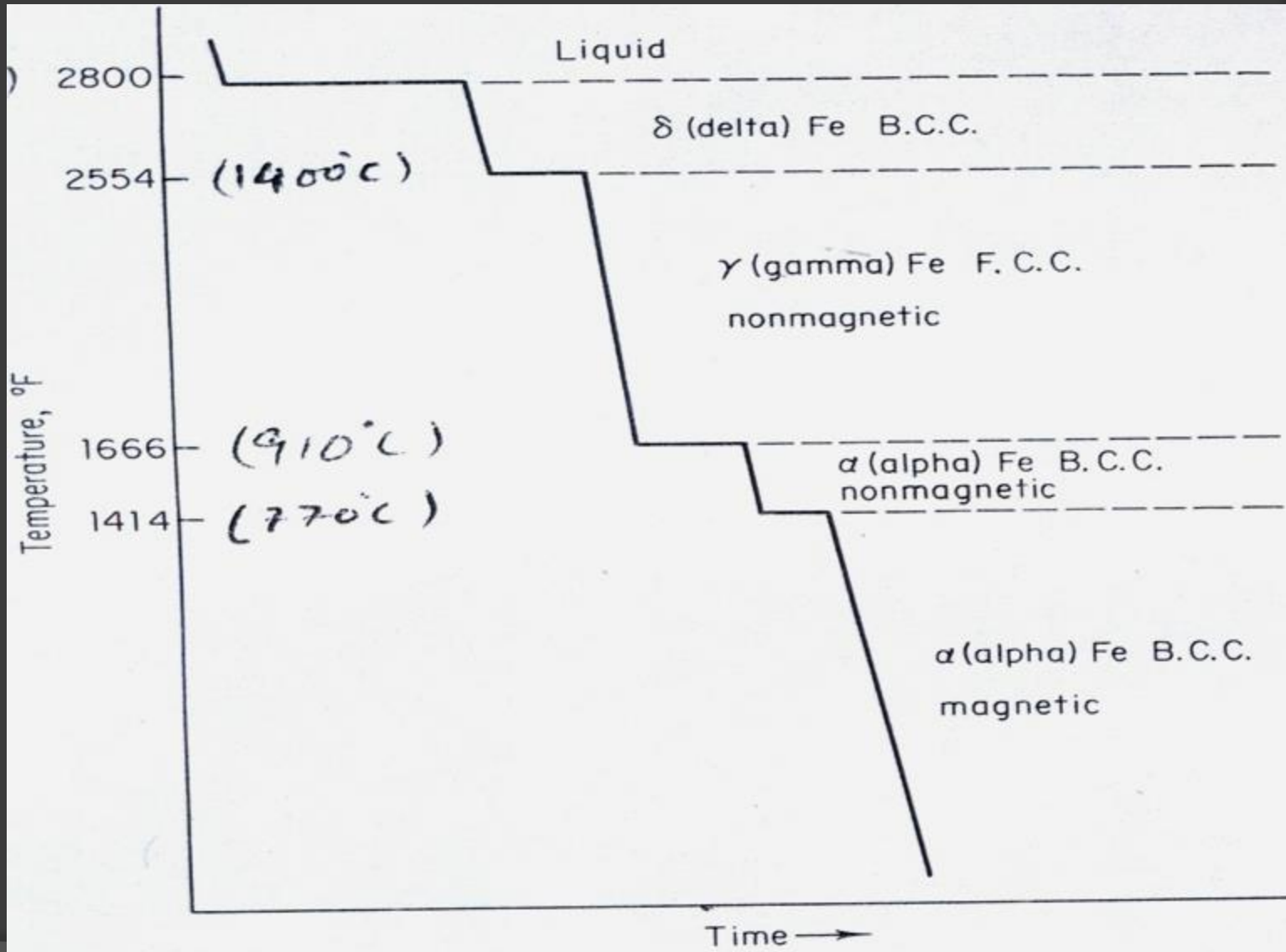
Fe-C alloy classification



IRON-CARBON PHASE DIAGRAM



Cooling curve for pure iron

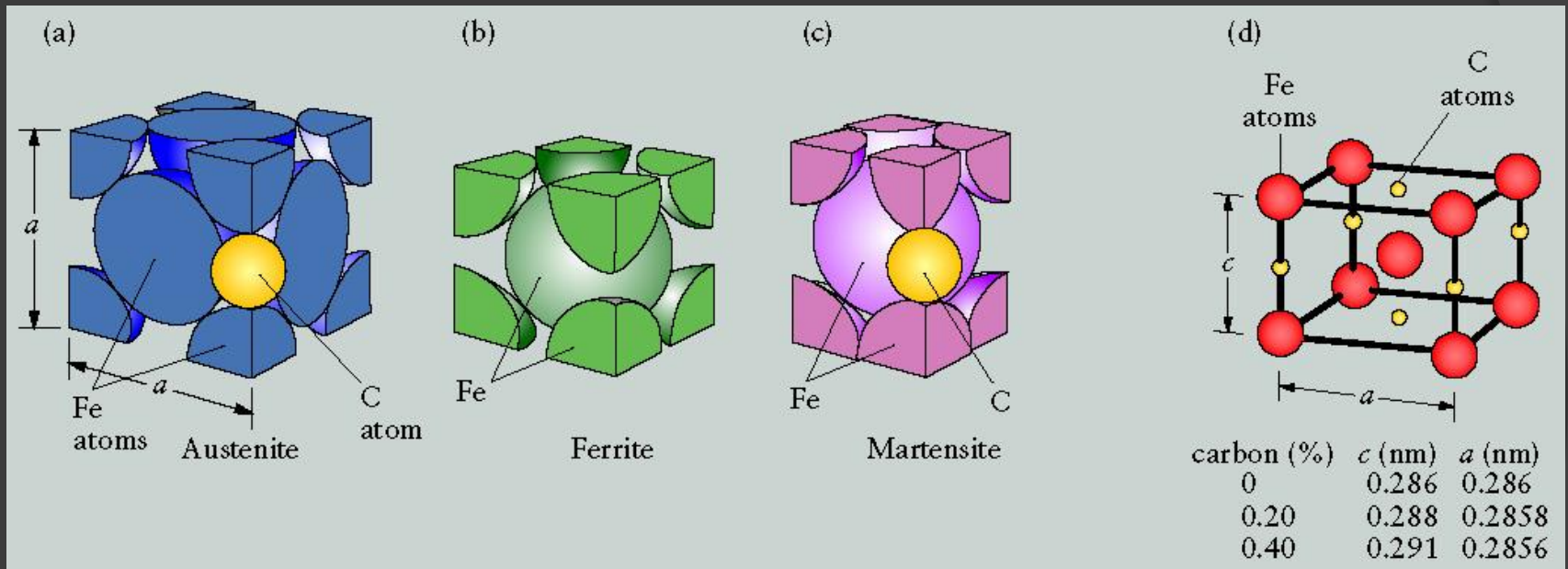


Definition of structures

Various phases that appear on the Iron-Carbon equilibrium phase diagram are as under:

- Austenite
- Ferrite
- Pearlite
- Cementite
- Martensite*
- Ledeburite

Unit Cells of Various Metals



- FIGURE - The unit cell for (a) austenite, (b) ferrite, and (c) martensite. The effect of the percentage of carbon (by weight) on the lattice dimensions for martensite is shown in (d). Note the interstitial position of the carbon atoms and the increase in dimension c with increasing carbon content. Thus, the unit cell of martensite is in the shape of a rectangular prism.

Microstructure of different phases of steel

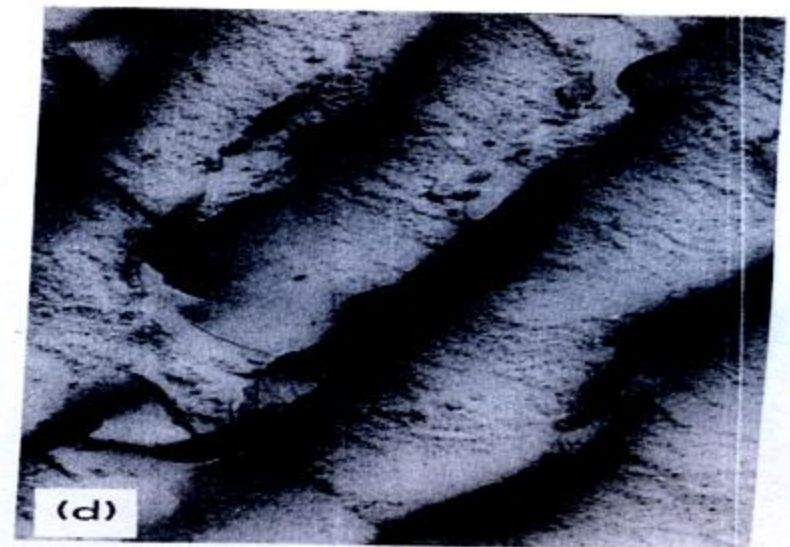
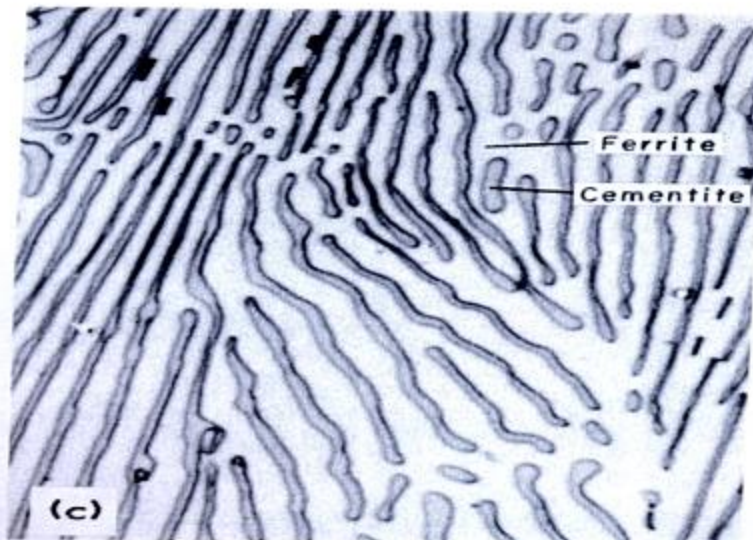
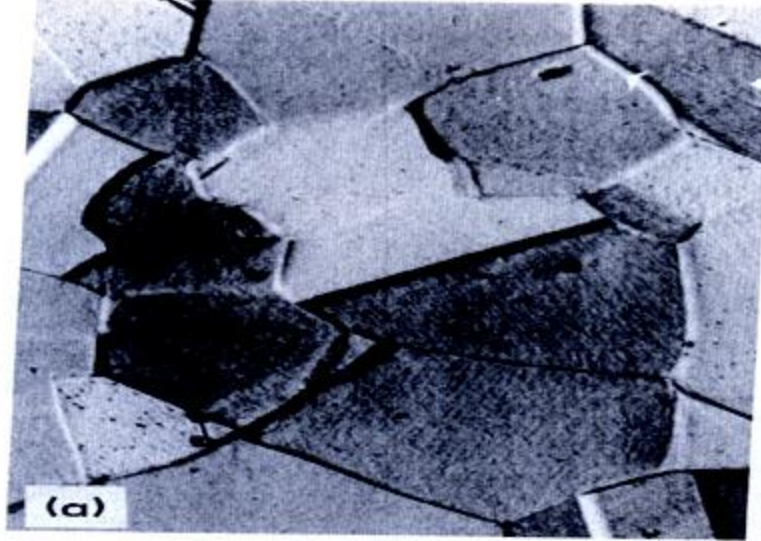


Fig. 7-8 The microstructure of (a) austenite, 500X; (b) ferrite, 100X; (c) pearlite, 2,500X; (d) pearlite, electron micrograph, 17,000X; enlarged 3X in printing. (a, b, and c, Research Laboratory, U.S. Steel Corporation.)

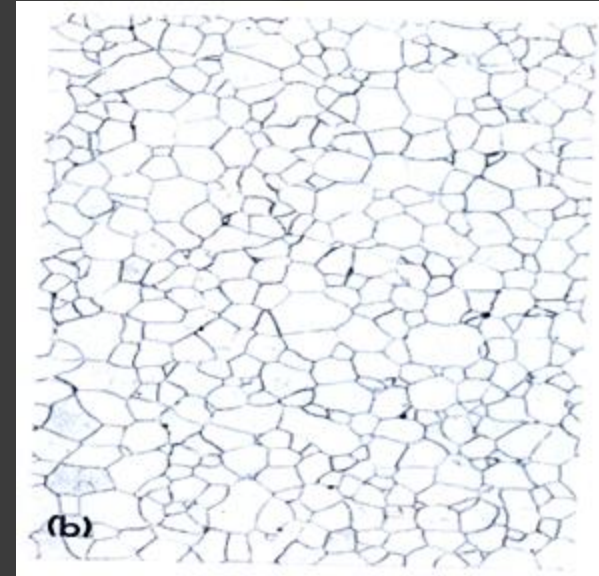
Definition of structures

- **Ferrite** is known as α solid solution.
- It is an interstitial solid solution of a small amount of carbon dissolved in α (BCC) iron.
- stable form of iron below 912 deg.C
- The maximum solubility is 0.025 % C at 723°C and it dissolves only 0.008 % C at room temperature.
- It is the softest structure that appears on the diagram.

Definition of structures

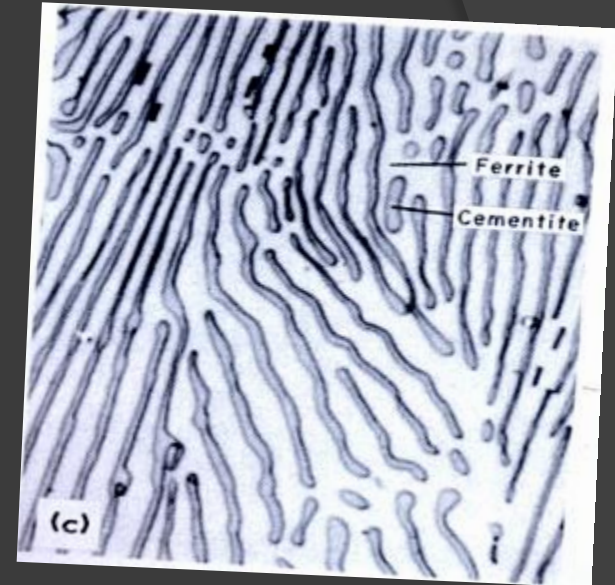
Ferrite

- ⦿ Average properties are:
 - Tensile strength = 40,000 psi;
 - Elongation = 40 % in 2 in;
 - Hardness > Rockwell C 0 or
> Rockwell B 90



Definition of structures

- **Pearlite** is the eutectoid mixture containing 0.80 % C and is formed at 723°C on very slow cooling.
- It is a very fine platelike or lamellar mixture of ferrite and cementite.
- The white ferritic background or matrix contains thin plates of cementite (dark).



Definition of structures

Pearlite

⦿ Average properties are:

- Tensile strength = 120,000 psi;
- Elongation = 20 % in 2 in.;
- Hardness = Rockwell C 20, Rockwell B 95-100, or BHN 250-300.

Definition of structures

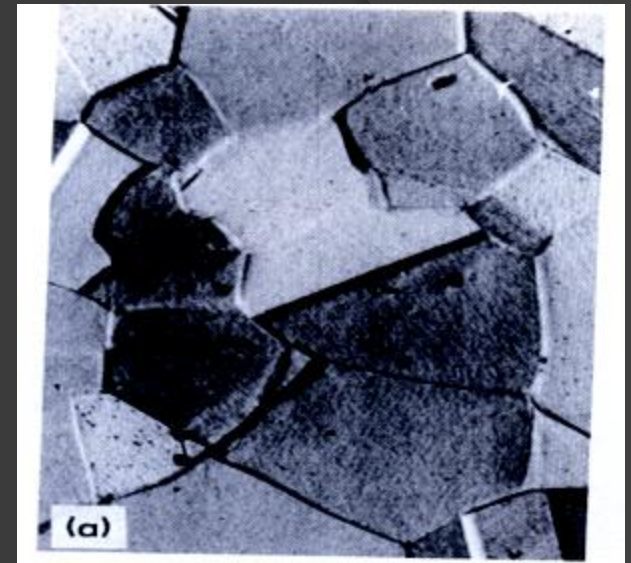
- ⦿ **Austenite** is an interstitial solid solution of Carbon dissolved in γ (F.C.C.) iron.
- ⦿ Maximum solubility is 2.0 % C at 1130°C.
- ⦿ High formability, most of heat treatments begin with this single phase.
- ⦿ It is normally not stable at room temperature. But, under certain conditions it is possible to obtain austenite at room temperature.

Definition of structures

Austenite

⦿ Average properties are:

- Tensile strength = 150,000 psi;
- Elongation = 10 percent in 2 in.;
- Hardness = Rockwell C 40, approx; and
- toughness = high



Definition of structures

- ◎ **Cementite** or iron carbide, is **very hard, brittle** intermetallic compound of iron & carbon, as Fe_3C , contains 6.67 % C.
- ◎ It is the hardest structure that appears on the diagram, exact melting point unknown.
- ◎ Its crystal structure is orthorhombic.
- ◎ It has
 - low tensile strength (approx. 5,000 psi), but
 - high compressive strength.

Definition of structures

- ◎ **Ledeburite** is the eutectic mixture of austenite and cementite.
- ◎ It contains 4.3 percent C and is formed at 1130°C.

Definition of structures

- ⑩ **Martensite** - a super-saturated solid solution of carbon in ferrite.
- ⑩ It is formed when steel is cooled so rapidly that the change from austenite to pearlite is suppressed.
- ⑩ The interstitial carbon atoms distort the BCC ferrite into a BC-tetragonal structure (BCT).; responsible for the hardness of quenched steel

The Iron-Iron Carbide Diagram

- ◎ A map of the temperature at which different phase changes occur on very slow heating and cooling in relation to Carbon, is called Iron- Carbon Diagram.
- ◎ Iron- Carbon diagram shows
 - the type of alloys formed under very slow cooling,
 - proper heat-treatment temperature and
 - how the properties of steels and cast irons can be radically changed by heat-treatment.

Various Features of Fe-C diagram

Phases present

L

α ferrite
BCC structure
Ferromagnetic
Fairly ductile

δ
BCC structure
Paramagnetic

γ austenite
FCC structure
Non-magnetic
ductile

Fe_3C cementite
Orthorhombic
Hard
brittle

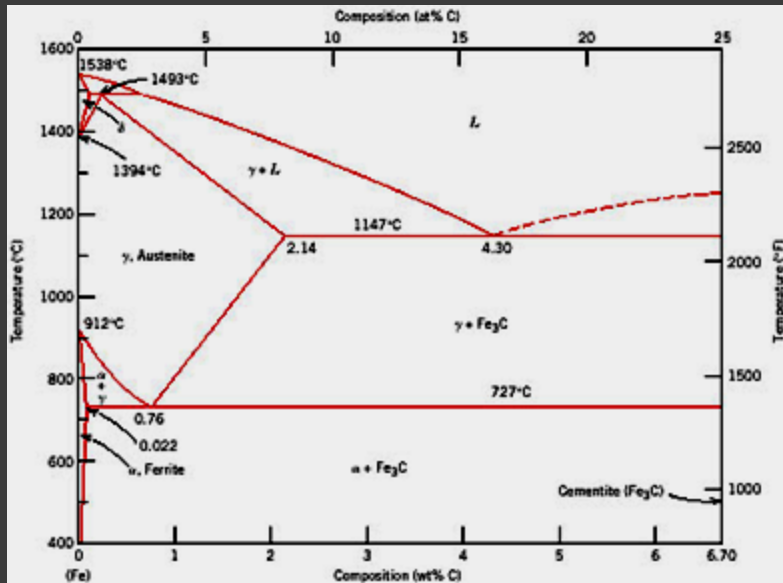


FIGURE 9.22 The iron-iron carbide phase diagram. (Adapted from *Binary Alloy Phase Diagrams*, 2nd edition, Vol. 1, T. B. Massalski, Editor-in-Chief, 1990. Reprinted by permission of ASM International, Materials Park, OH.)

Reactions

Peritectic $L + \delta = \gamma$

Eutectic $L = \gamma + \text{Fe}_3\text{C}$

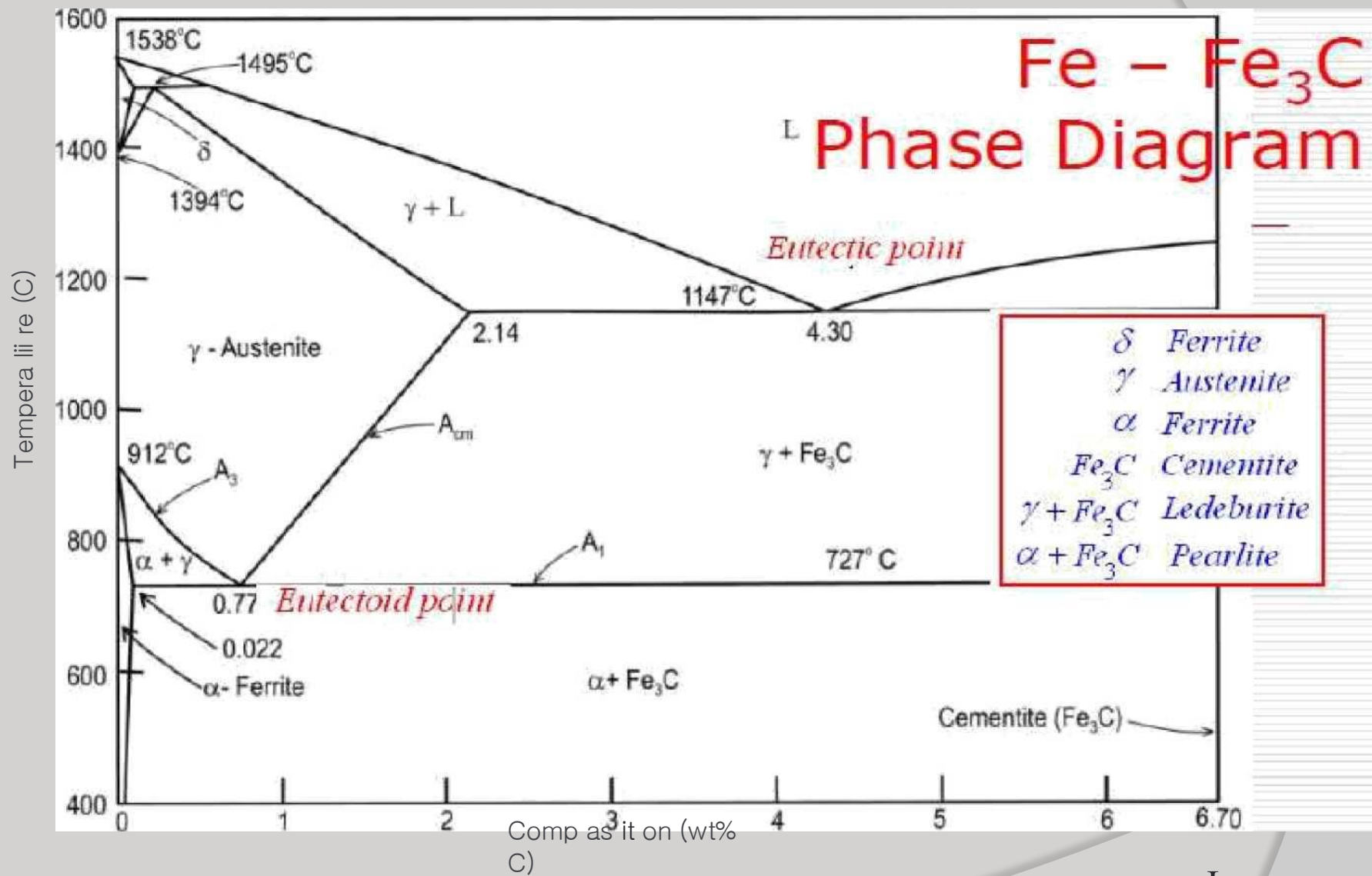
Eutectoid $\gamma = \alpha + \text{Fe}_3\text{C}$

Max. solubility of C in ferrite=0.022%

Max. solubility of C in austenite=2.11%

Three Phase Reactions

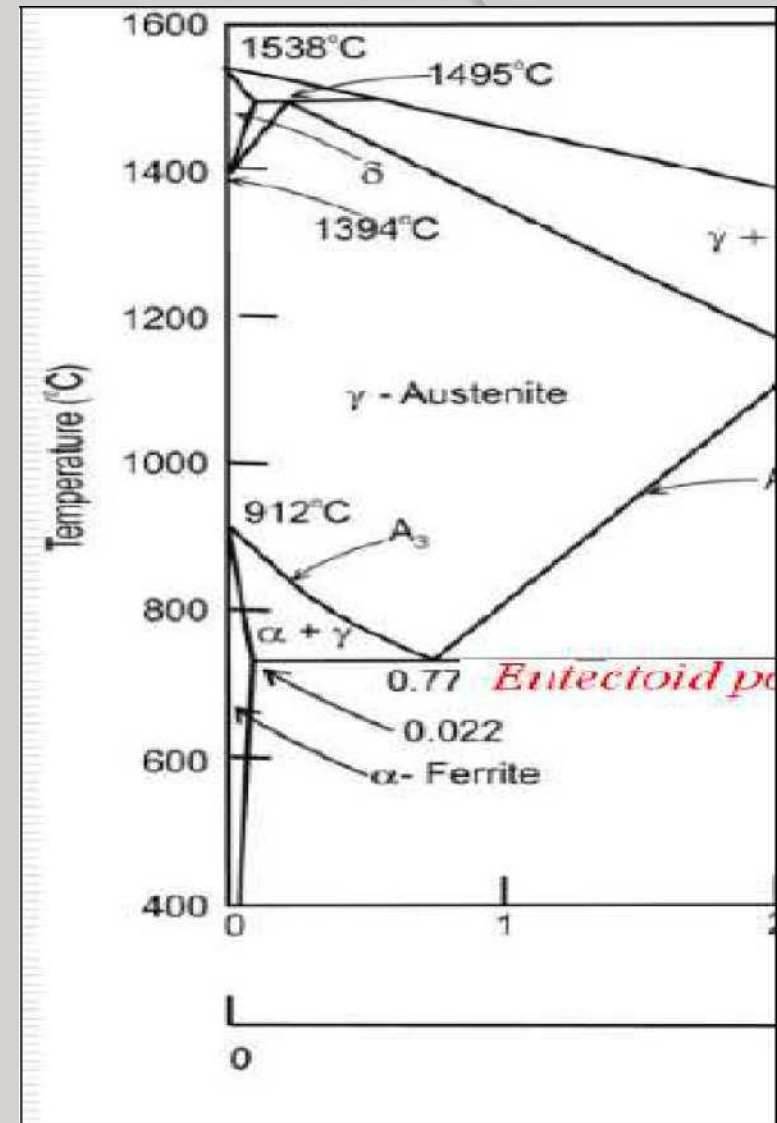
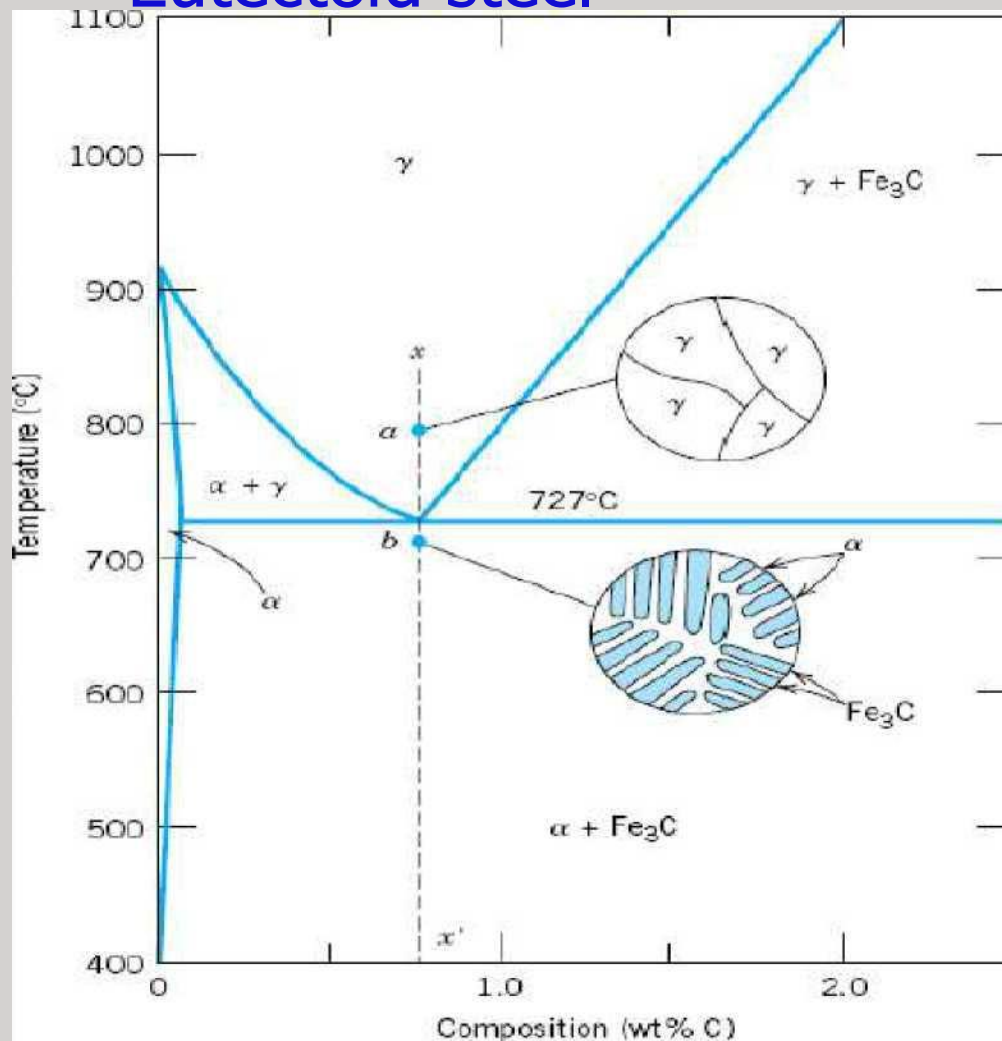
- ⦿ **Peritectic**, at 1490 deg.C, with low wt% C alloys (almost no engineering importance).
- ⦿ **Eutectic**, at 1130 deg.C, with 4.3wt% C, alloys called **cast irons**.
- ⦿ **Eutectoid**, at 723 deg.C with eutectoid composition of 0.8wt% C, two-phase mixture (ferrite & cementite). They are **steels**.



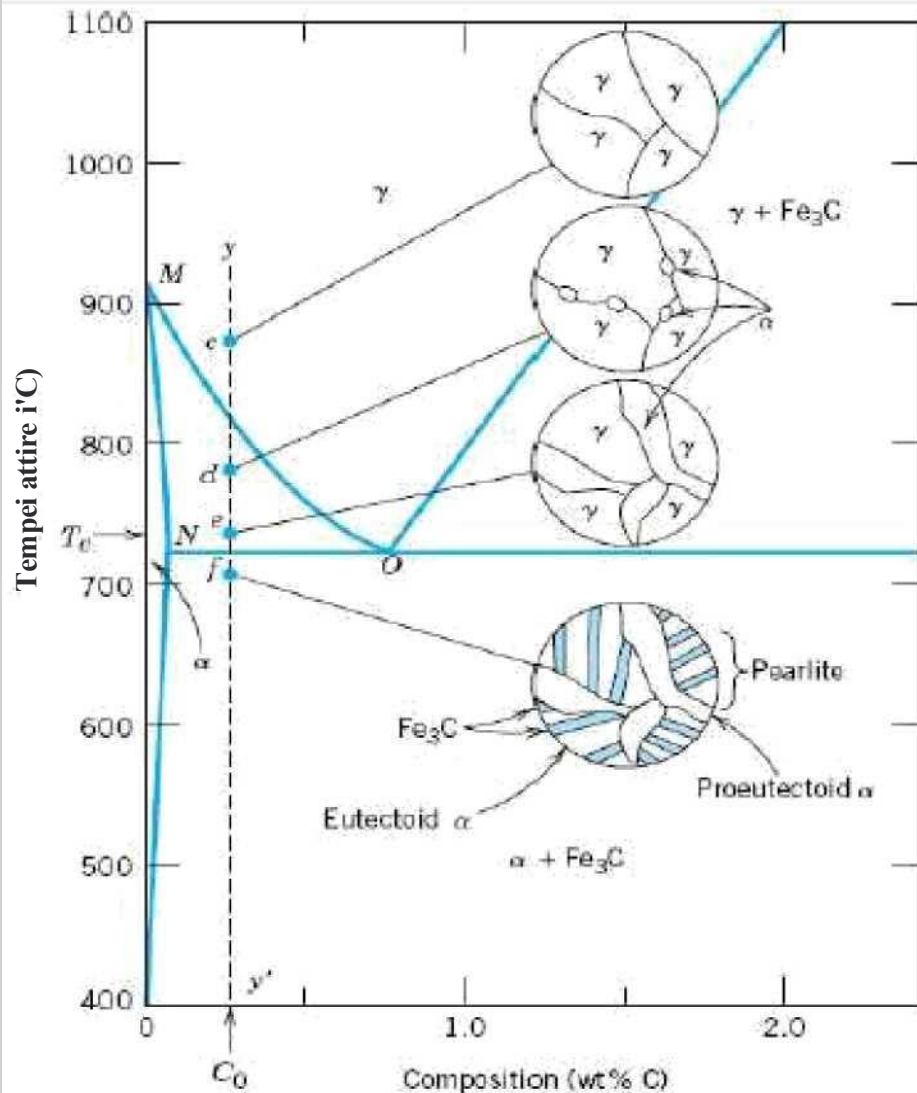
I 5a
Composition fw(V_c Fe C)

J -
too

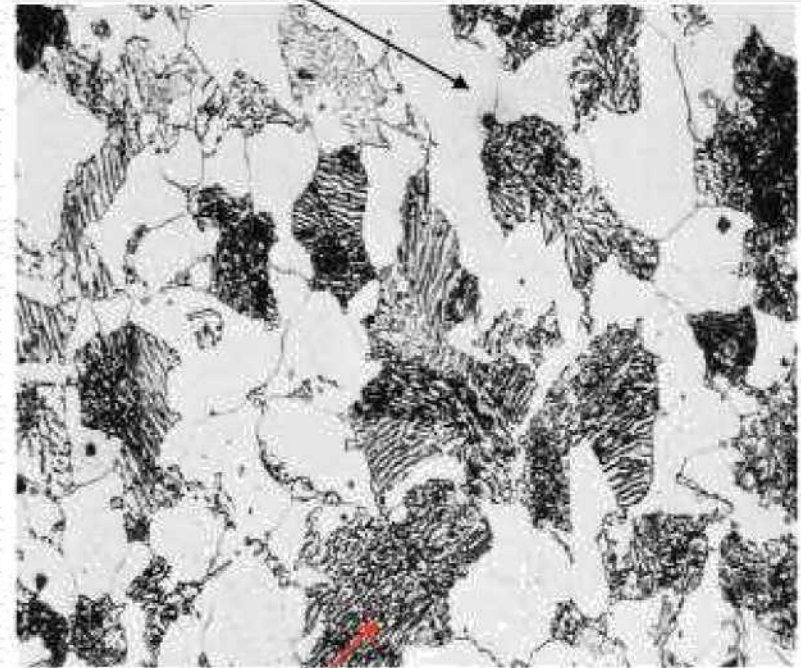
Eutectoid steel



From previous fig. of FE-C Diagram



Proeutectoid
Ferrite



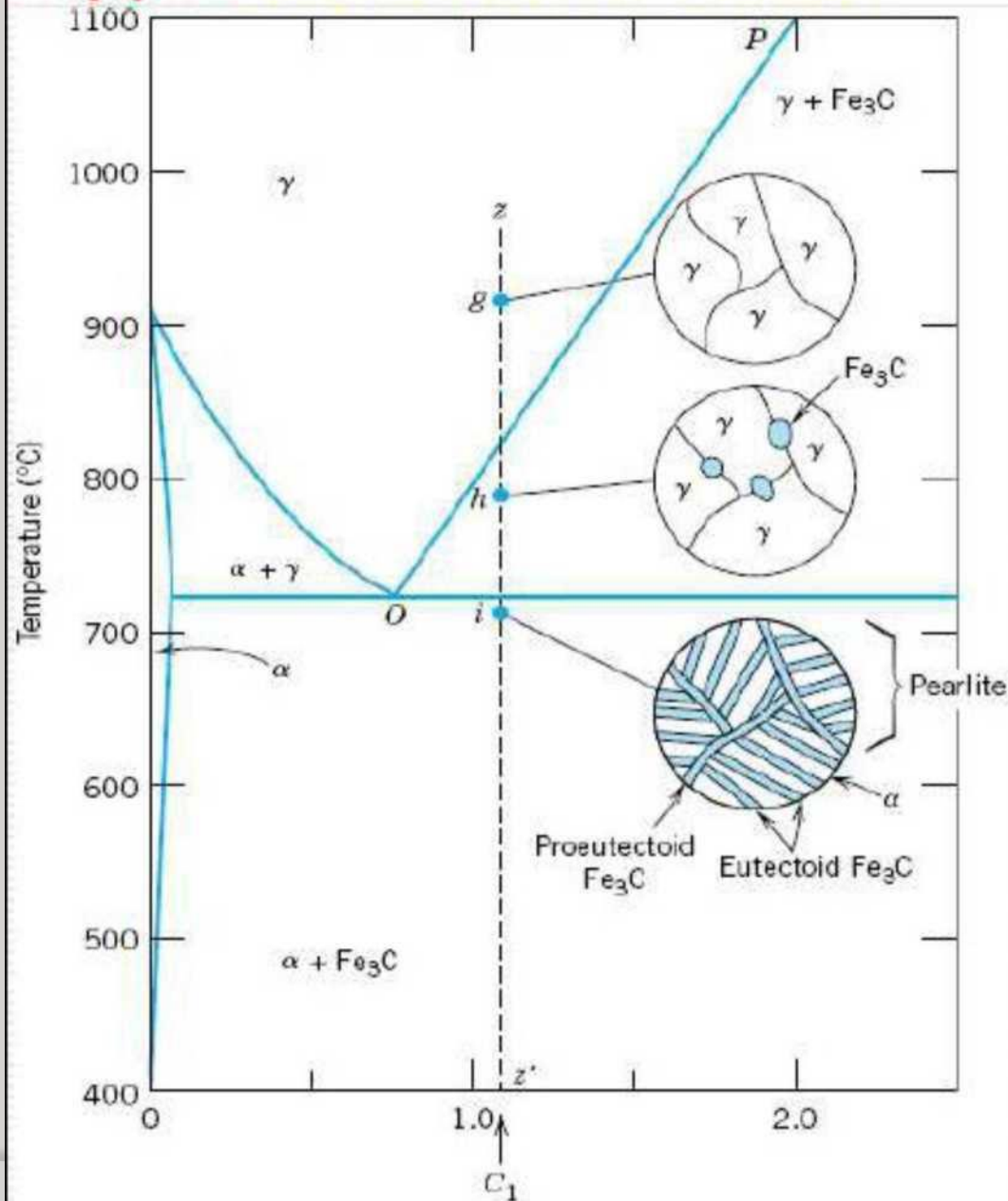
Pearlite

Microstructure of 0.38 wt% C

hypoeutectoid steel

- **Hypo-eutectoid steels:** Steels having less than 0.8% carbon are called *hypo-eutectoid steels* (*hypo* means "less than").
- Consider the cooling of a typical hypo-eutectoid alloy along line $y-y$.
- At high temperatures the material is entirely austenite.
- Upon cooling it enters a region where the stable phases are ferrite and austenite.
- The low-carbon ferrite nucleates and grows, leaving the remaining austenite richer in carbon.
- At 723°C , the remaining austenite will have assumed the eutectoid composition (0.8% carbon), and further cooling transforms it to pearlite.
- The resulting structure, is a mixture of *primary or proeutectoid ferrite* (ferrite that forms before the eutectoid reaction) and regions of pearlite.

Hypereutectoid steel



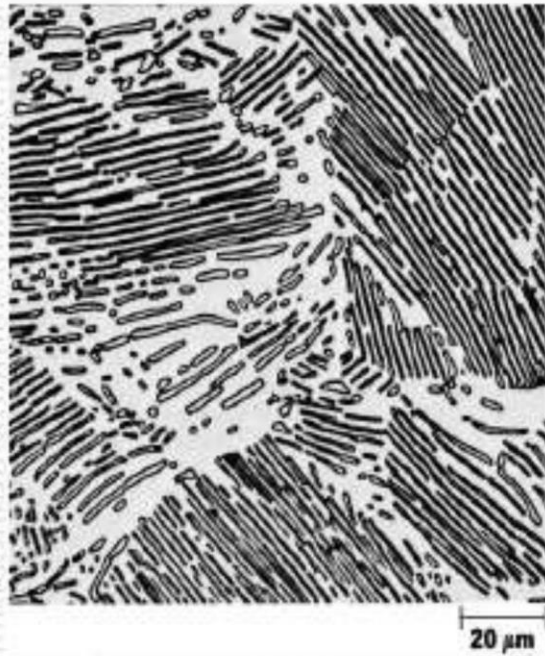
Pearlite



Proeutectoid
cementite

Microstructure of 1.4 wt% C
hypereutectoid steel

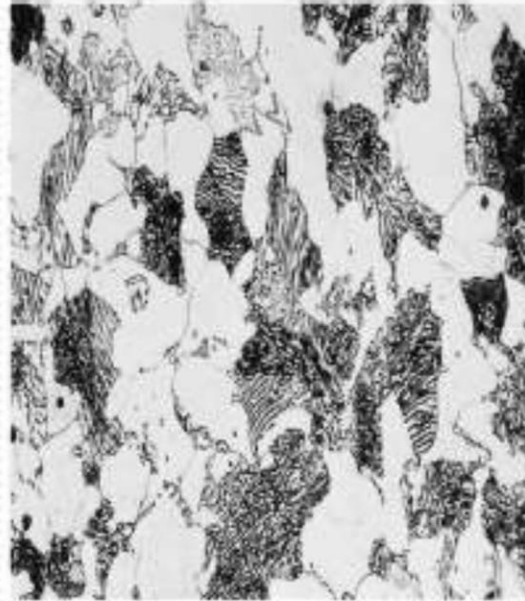
- **Hyper-eutectoid steels** (*hyper means* "greater than") are those that contain more than the eutectoid amount of Carbon.
- When such a steel cools, as along line $z-z'$, the process is similar to the hypo-eutectoid steel, except that the primary or pro-eutectoid phase is now cementite instead of ferrite.
- As the carbon-rich phase nucleates and grows, the remaining austenite decreases in carbon content, again reaching the eutectoid composition at 723°C.
- This austenite transforms to pearlite upon slow cooling through the eutectoid temperature.
- The resulting structure consists of primary cementite and pearlite.
- The continuous network of primary cementite will cause the material to be extremely brittle.



Eutectoid steel

$\alpha + \text{Fe}_3\text{C}$

Pearlite



Hypoeutectoid steel

$\alpha + \text{Fe}_3\text{C}$

Pearlite +
proeutectoid ferrite



Hypereutectoid steel

$\alpha + \text{Fe}_3\text{C}$

Pearlite +
proeutectoid cementite

Three invariant reactions

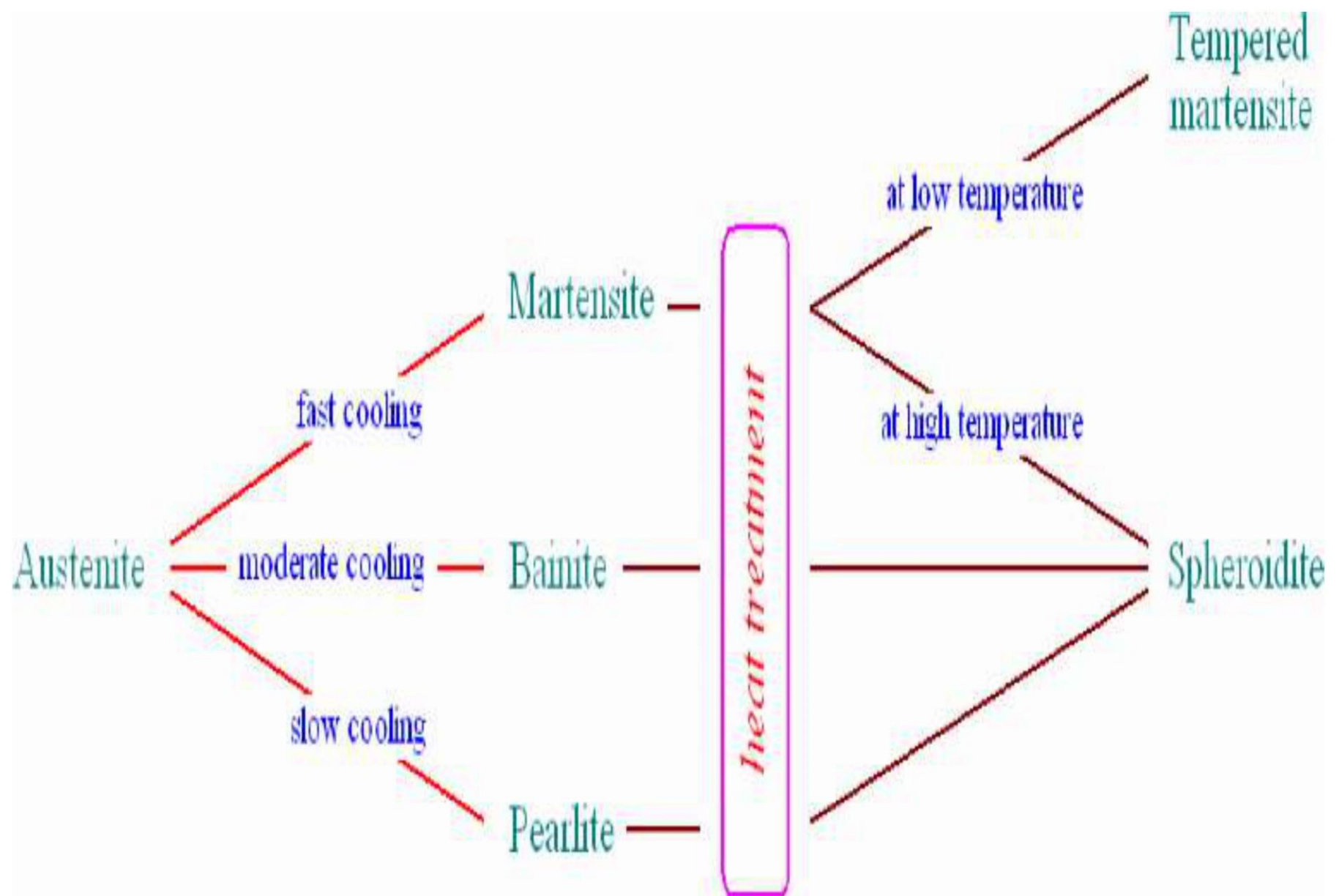
A horizontal line always indicates an invariant reaction in binary phase diagrams

- Peritectic reaction at 1495°C and $0.18\%\text{C}$,
 - $\delta\text{-ferrite} + L \leftrightarrow \gamma\text{-iron (austenite)}$
- Eutectic reaction at 1147°C and $4.3\%\text{C}$,
 - $L \leftrightarrow \gamma\text{-iron} + \text{Fe}_3\text{C (cementite) [ledeburite]}$
- Eutectoid reaction at 727°C and $0.77\%\text{C}$,
 - $\gamma\text{-iron} \leftrightarrow \alpha\text{-ferrite} + \text{Fe}_3\text{C (cementite) [pearlite]}$

DEFINITION OF STRUCTURES

Various phases that appear on the **Iron-Carbon equilibrium phase diagram** are as under:

- Austenite
- Ferrite
- Pearlite
- Cementite
- Martensite*
- Ledeburite



Austenite

Austenite is an interstitial solid solution of Carbon dissolved in F.C.C. iron.

Maximum solubility is 2.0 % C at 1130°C.

High formability, most of heat treatments begin with this single phase. It is normally not stable at room temperature. But, under certain conditions it is possible to obtain austenite at room temperature.

The Austenite to ferrite / cementite transformation in relation to Fe-C diagram

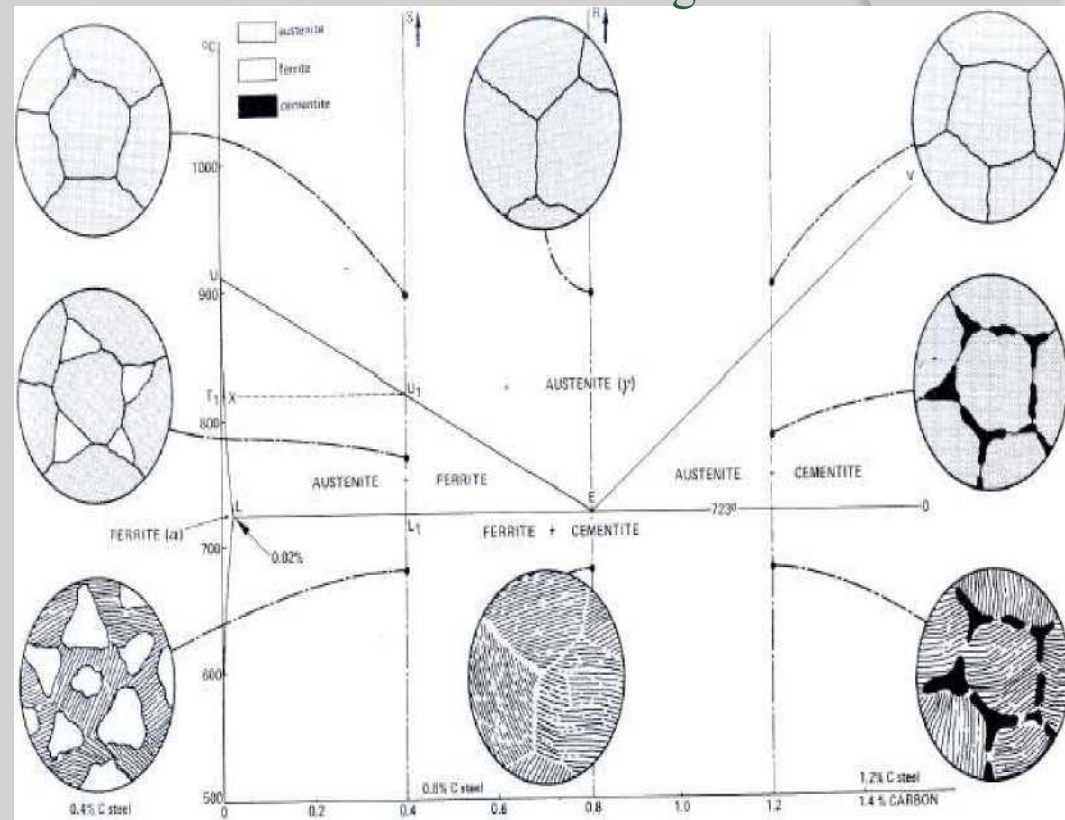


Fig 9.3 The austenite - ferrite / cementite transformation in relation to the iron carbon diagram.

Ferrite

- Ferrite is known as a solid solution.
- It is an interstitial solid solution of a small amount of carbon dissolved in a (BCC) iron. stable form of iron below 912 °C.
- The maximum solubility is 0.025 % C at 723°C and it dissolves only 0.008 % C at room temperature.
- It is the softest structure that appears on the diagram.

Pearlite

- Pearlite is the eutectoid mixture containing 0.80 % C and is formed at 723°C on very slow cooling.
- It is a very fine plate like or lamellar mixture of ferrite and cementite.
- The white ferrite background or matrix contains thin plates of cementite (dark).

Cementite or iron carbide

- Cementite or iron carbide, is very hard, brittle intermetallic compound of iron & carbon, as Fe_3C , contains 6.67 % C.
- It is the hardest structure that appears on the diagram, exact melting point unknown. Its crystal structure is orthorhombic. It has low tensile strength (approx. 5,000 psi), but high compressive strength.

Ledeburite

- Ledeburite is the eutectic mixture of austenite and cementite. It contains 4.3 percent C and is formed at 1130°C .

Martensite

- Martensite - a super-saturated solid solution of carbon in ferrite. It is formed when steel is cooled so rapidly that the change from austenite to pearlite is suppressed.
- The interstitial carbon atoms distort the BCC ferrite into a BC-tetragonal structure (BCT).; responsible for the hardness of quenched steel.

TTT DIAGRAM

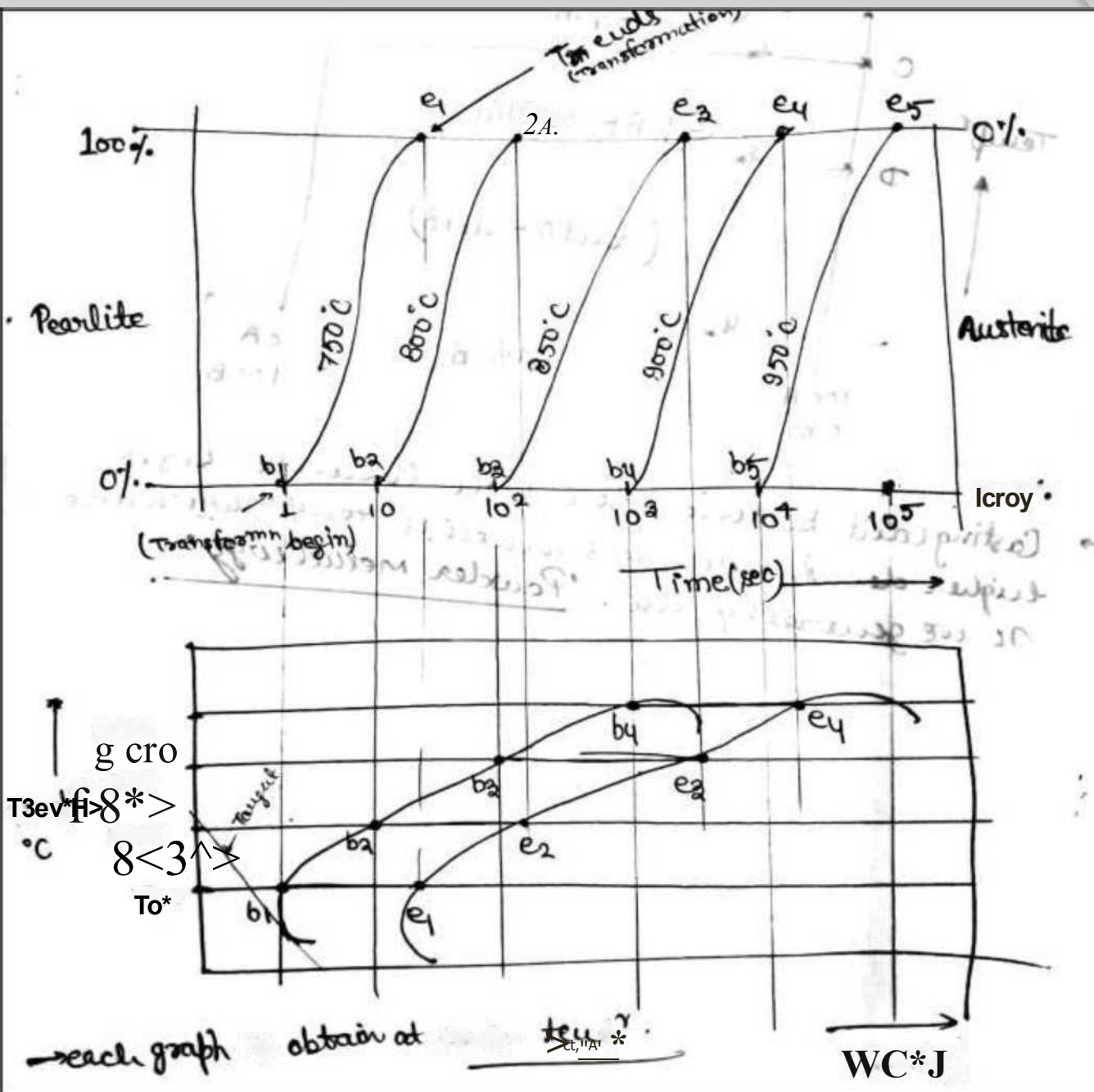
- T.T.T. shows relation between temperature & time taken for decomposition transformations to take place in a metal when the transformation is isothermal.
- Assess decomposition of austenite in a heat treatable steel.
- Provides information for the process of austenite decomposition under non-equilibrium conditions. (Transformation of austenite to the time & temperature conditions.)

DIFFERENCE BETWEEN IRON-CARBON & TTT

- Study of Fe-C diagram shows study of cooled steels under non equilibrium conditions.
- Doesn't involve **reaction condition** during heat treatment of steel.
- It only shows **phases & resulting microstructure** corresponding to equilibrium conditions.
- Fixing to **austenitizing temperature & predicting phases eventually obtained at given % of C & temperature.**
- Microstructure & properties of steel depends upon rate of cooling.
- As **cooling rate increases** transformation temperatures are **lowered & metastable (non-equilibrium) phases** are formed.
- At a very high rate of cooling of steel produces **Martensite (nonequilibrium phase)**

STEPS TO CONSTRUCT TTT DIAGRAM

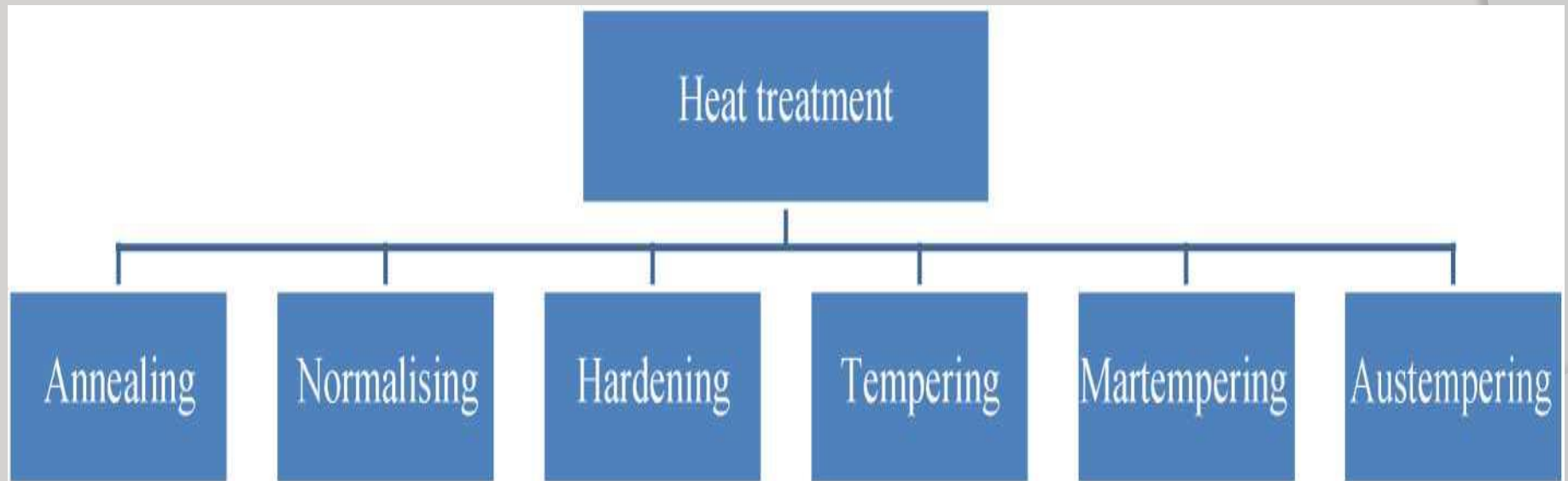
- Obtain large number of relatively small specimens. Place the sample in a molten salt bath held at the austenitizing temperature of 1080°C . Specimen are kept in a salt bath for a long period of time to form complete austenite.
- When austenitized, specimen is transferred to other salt bath at temperature of 810°C .
- After specimen react isothermally, quenched in cold water/ iced brine.
- As the specimen is quenched in cooled water, isothermal reaction stops & remaining austenite suddenly transforms into martensite.
- Reaction curve forms when large no. of specimen isothermally reacted for varying time periods.
- Finally **data obtained from a series of isothermal reaction curves (TTT)** for the whole temperature range of austenite instability for a given composition of steel.



- The transformation rate is inversely proportional to time at any temperature.
- Tangent to C-curves (TTT Curves) gives us the cooling rate.
- Based on the above curves the time required to transformation of Austenite-pearlite can be determined.
- TTT diagram denotes that it is drawn at different cooling rates, hence it is not recorded as an equilibrium diagram as Iron carbide diagram.

HEAT TREATMENT

- An operation or combination of operations which involves heating & cooling of a metal/alloy in solid state to obtain desirable conditions & properties.



PURPOSE OF HEAT TREATMENT

Heat treatment is carried out to

- (1) Cause relief of internal stresses developed during cold working, welding, casting, forging etc.
- (2) Harden & strengthen metals
- (3) Improve machinability
- (4) Change grain Size
- (5) Soften metals for further working as in wire drawing or cold rolling
- (6) Improve ductility & toughness
- (7) Increase heat, wear & corrosion resistance of materials
- (8) Improve electrical & magnetic properties
- (9) Homogenize the structure

Heat treatment techniques include

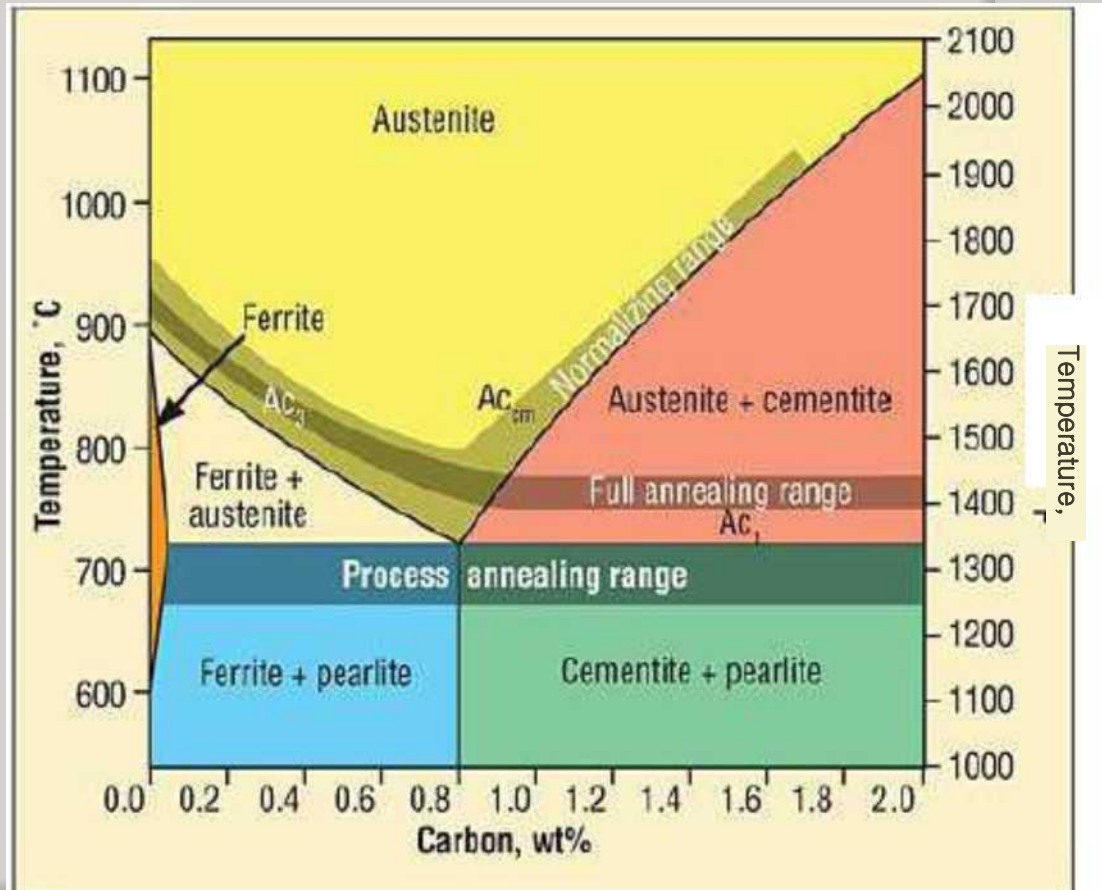
Annealing,
Hardening,
Tempering,
Quenching.

Annealing: Process Annealing

In this treatment, steel (or any material) is heated to a temperature below the lower critical temperature, and is held at this temperature for sufficient time and then cooled.

Cooling rate is of little importance as the process is being done at sub critical temperatures.

The purpose of this treatment is to reduce hardness and to increase ductility of cold-worked steel so that further working may be carried easily.



Annealing: Stress Relieving

As the name suggests, this process is employed to relieve internal stresses. No microstructural changes occur during the process.

Internal stresses are those stresses which can exist within a body in the absence of external forces. These are also known as residual stresses are locked-in stresses.

These stresses are developed in operations like:

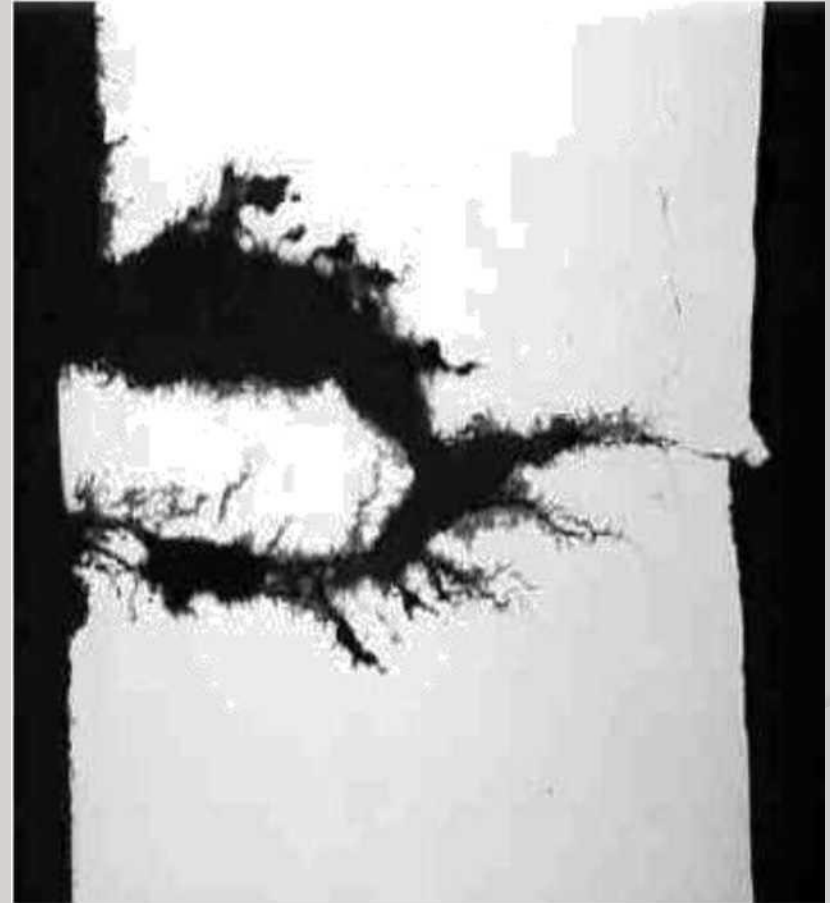
Solidification of castings, welding, machining, grinding, shot peening, surface hammering, cold working, case hardening, electroplated coatings, precipitation and phase transformation.

These internal stresses under certain conditions can have adverse effects: example: Steels with residual stresses under corrosive environment fail with stress corrosion cracking.

These stresses also enhance the tendency of steels towards warpage and dimensional instability.

Fatigue strength is reduced considerably when residual tensile stresses are present in steel.

The problems associated with internal stresses are more difficult in brittle materials than in ductile materials.



Stress - Corrosion Cracking

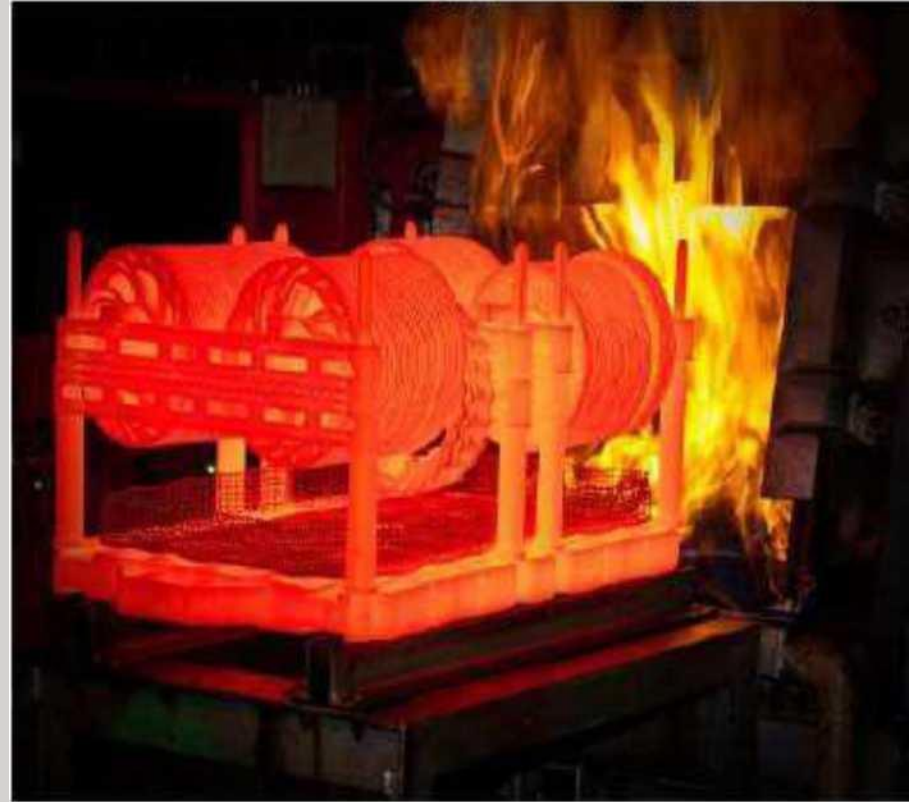
The process of stress relieving consists of heating materials uniformly to a temperature below the lower critical temperature, holding at this temperature for sufficient time, followed by uniform cooling

Uniform cooling is of utmost importance as non-uniform cooling will itself result in the development of internal stresses. Thus the main purpose of stress relieving will be lost.

Plain carbon steels and low alloy steels generally temperature is limited to 600 °C. Higher temperature is used for high alloy steels.

S J

The extent of the stresses relieved depend upon the temperature employed.

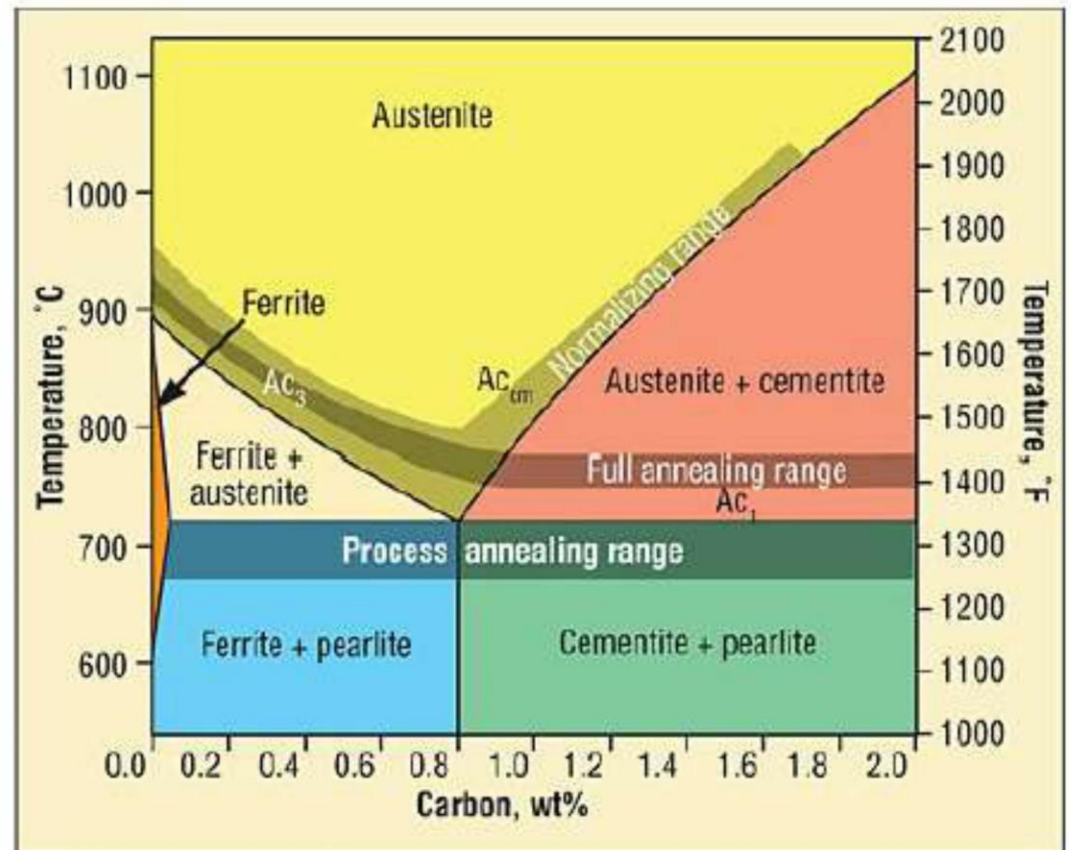


Annealing: Normalizing

➤ Normalizing is similar to full annealing, except steel is generally cooled in still air.

➤ The normalizing consists of heating steel to about 40-55 °C above critical temperature (Ac_3 or Ac_{cm}), and holding for proper item and then cooling in still air or slightly agitated air to room temperature.

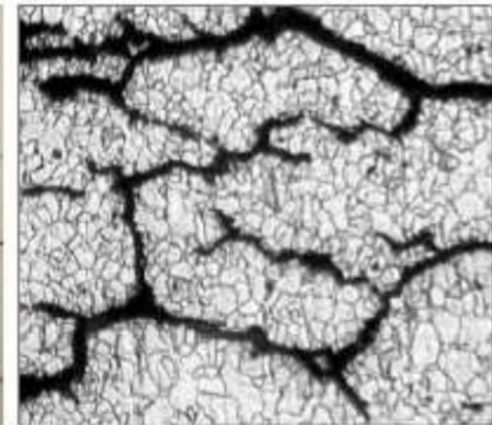
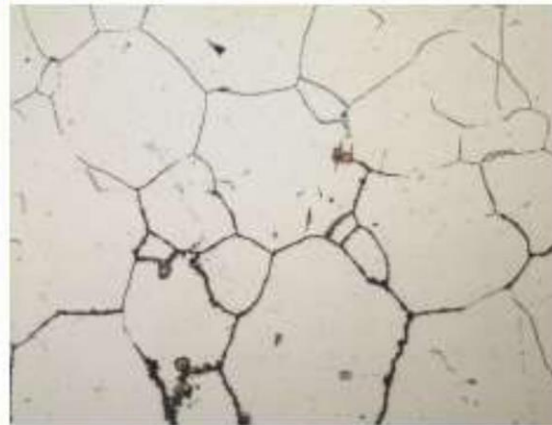
➤ In some special cases, cooling rates can be controlled by either changing air temperature or air volume.



➤ Normalized steels are generally stronger and harder than fully annealed steels.

➤ Steels are soft in annealed condition and tend to stick during machining. By normalizing, an optimum combination of strength and softness is achieved, which results in satisfactory level of Machinability in steels.

➤ Normalizing is the effective way to eliminate the carbide network.



➤ Normalized treatment is frequently applied to steel in order to achieve any one or more of the objectives, namely:

- To refine the grain structure,
- To obtain uniform structure,
- To decrease residual stresses,
- To improve Machinability.



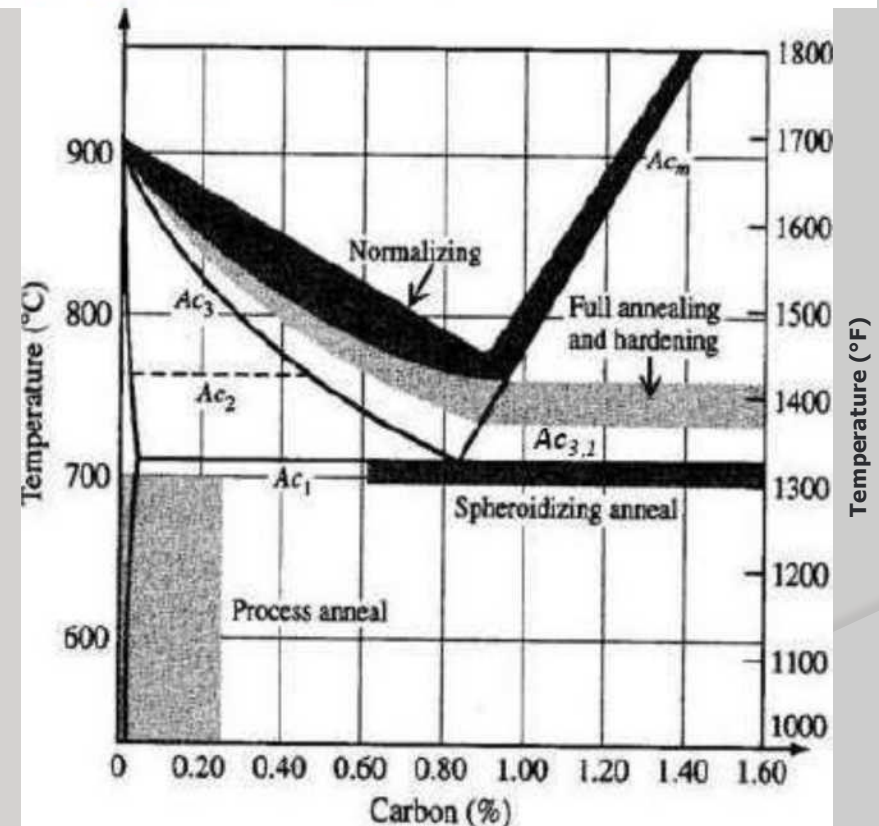
Hardening

Hardening and Hardness are two very different things. One is a process of heat treatment and other is an extrinsic property of a material.

Hardening is a heat treatment process in which steel is rapidly cooled from austenitising temperature. As a result to hardening, the hardness and wear resistance of steel are improved.

Hardening treatment generally consists of heating to hardening

temperature, holding at that temperature, followed by rapid cooling such as quenching in oil or water or salt baths.



Temperature Range for Heat Treatment of Carbon Steels

The high hardness developed by this process is due to the phase transformation accompanying rapid cooling. Rapid cooling results in the transformation of austenite at considerably low temperature into non equilibrium products.

The hardening temperature depends on chemical composition. For plain carbon steels, it depends on the carbon content alone. Hypoeutectoid steels are heated to about 30 - 50 °C above the upper critical temperature, whereas eutectoid and hyper eutectoid steels are heated to about 30 - 50 °C above lower critical temperature.

Ferrite and pearlite transform to austenite at hardening temperature for hypoeutectoid steel. This austenite transforms to martensite on rapid quenching from hardening temperature. The presence of martensite accounts for high hardness of quenched steel.

Hardening is applied To cutting tools and machine pans where high hardness and wear resistance are important.

The Process Variables:

Hardening Temperature: The steel should be heat treated to optimum austenitising temperature. A lower temperature results lower hardness due to incomplete transformation austenite. If this temperature is too high will also results lower hardness due to a coarse grained structure.

Soaking Time: Soaking time at hardening temperature should be long enough to transform homogenous austenite structure. Soaking time increases with increase in section thickness and the amount of alloying element.

Delay in quenching: After soaking, the steel is immediately quenched. Delay in quenching may reduce hardness due to partial transformation of austenite.

Type of quenching medium also has a profound effect, which will be discussed briefly

The main purpose of hardening tool steel is to develop high hardness. This enables tool steel to cut other metals. High hardness developed by this process also improves wear resistance. Gears, shafts and bearings. Tensile strength and yield strength are improved considerably by hardening structural steels.

Because of rapid cooling, high internal stresses are developed in the hardened steel. Hence these steels are generally brittle.

Hardening in general is followed by another treatment known as tempering which reduces internal stresses and makes the hardened steel relatively stable.

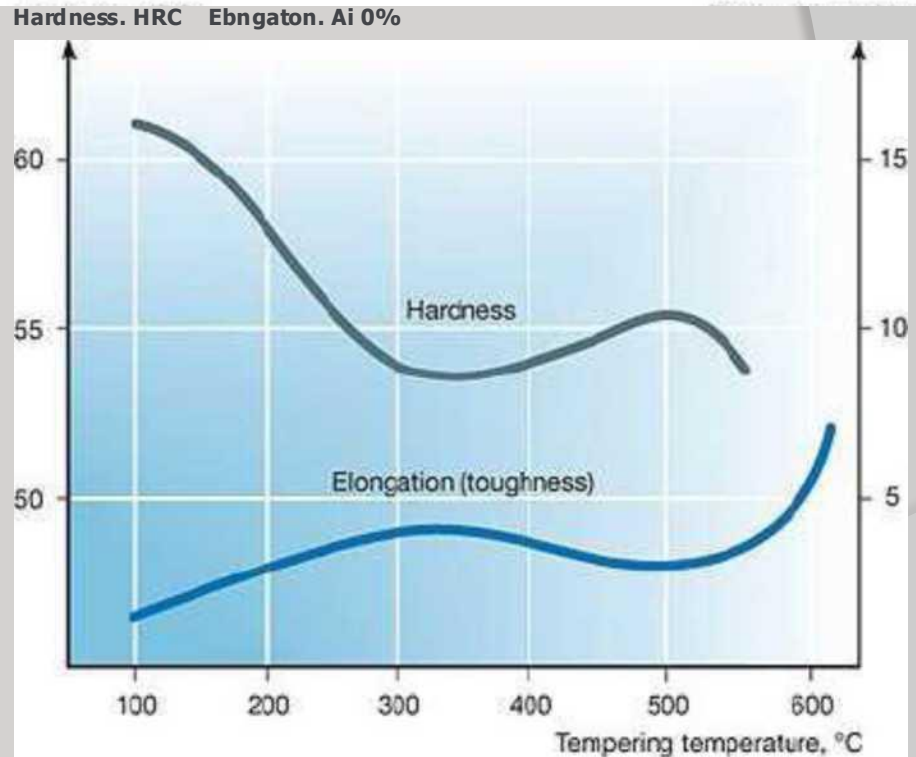


Tempering

Hardened steels are so brittle that even a small impact will cause fracture. Toughness of such a steel can be improved by tempering. However there is small reduction in strength and hardness.

Tempering is a sub-critical heat treatment process used to improve the toughness of hardened steel.

Tempering consists of reheating of hardened steel to a temperature below Lower critical temperature and is held for a period of time, and then slowly cooled in air to room temperature.

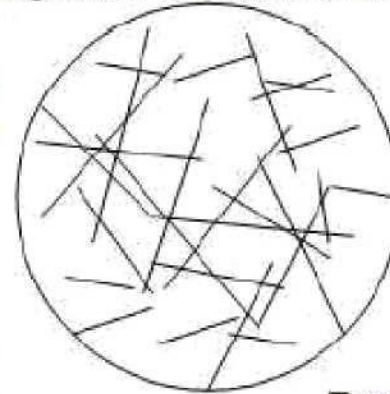


At tempering temperature, carbon atoms diffuse out and form fine cementite and softer ferrite structure left behind. Thus the structure of tempered steel consists of ferrite and fine cementite.

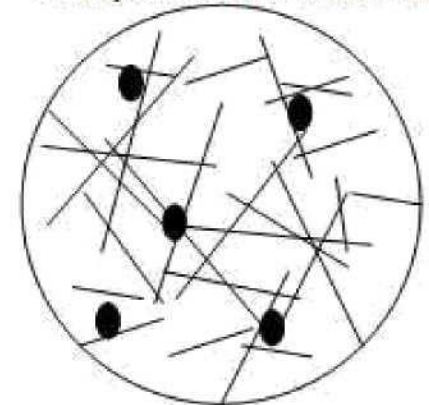
➤ Thus tempering allows to precipitate carbon as very fine carbide and allow the microstructure to return to BCC

➤ The temperatures are related to the function of the parts. Cutting tools are tempered between 230 – 300 °C. If greater ductility and toughness are desired as in case of shafts and high strength bolts, the steel is tempered in the range of 300 – 600 °C.

Toughened Microstructure



Tempered Microstructure



THANK YOU

Metal Forming

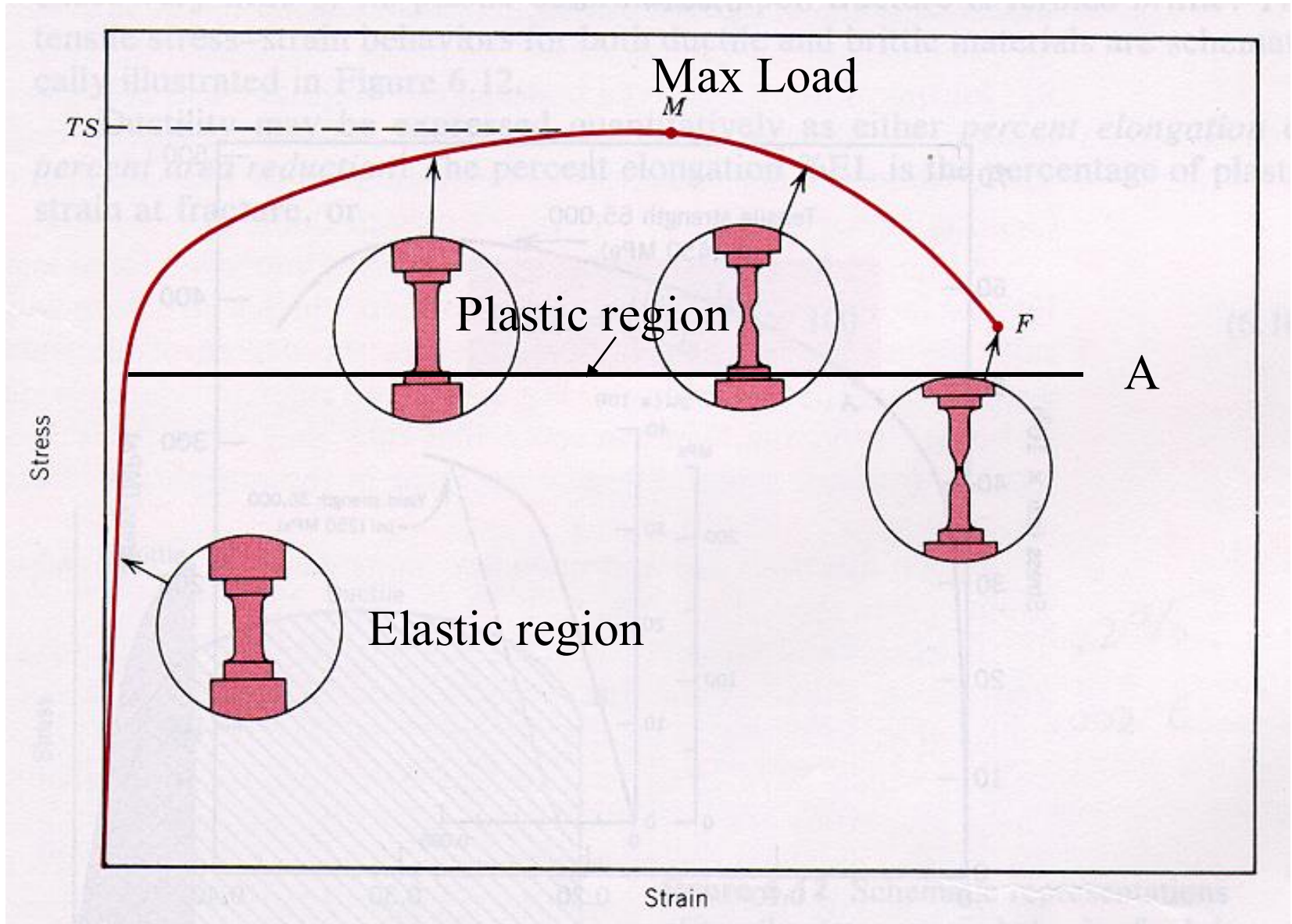


Outline

- Fundamentals of Metal Forming
- Bulk deformation
- Sheet metalworking



Plastic Deformation



Forming Operations

Hot Working

- Above recrystallization temperature
- Allows for large deformations
- Less energy required than for cold working
- May result in surface oxidation

Cold Working

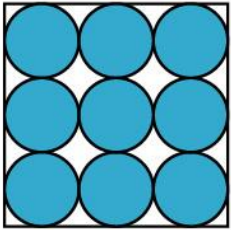
- Below recrystallization temperature
- Increases strength, but decreases ductility and corrosion resistance
- Better control of dimensions
- Better surface finish

Temperature in metal forming

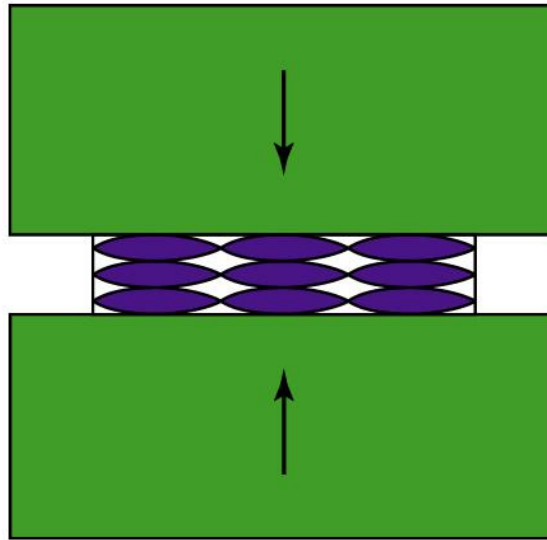
- Better accuracy
- Better Surface finishing
- Strain hardening increase strength
- Costs

Preferred Orientation

(a)

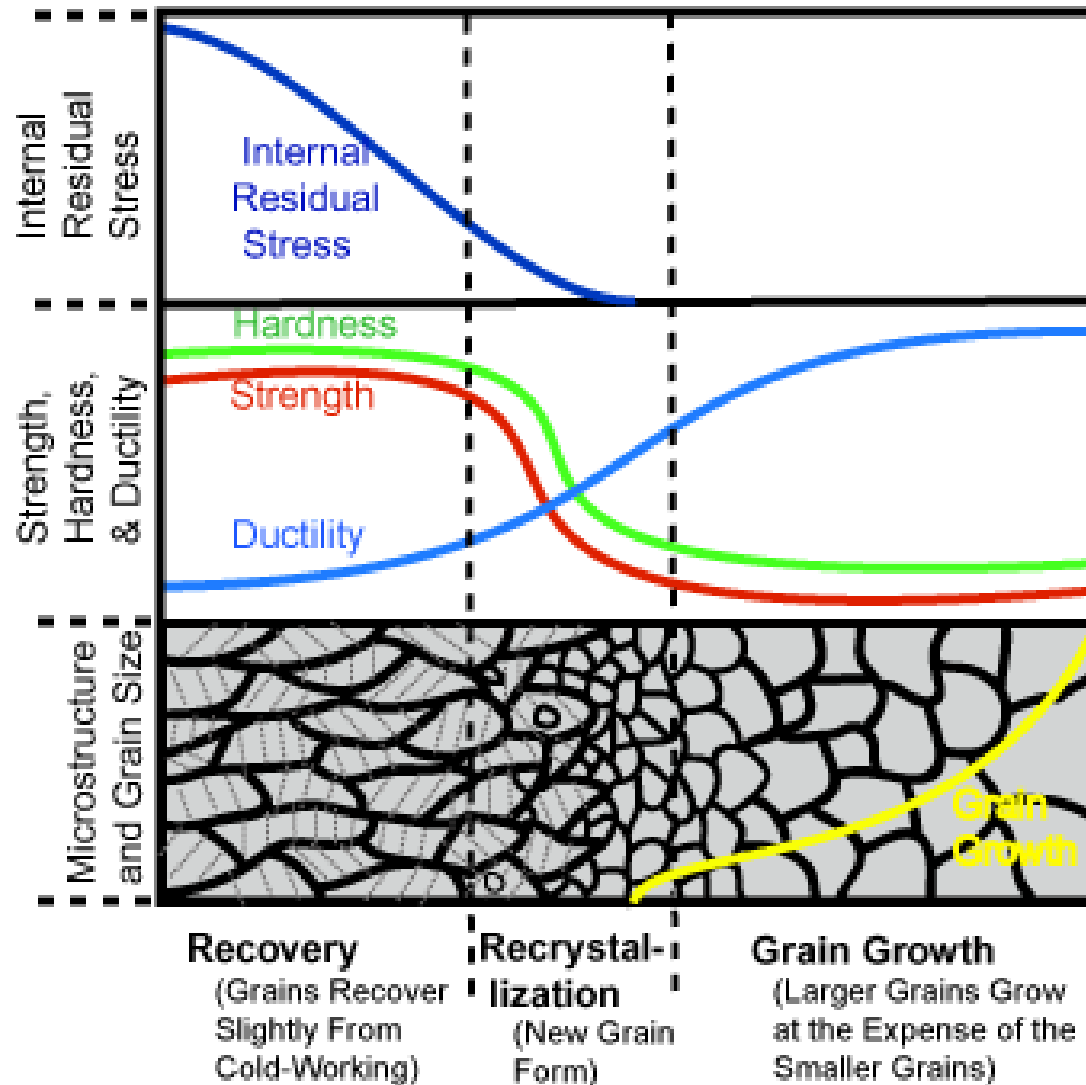


(b)



Plastic deformation of idealized (equiaxed) grains in a specimen subjected to compression (such as occurs in the rolling or forging of metals): (a) before deformation; and (b) after deformation. Note the alignment of grain boundaries along a horizontal direction; this effect is known as preferred orientation.

Recrystallization



Cold Working

- Metal forming performed at room temp or slightly above (below their recrystallisation temperature)

Advantage

Better accuracy

Better Surface finishing

Strain hardening increase strength

Costs

Disadvantage

Higher forces are required

Care the surface

Ductility is limited.

Cracks

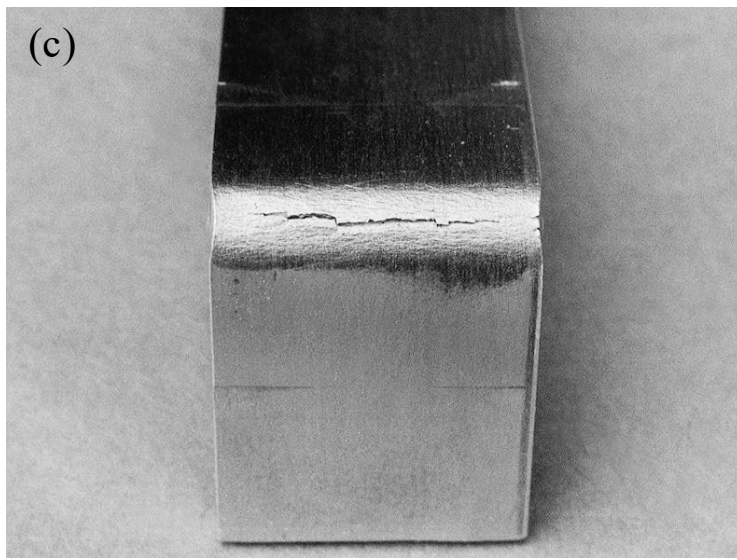
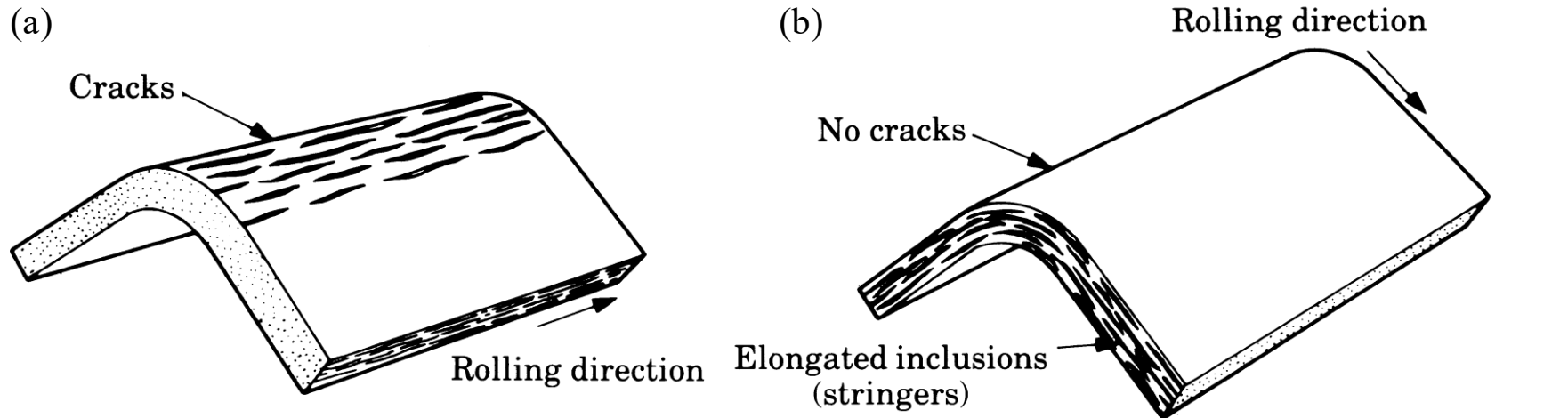


Figure (a) and (b) The effect of elongated inclusions (stringers) on cracking, as a function of the direction of bending with respect to the original rolling direction of the sheet. (c) Cracks on the outer surface of an aluminum strip bent to an angle of 90° . Note the narrowing of the top surface due to the Poisson effect.

Hot working

- T above Recrystallization Temp.
- $0.5T_m < T < 0.75T_m$

Advantage

- The shape can be significantly altered
- Lower forces are required
- Fracture can be reduced
- Strength properties are generally isotropic
- No strengthening of the part occurs from work hardening

Disadvantage

Lower dimensional accuracy, higher energy, poorer surface finish
Shorter tool life

Warm Working

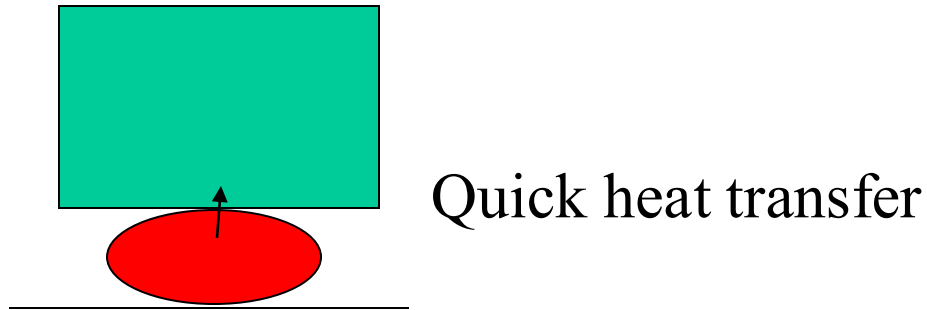
- $0.3T_m < T < 0.5T_m$

Advantage

- Lower forces are required
- More intricate work geometries possible
- The need for annealing may be reduced

Isothermal

- For high speed steel, titanium alloy, good hot hardness



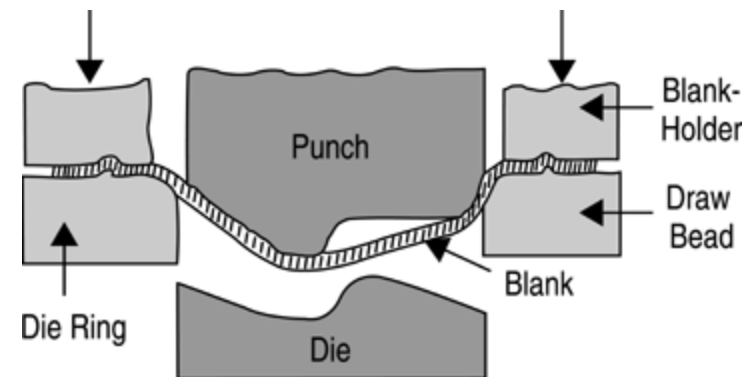
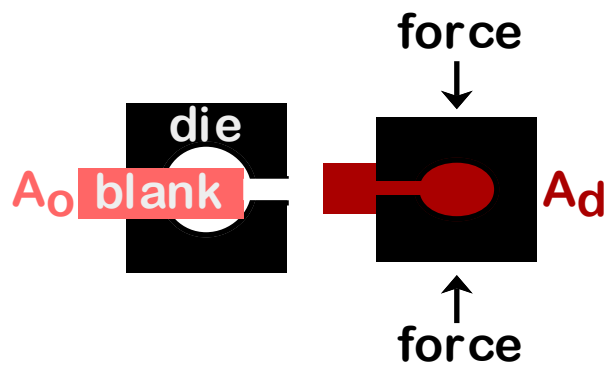
- Raising strength leading to high residual stress



Cracking

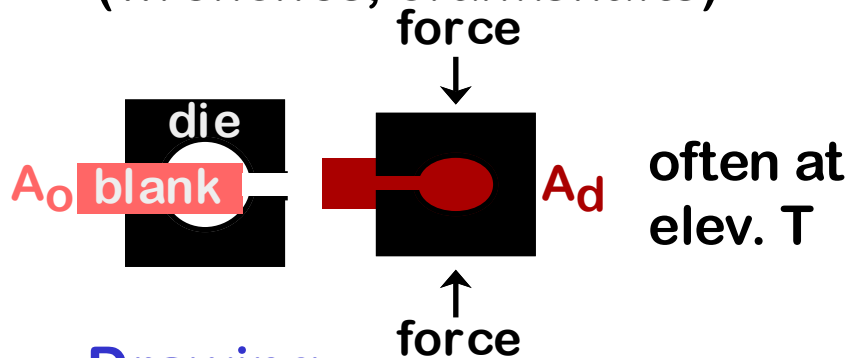
Metal forming process

- 1. Bulk deformation processes
- 2. Sheet metal working processes

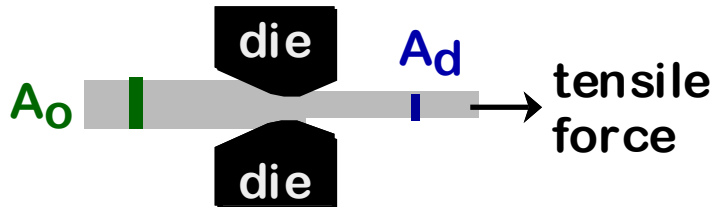


Bulk Deforming Processes

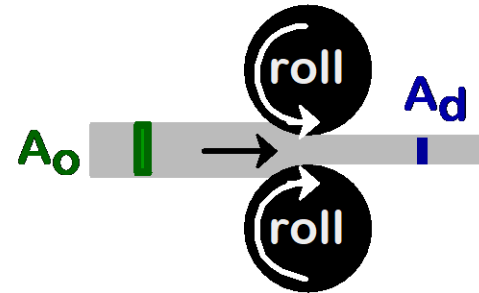
- **Forging**
(wrenches, crankshafts)



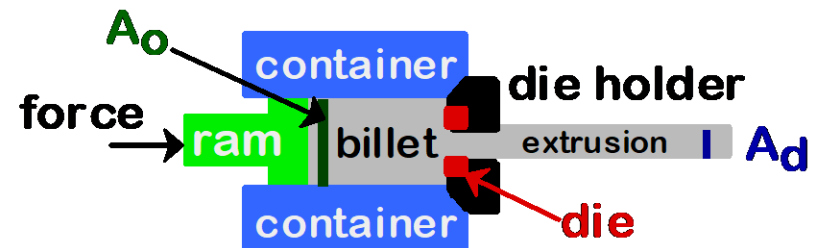
- **Drawing**
(rods, wire, tubing)



- **Rolling**
(I-beams, rails)

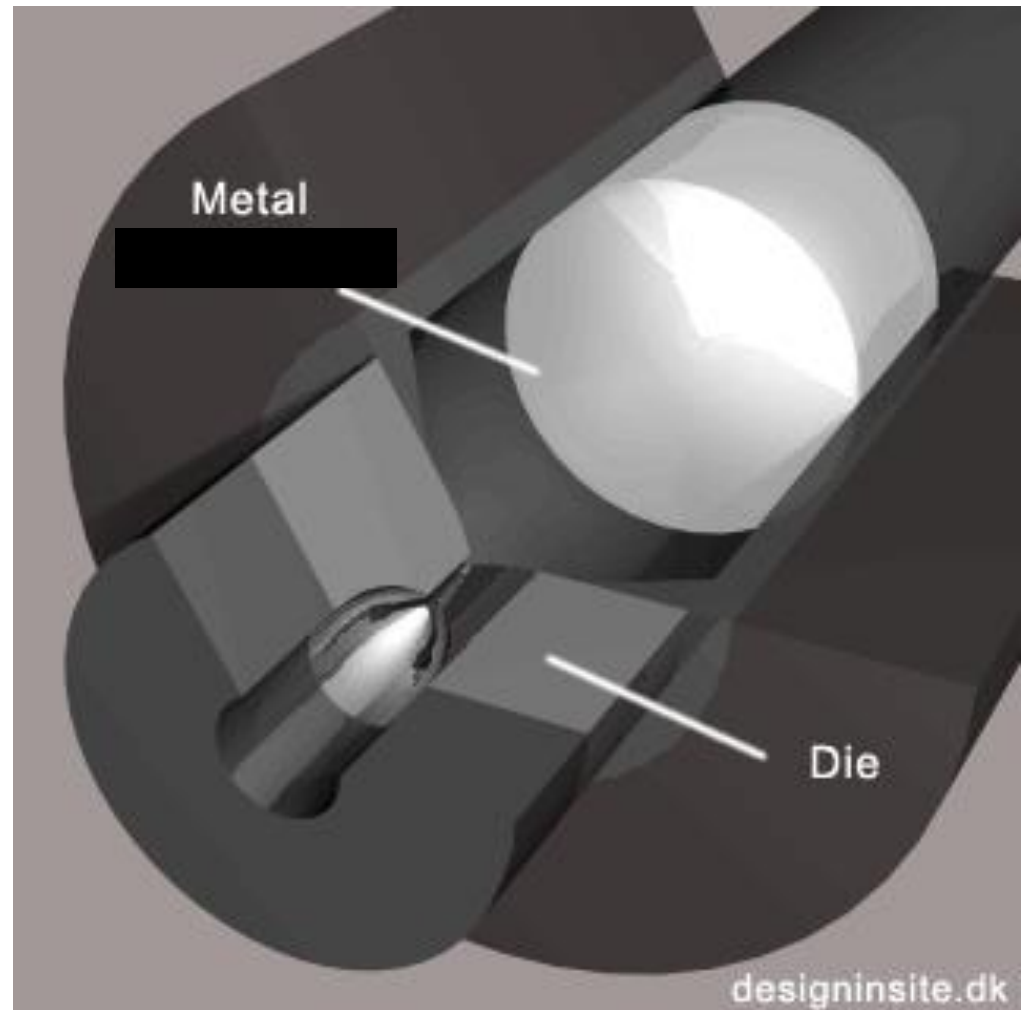


- **Extrusion**
(rods, tubing)



Extrusion

Metal forced/squeezed out through a hole (die)

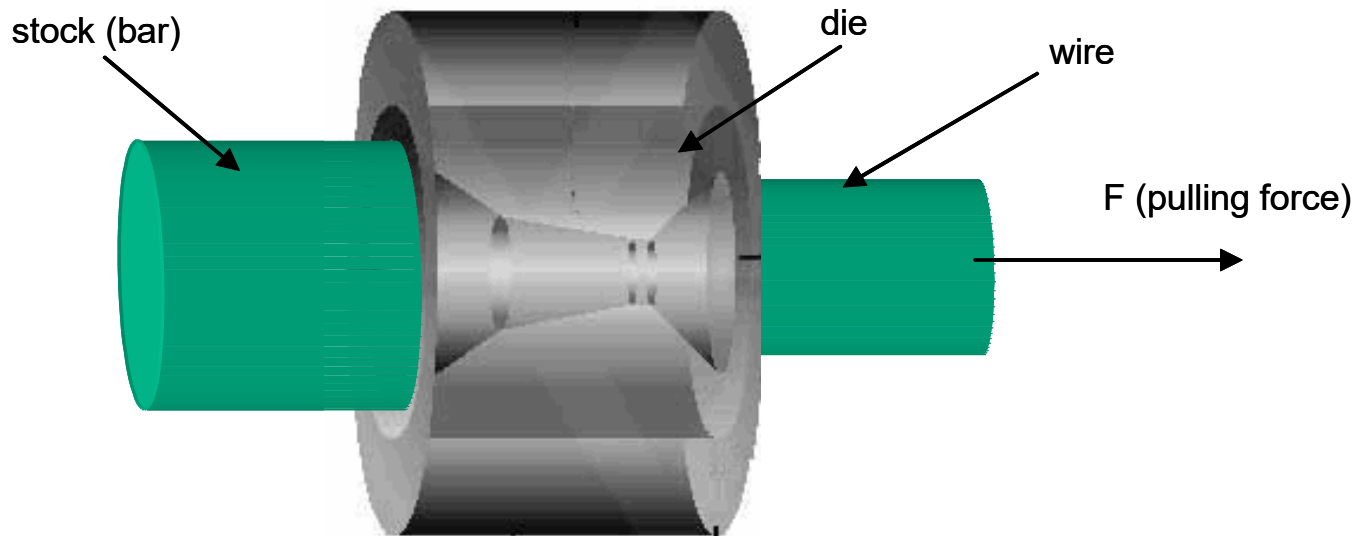


Typical use: ductile metals (Cu, Steel, Al, Mg), Plastics, Rubbers
Common products:

Al frames of white-boards, doors, windows, ...

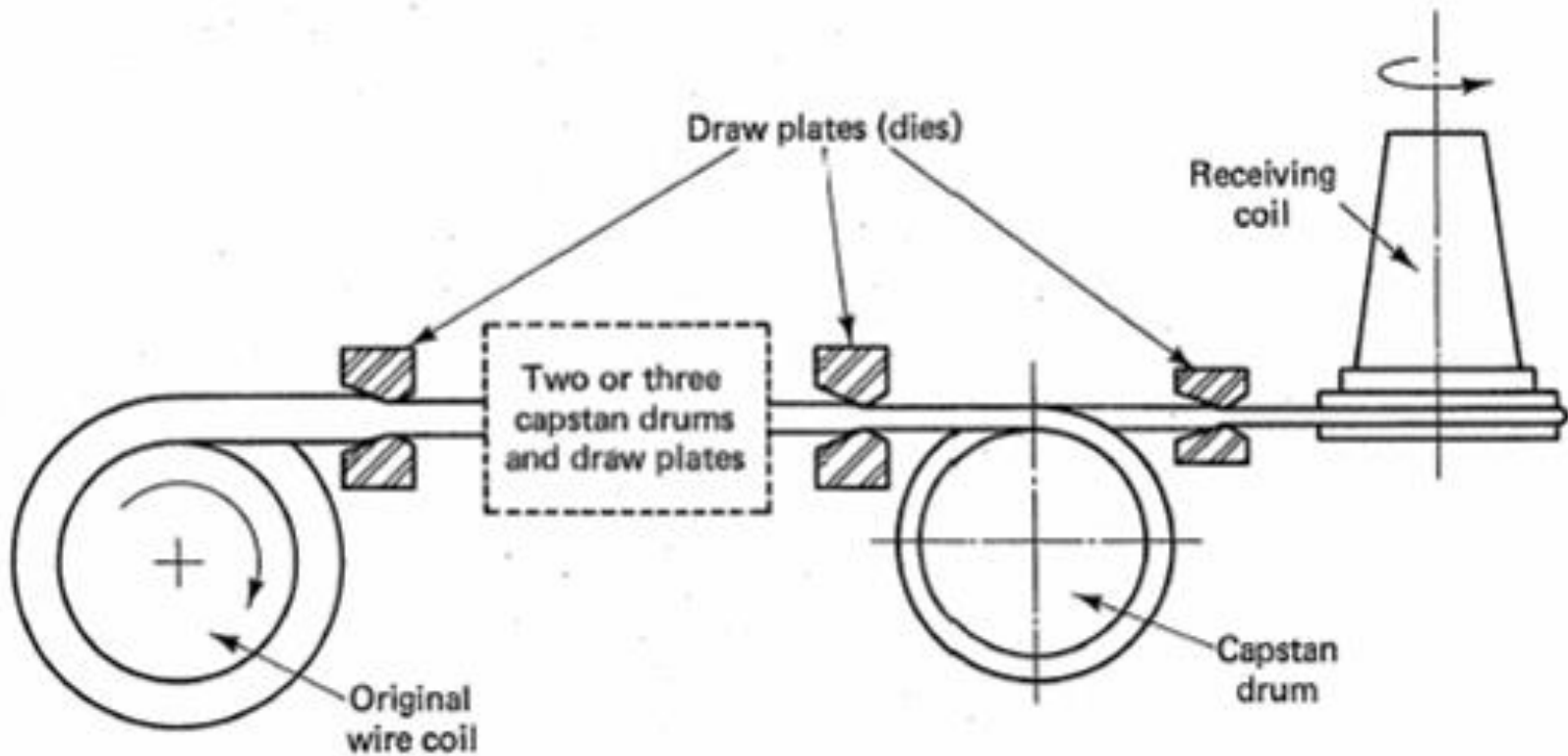
Drawing

Similar to extrusion, except: *pulling force* is applied



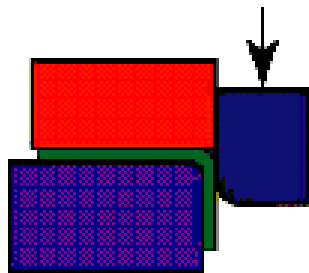
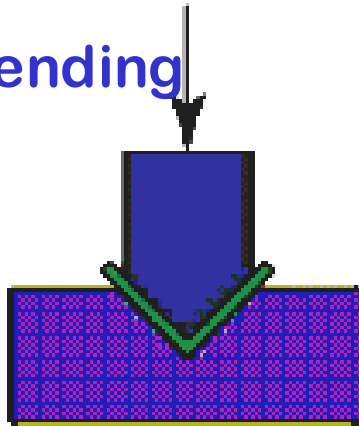
Commonly used to make wires from round bars

Continuous Drawing

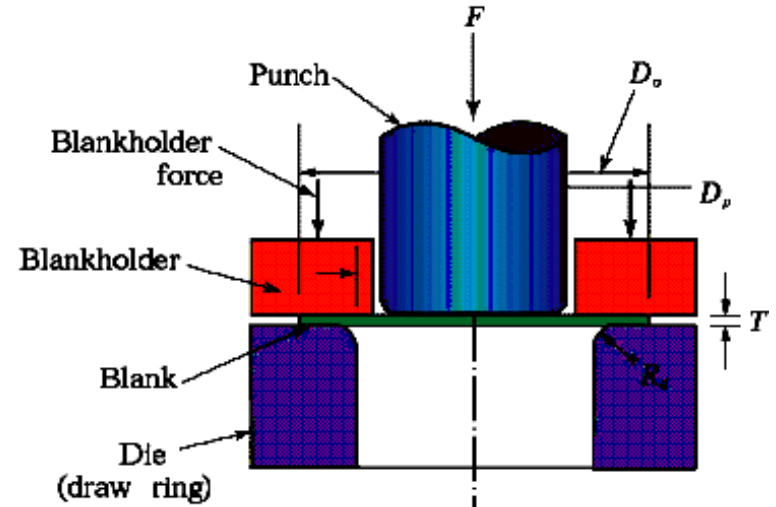


Sheet Metalworking

- Bending



- Deep drawing



- Shearing

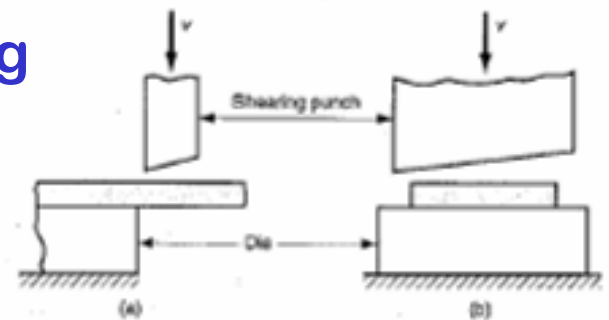
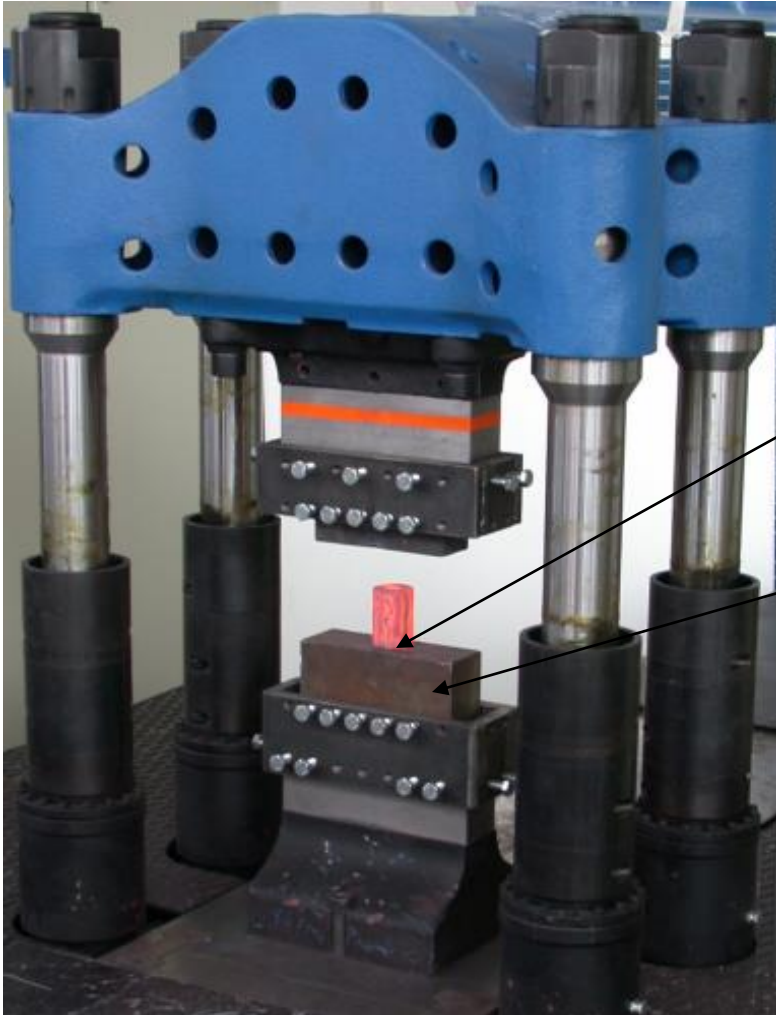


FIGURE 22.3 Shearing operation: (a) side view of the shearing operation; (b) front view of power shears equipped with inclined upper cutting blade. Symbol v indicates motion.

Forging

[Heated] metal is beaten with a heavy hammer to give it the required shape



Hot forging,

open-die

Forging

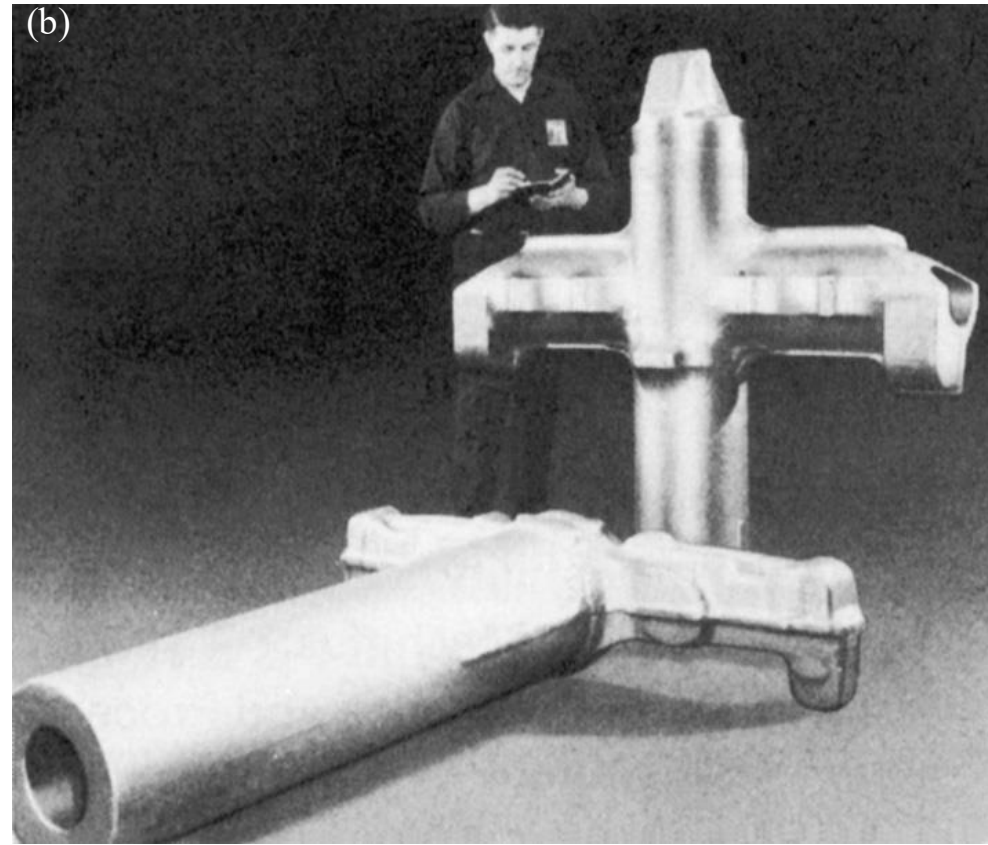
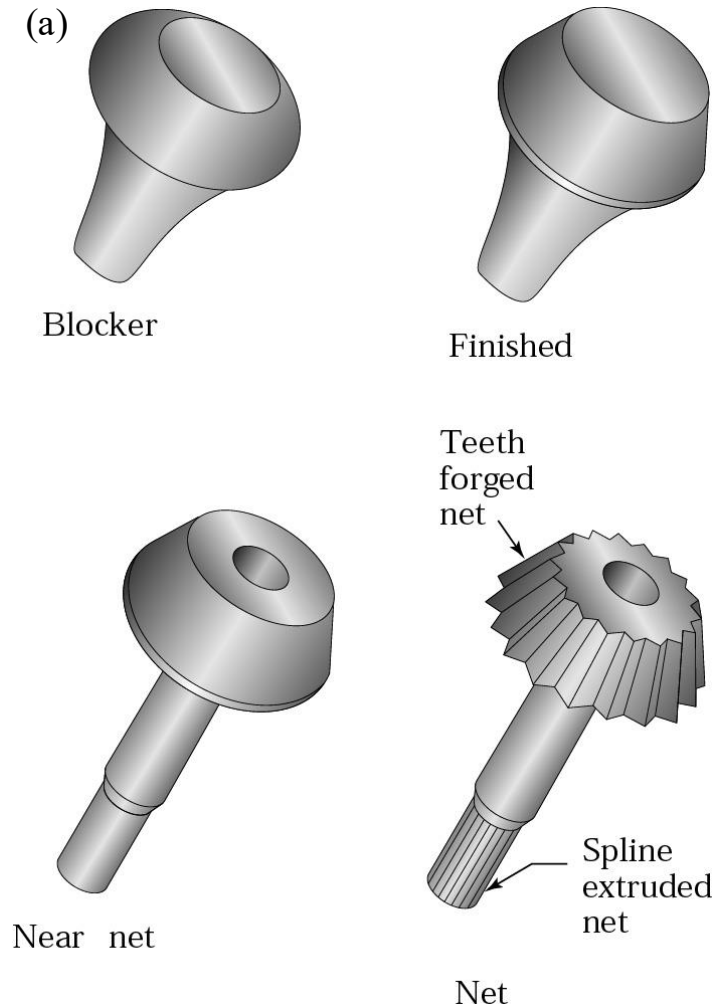
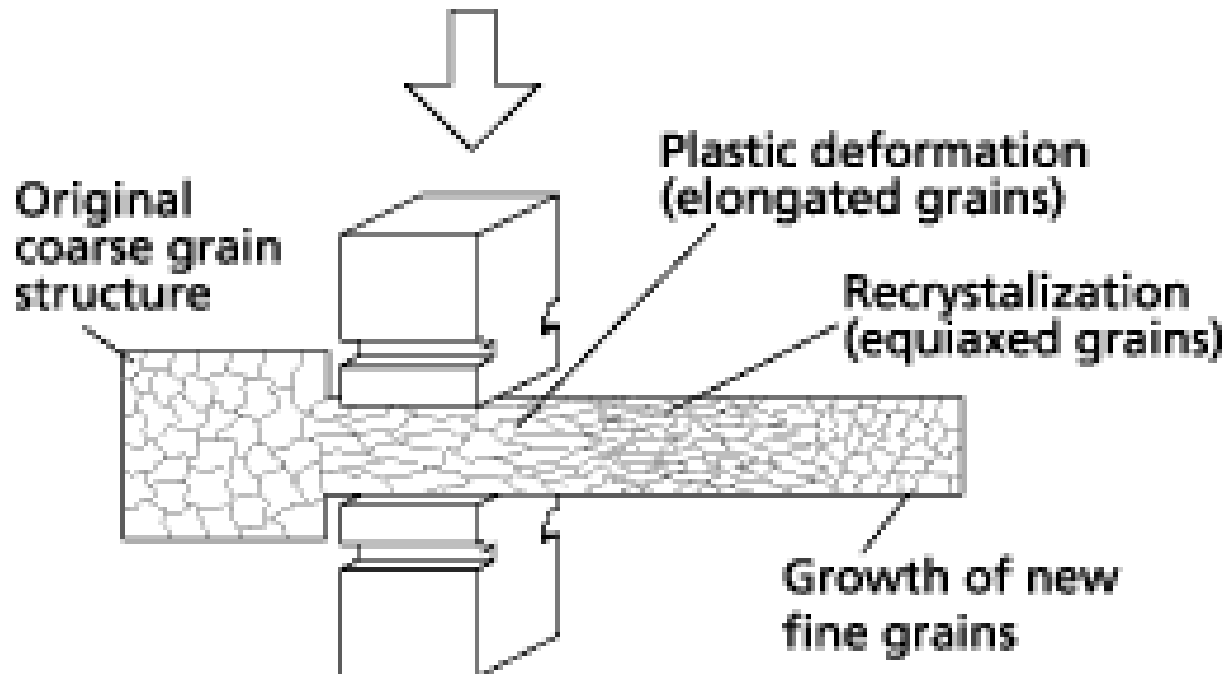


Figure 14.1 (a) Schematic illustration of the steps involved in forging a bevel gear with a shaft. *Source:* Forging Industry Association. (b) Landing-gear components for the C5A and C5B transport aircraft, made by forging. *Source:* Wyman-Gordon Company.

Quality of forged parts

Surface finish/Dimensional control:
Better than casting (typically)

Stronger/tougher than cast/machined parts of same material



Grain Flow Pattern of Pierced Round Billet

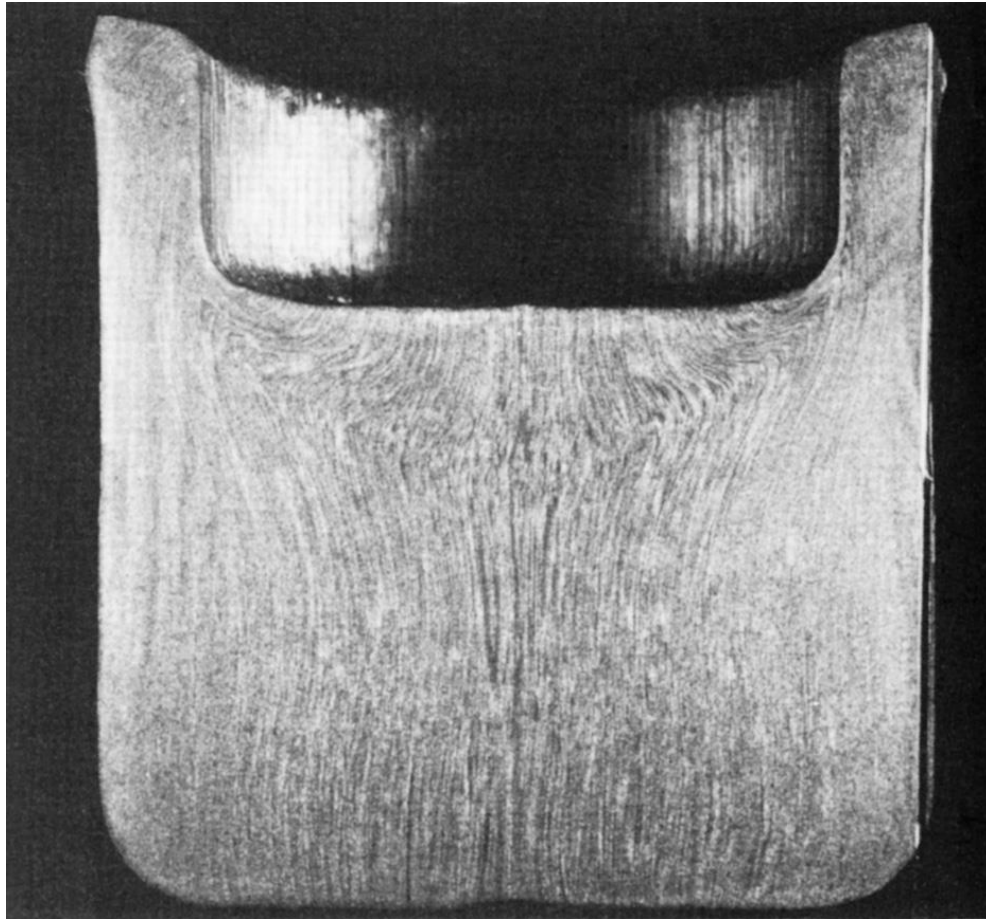


Figure 14.12 A pierced round billet, showing grain flow pattern. *Source:* Courtesy of Ladish Co., Inc.

Grain Flow Comparison

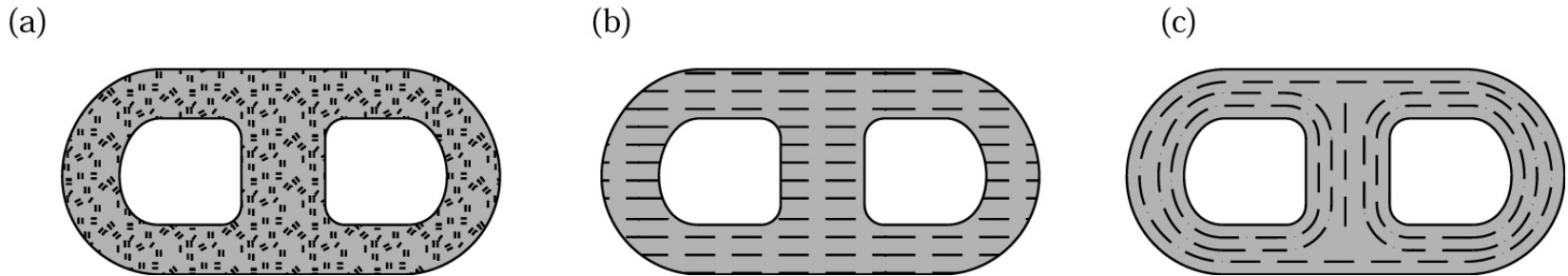


Figure 14.3 A part made by three different processes, showing grain flow. (a) casting, (b) machining, (c) forging. *Source:* Forging Industry Association.

Upsetting

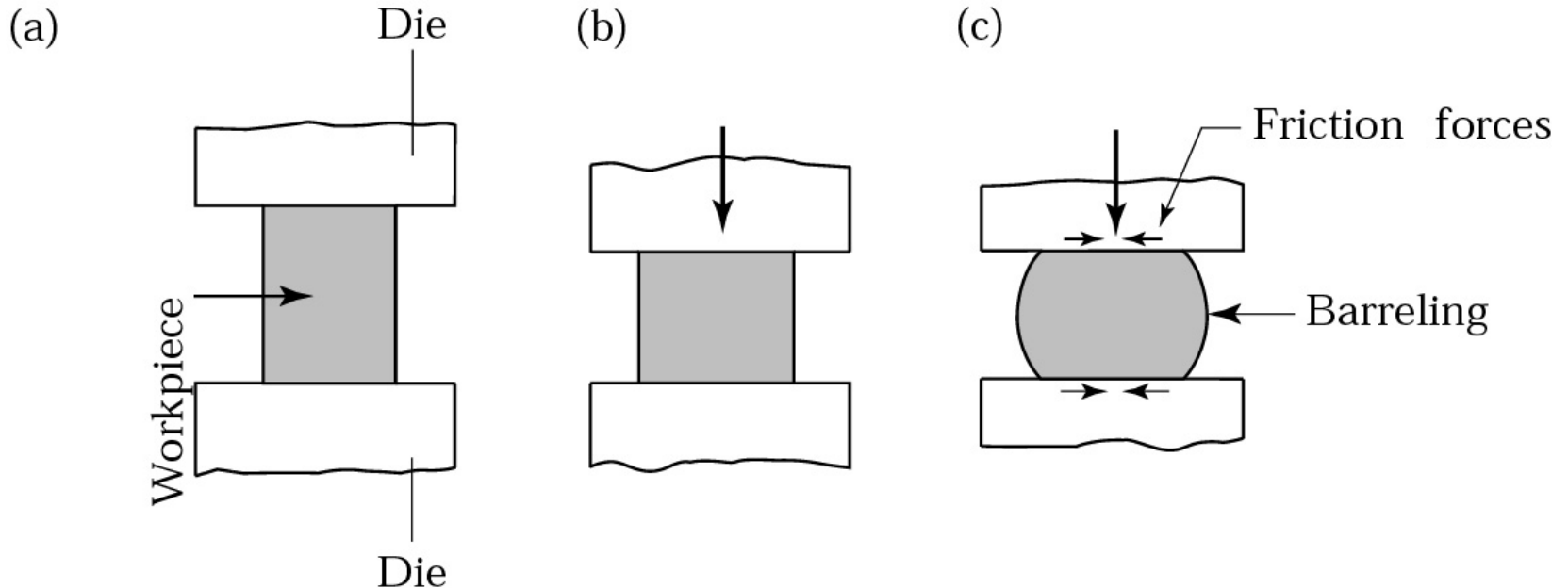


Figure (a) Solid cylindrical billet upset between two flat dies. (b) Uniform deformation of the billet without friction. (c) Deformation with friction. Note barreling of the billet caused by friction forces at the billet-die interfaces.

Impression-Die Forging

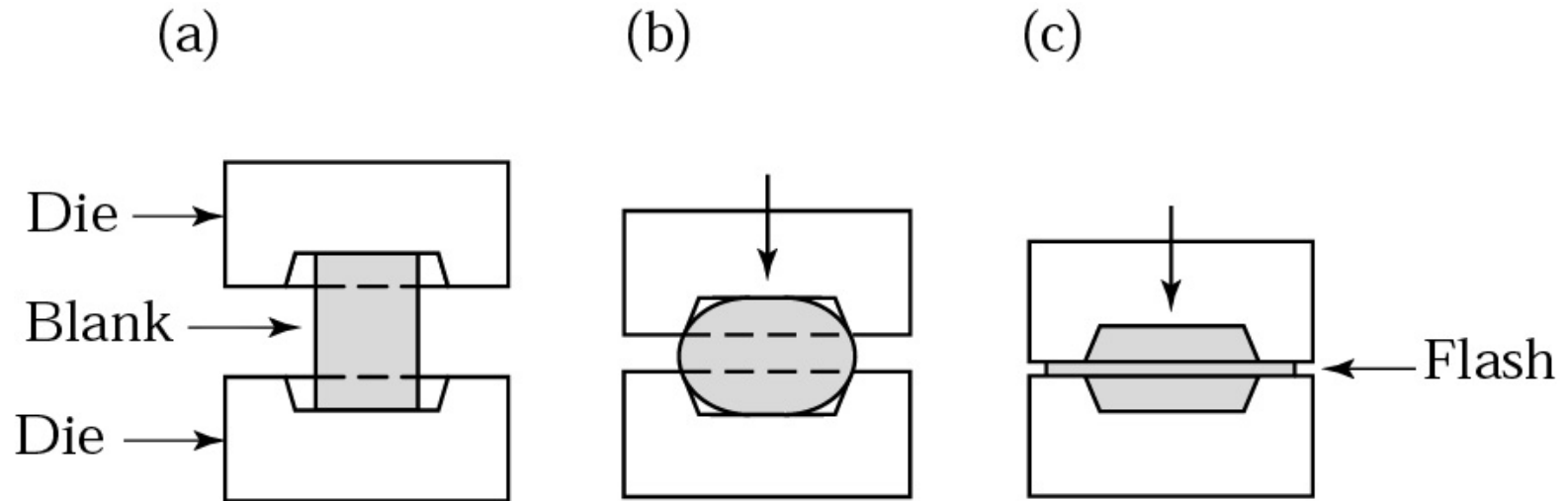
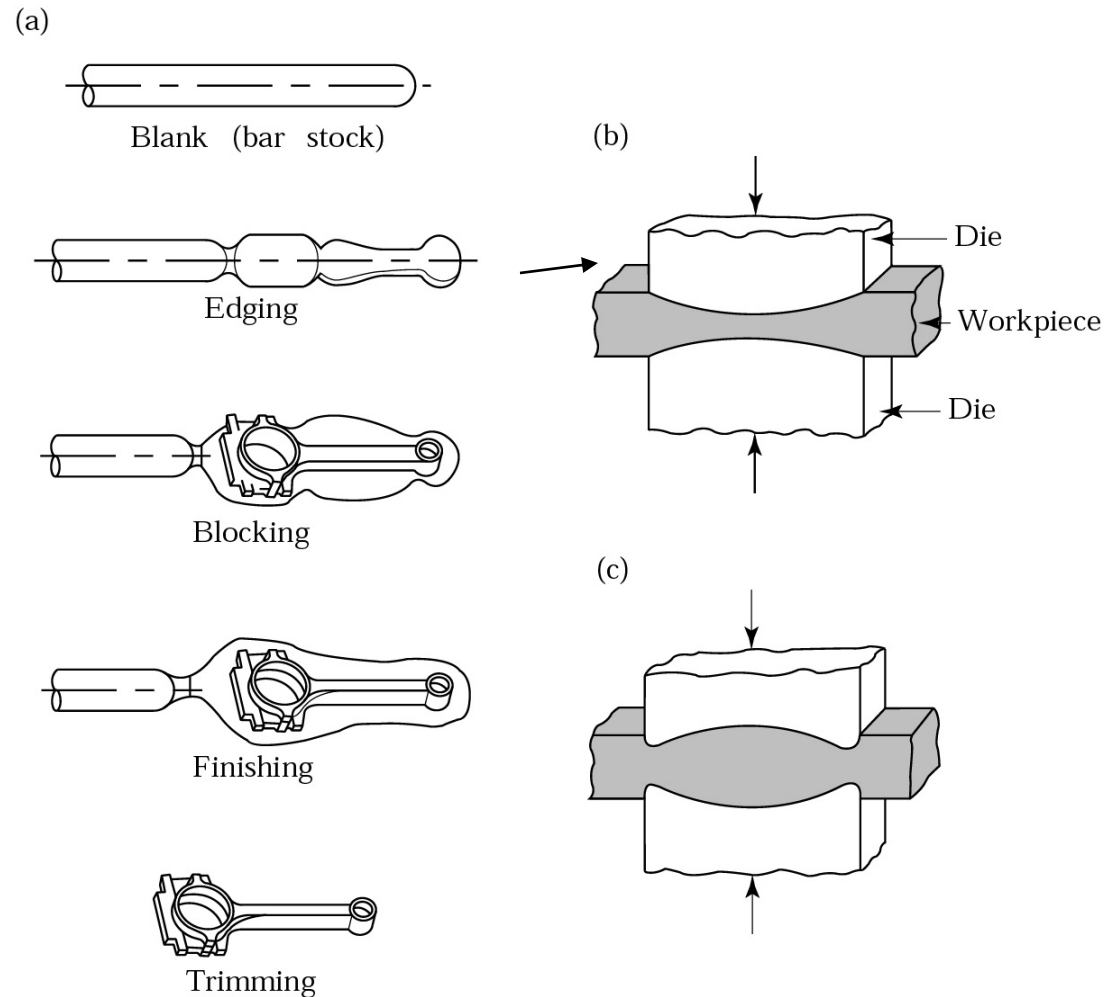


Figure 14.6 Stages in impression-die forging of a solid round billet. Note the formation of flash, which is excess metal that is subsequently trimmed off (see Fig. 14.8).

Forging a Connecting Rod

Figure (a) Stages in forging a connecting rod for an internal combustion engine. Note the amount of flash required to ensure proper filling of the die cavities. (b) Fullering, and (c) edging operations to distribute the material when preshaping the blank for forging.



Trimming Flash from a Forged Part

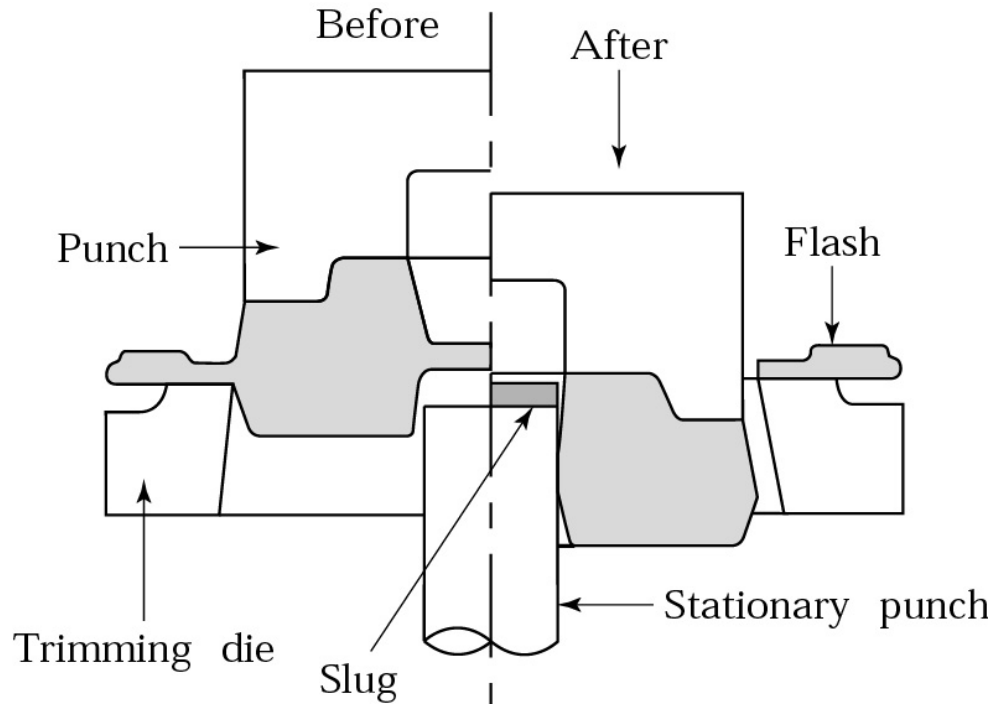


Figure. Trimming flash from a forged part. Note that the thin material at the center is removed by punching.

Flashless forging

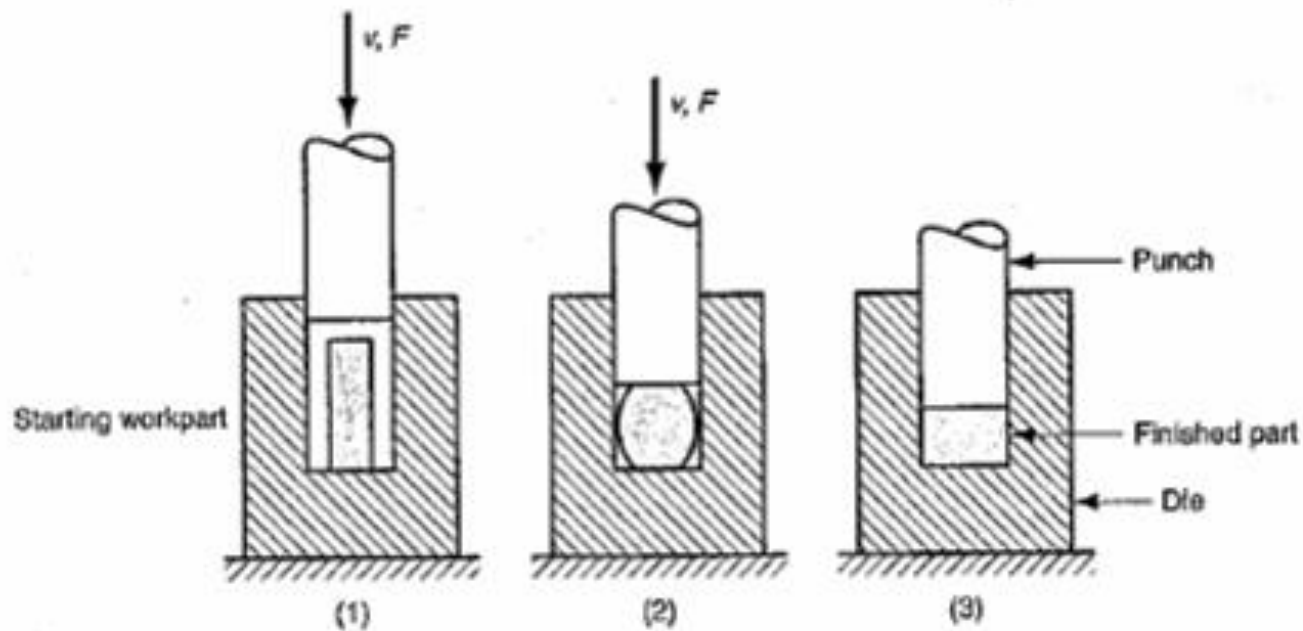
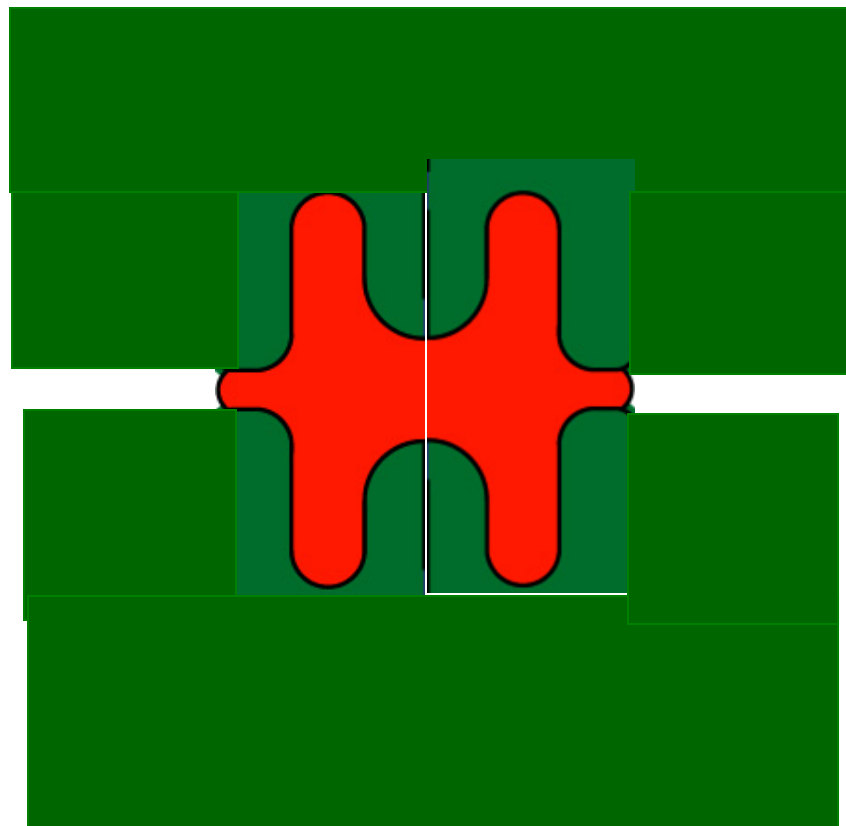
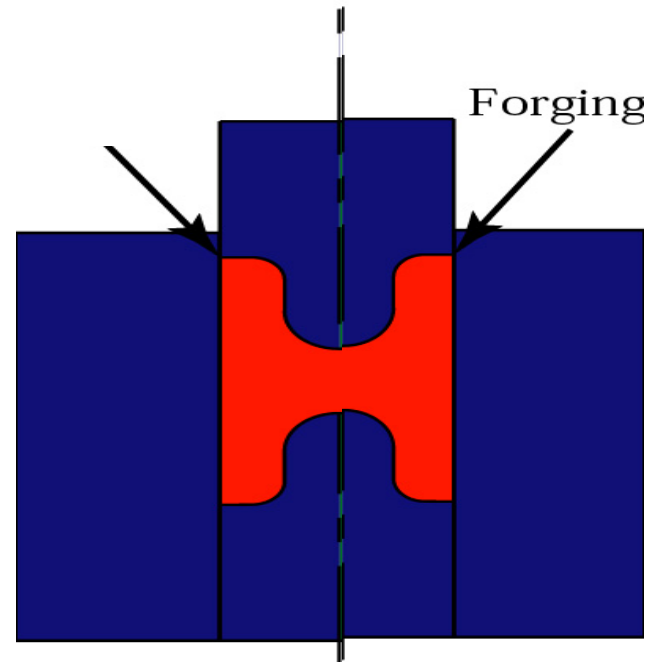
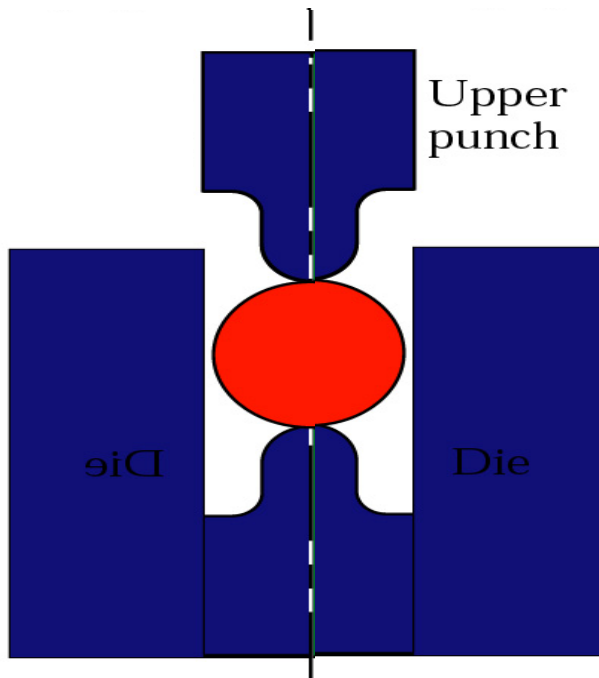


FIGURE 21.18 Flashless forging: (1) just before initial contact with workpiece, (2) partial compression, and (3) final punch and die closure. Symbols v and F indicate motion (v = velocity) and applied force, respectively.

Normal



Flashless Forging



of stroke

Comparison of Forging With and Without Flash

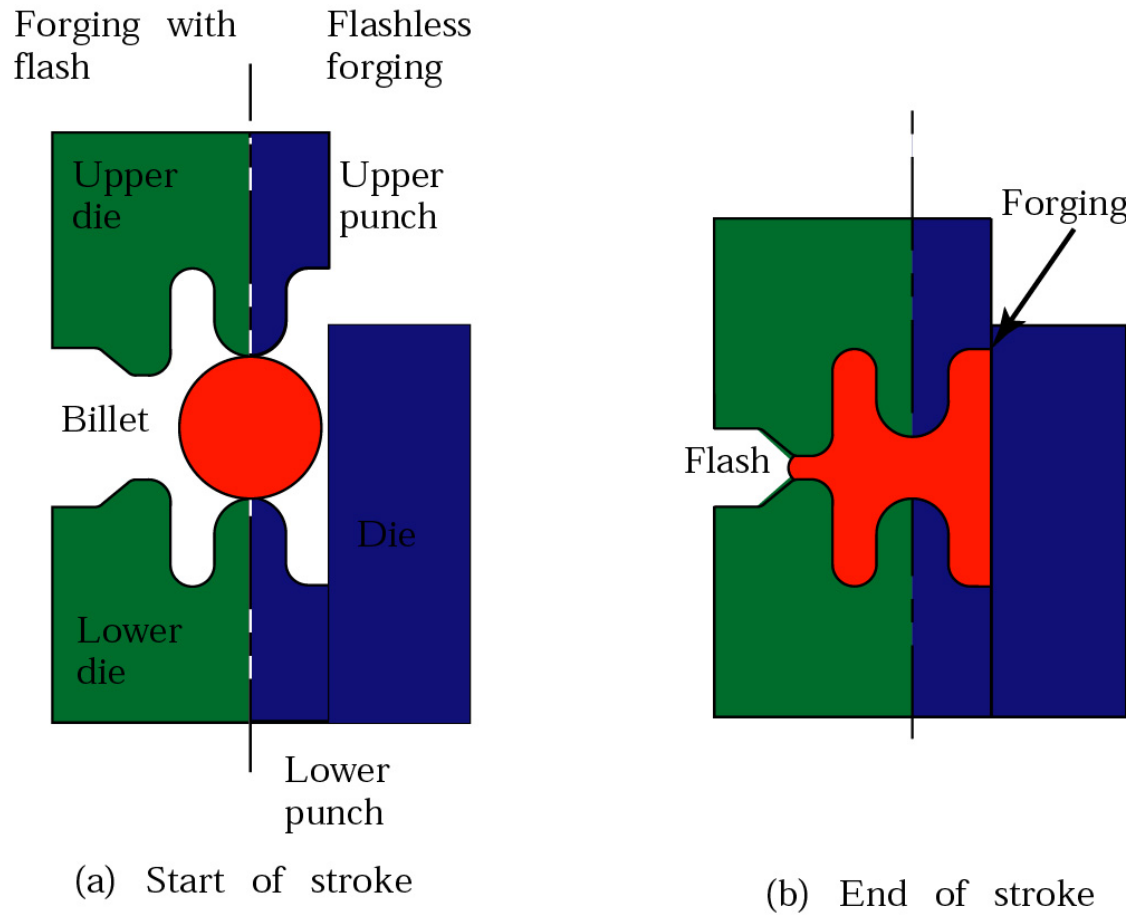


Figure 14.9
Comparison of closed-die forging to precision or flashless forging of a cylindrical billet.
Source: H. Takemasu, V. Vazquez, B. Painter, and T. Altan.

Unit Cost in Forging

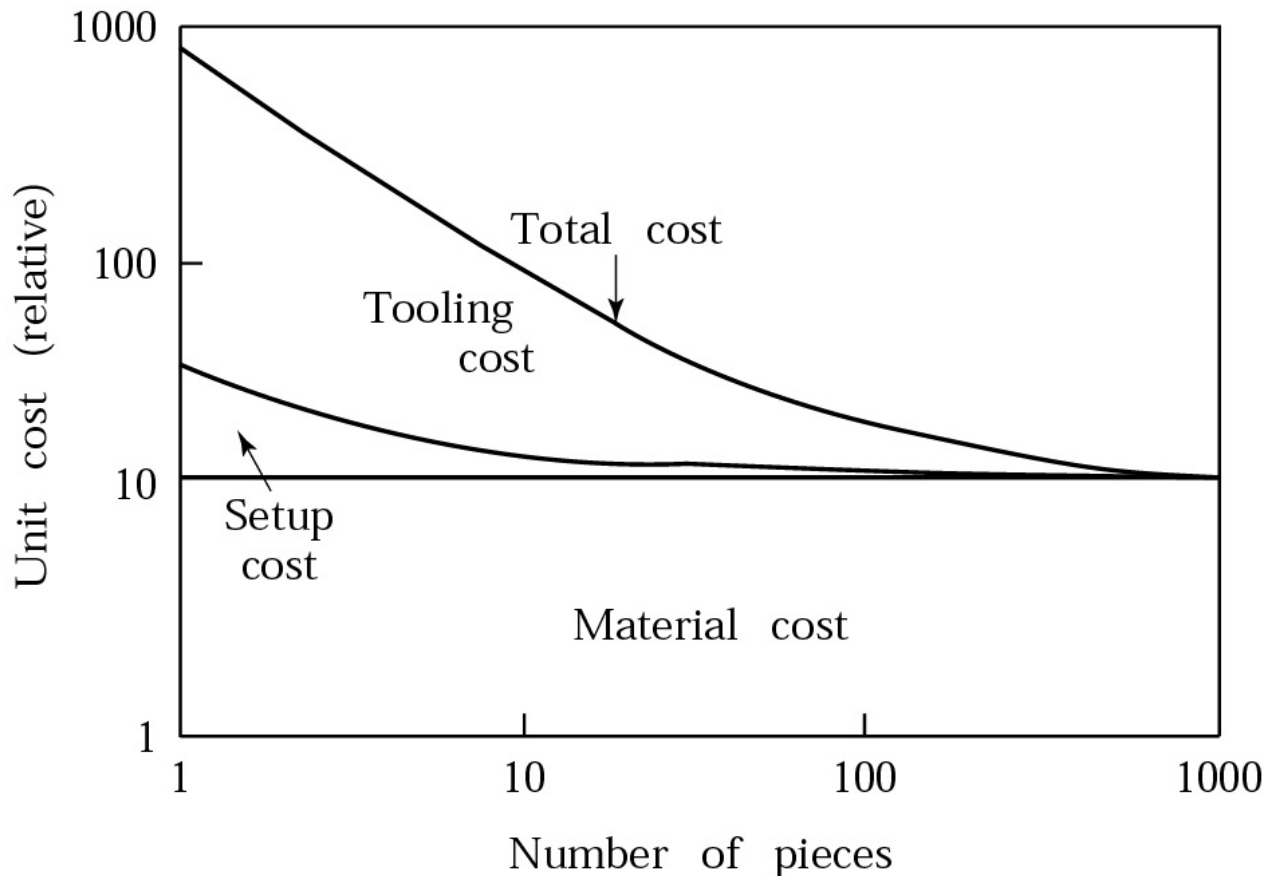
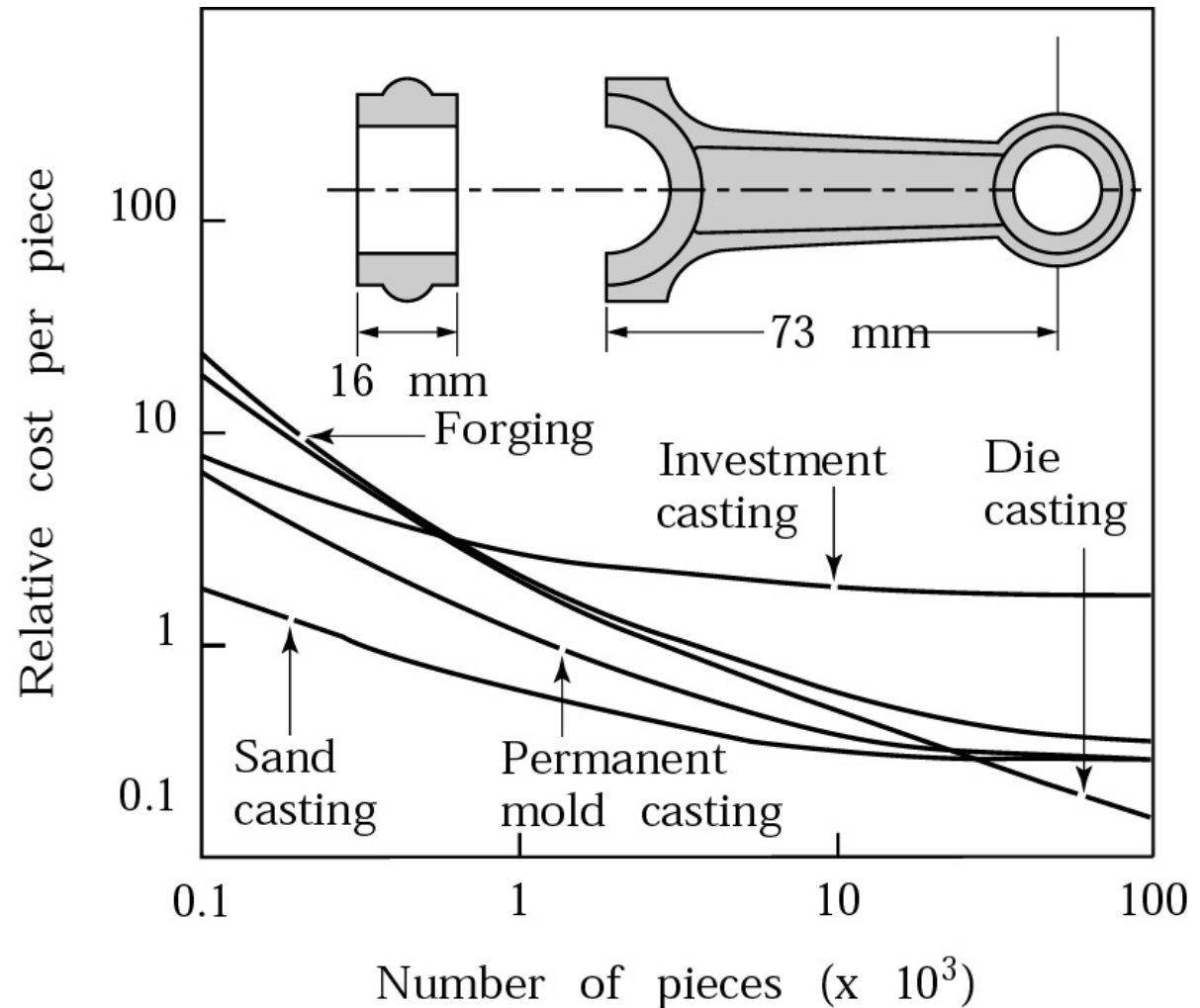


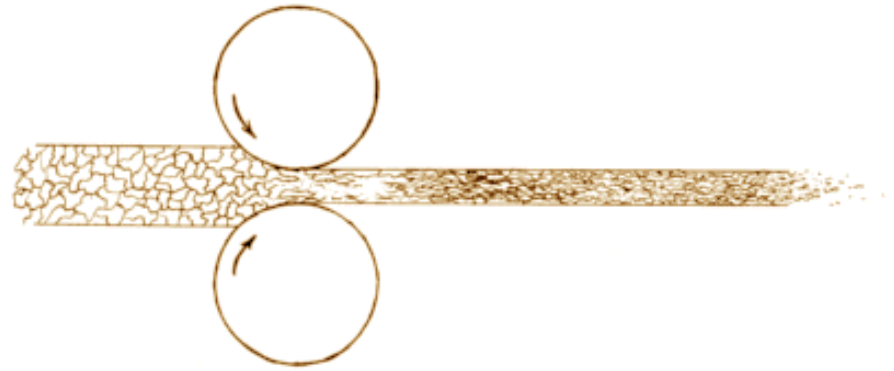
Figure Typical unit cost (cost per piece) in forging; note how the setup and the tooling costs per piece decrease as the number of pieces forged increases, if all pieces use the same die.

Relative Unit Costs of a Small Connecting Rod

Figure .
Relative unit costs of a small connecting rod made by various forging and casting processes. Note that, for large quantities, forging is more economical. Sand casting is the more economical process for fewer than about 20,000 pieces.



Rolling

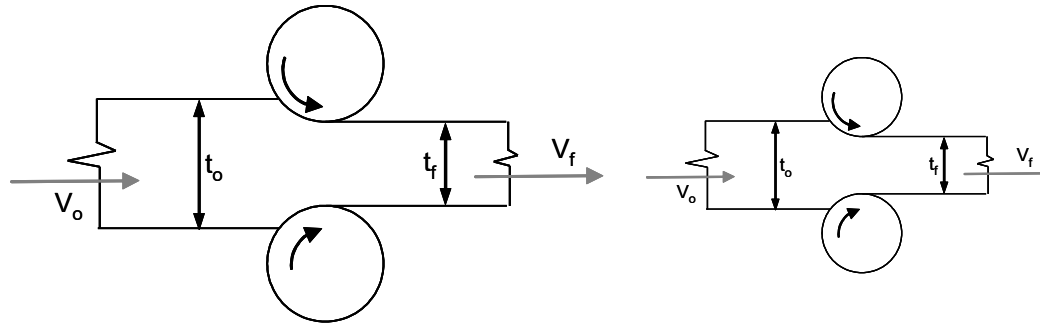


ROLLING PROCESS

Important Applications:

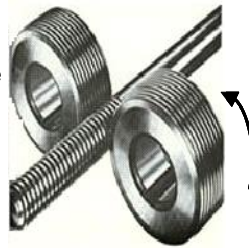
Steel Plants,
Raw stock production (sheets, tubes, Rods, etc.)
Screw manufacture

Rolling Basics



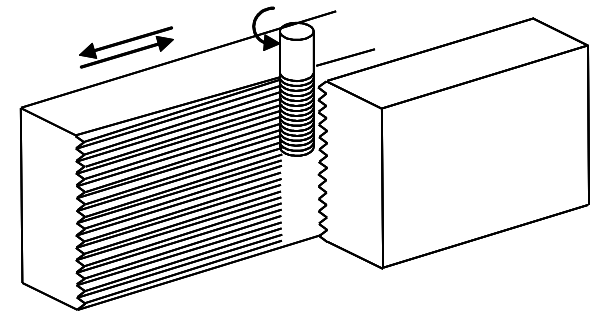
Screw manufacture:

stationary die



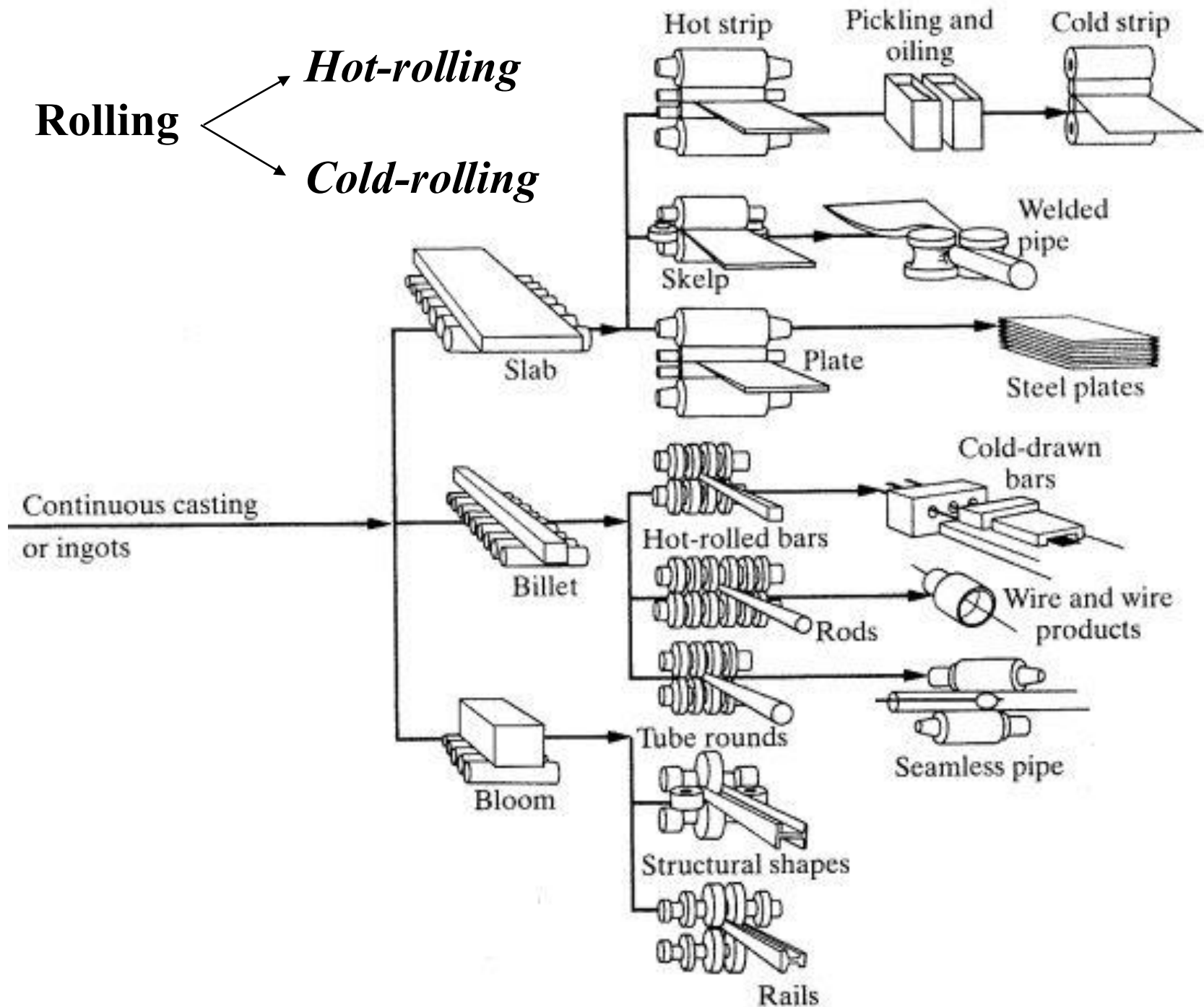
rolling die

thread rolling machine

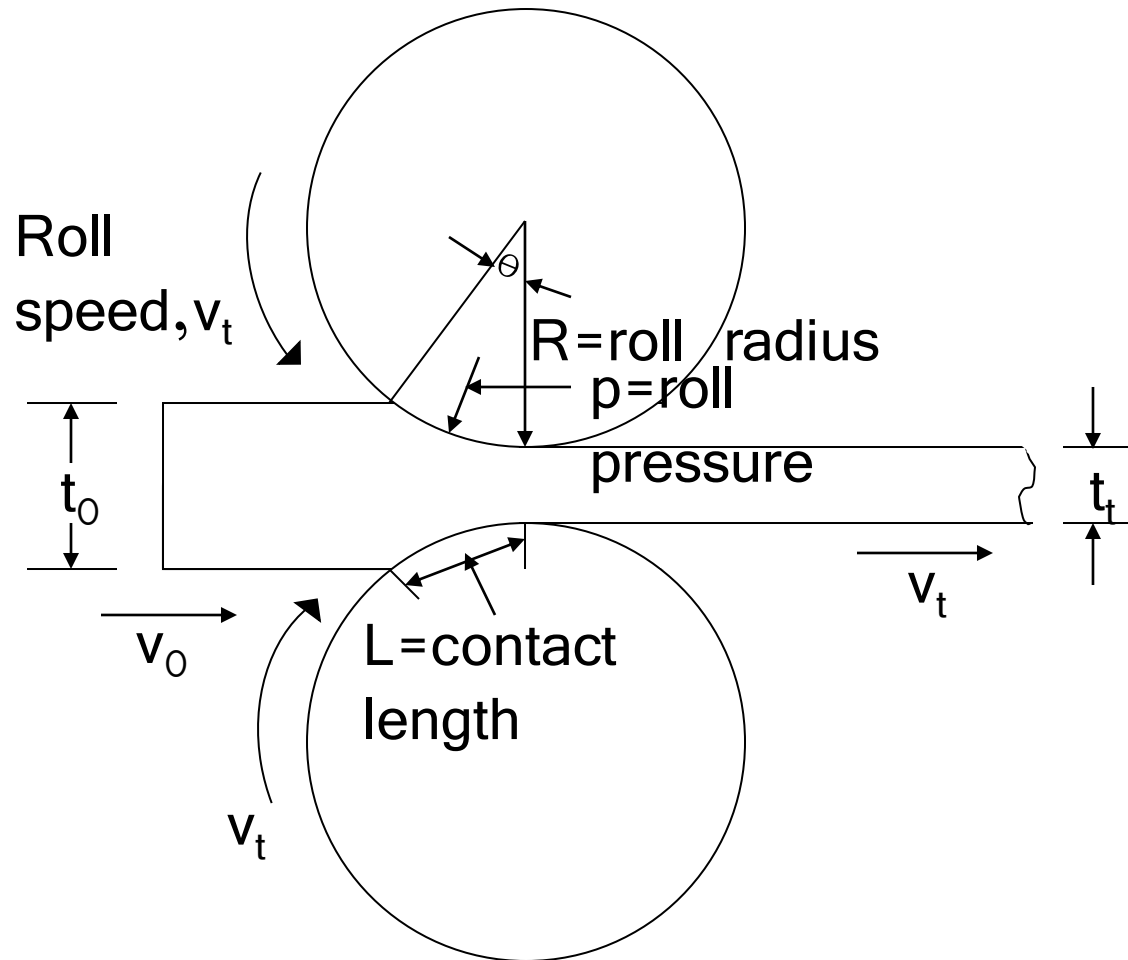


Reciprocating flat thread-rolling dies

Rolling *Hot-rolling*
Cold-rolling



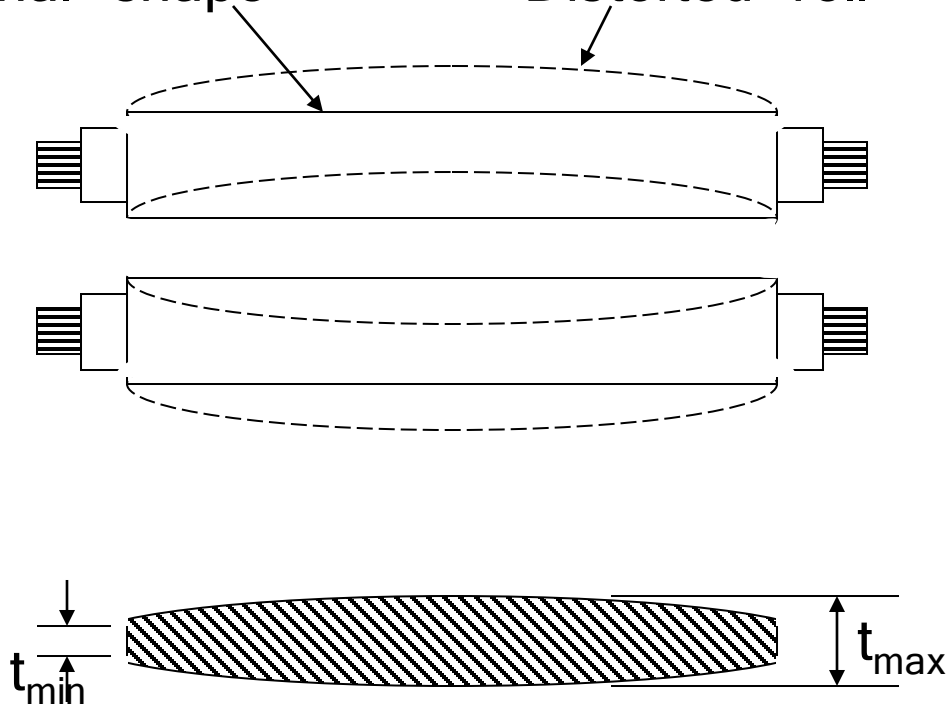
Rolling Theory



Distorted roll

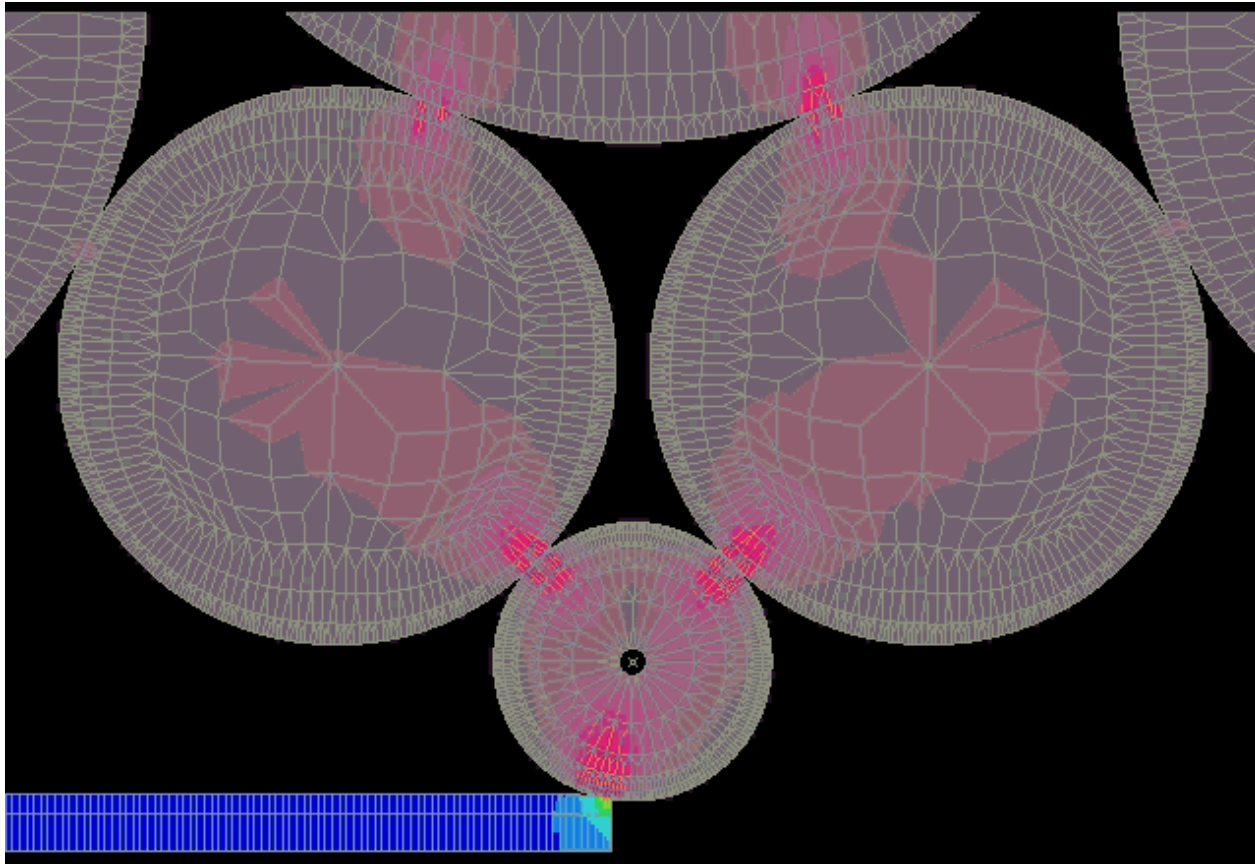
Original shape

Distorted roll

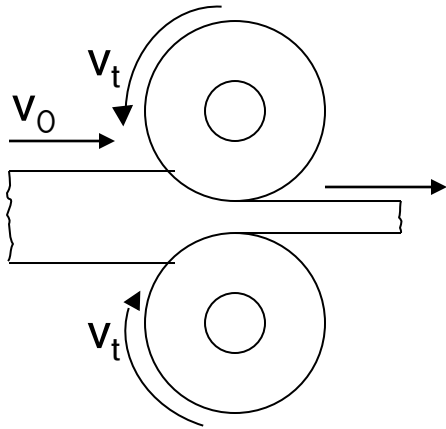


Cross section of the sheet

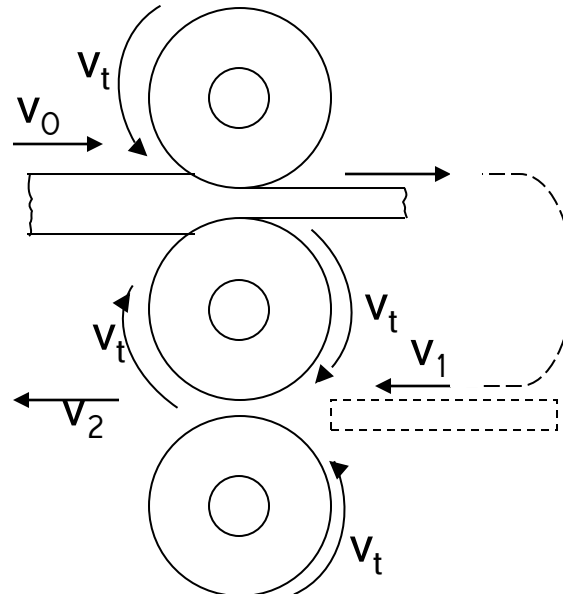
Rolling force



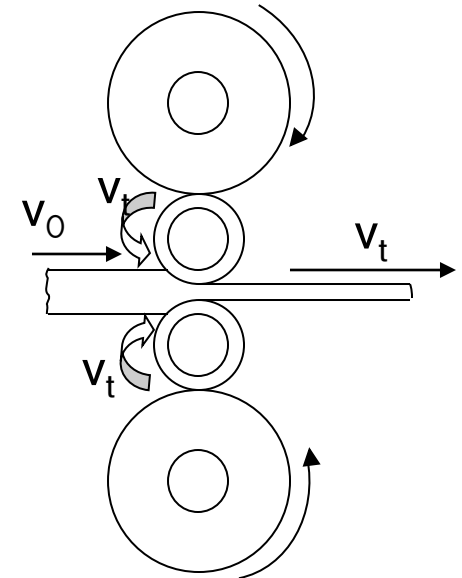
Various Configurations



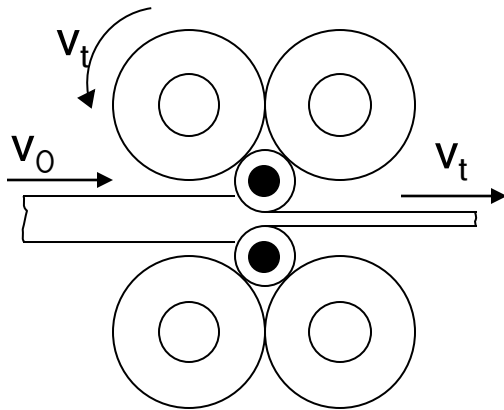
Two high



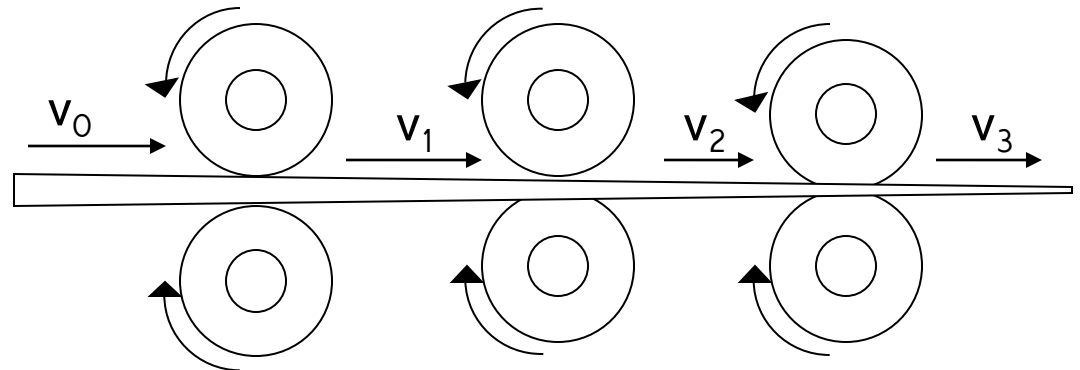
Three high



Four high



Cluster Mill



Tandem rolling mill

Cluster Mill

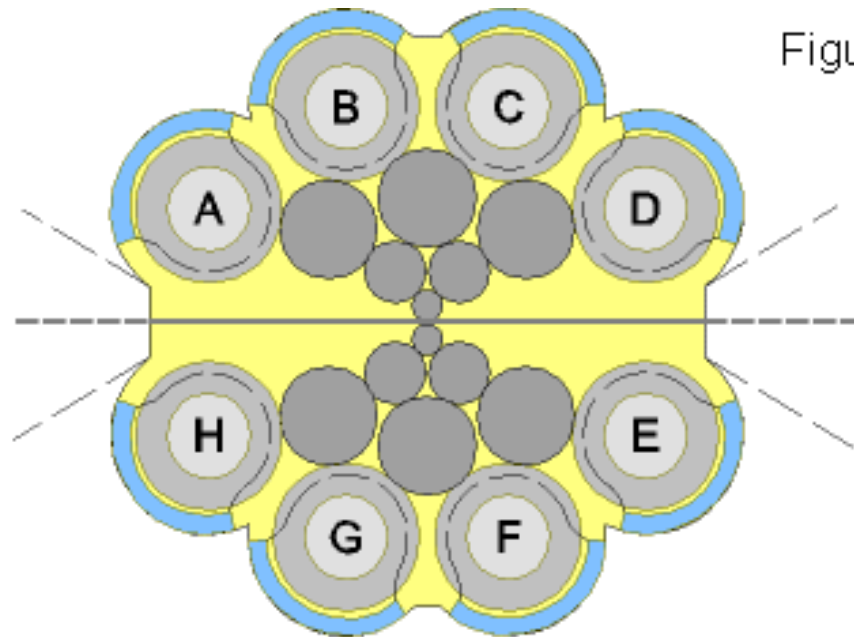


Figure 1

Shapes Produced by rolling



Square



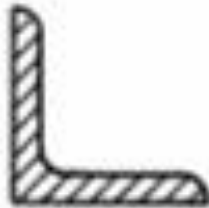
Slab



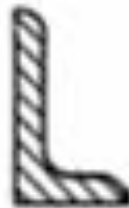
Hexagonal



Round



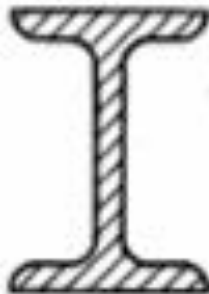
Equal-sided angle



L-section



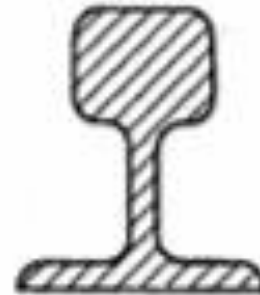
T-section



I beam



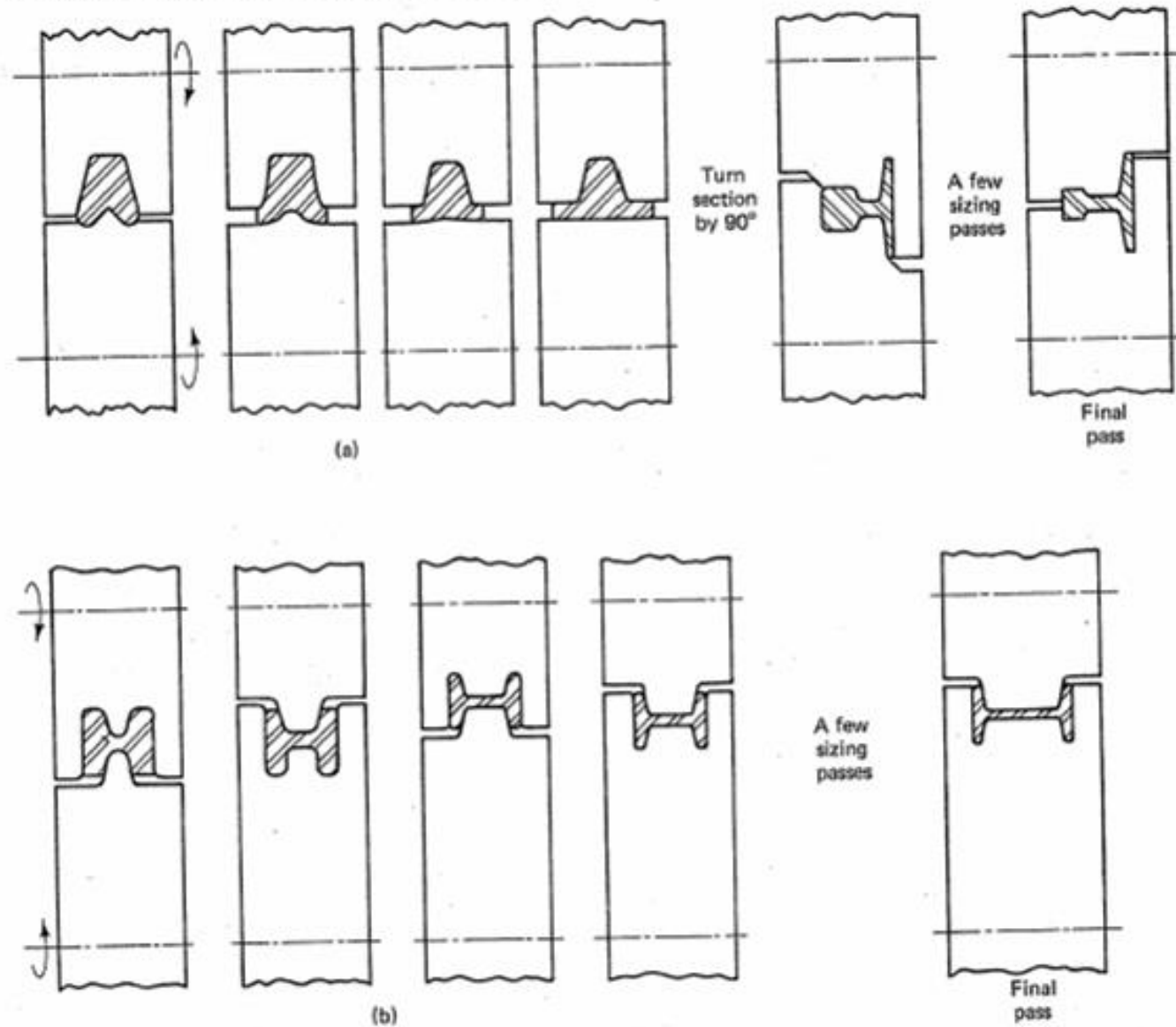
Channel beam



Rail

FIGURE 5.10

Roll passes: (a) for producing rails; (b) for producing an I beam



Roll-Forging

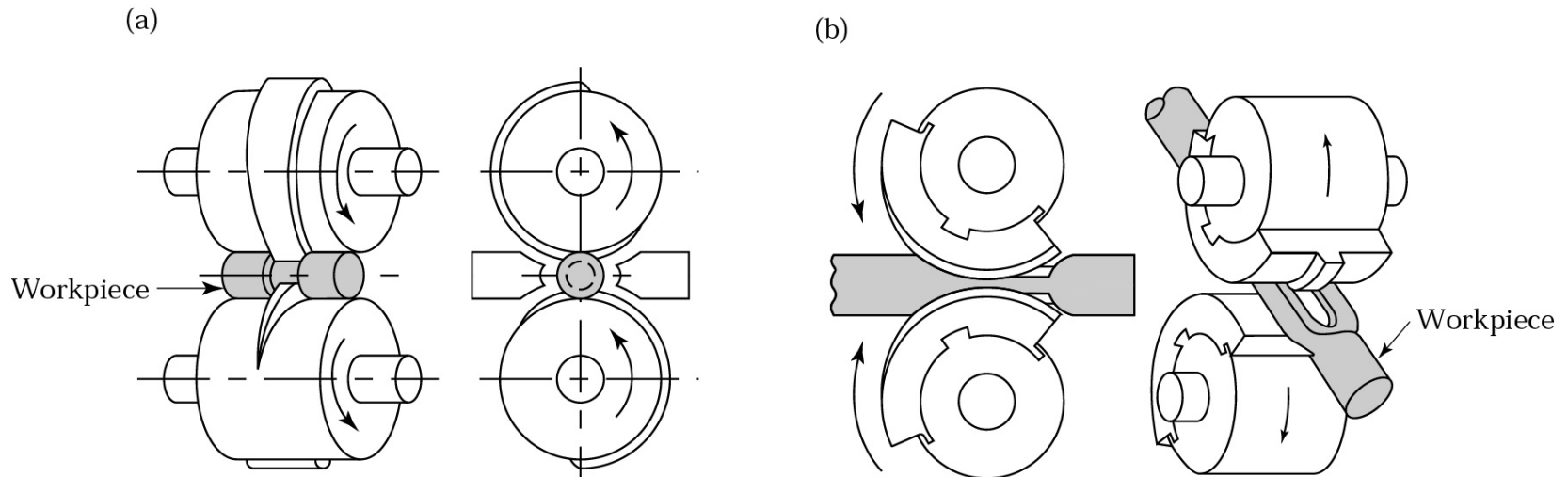
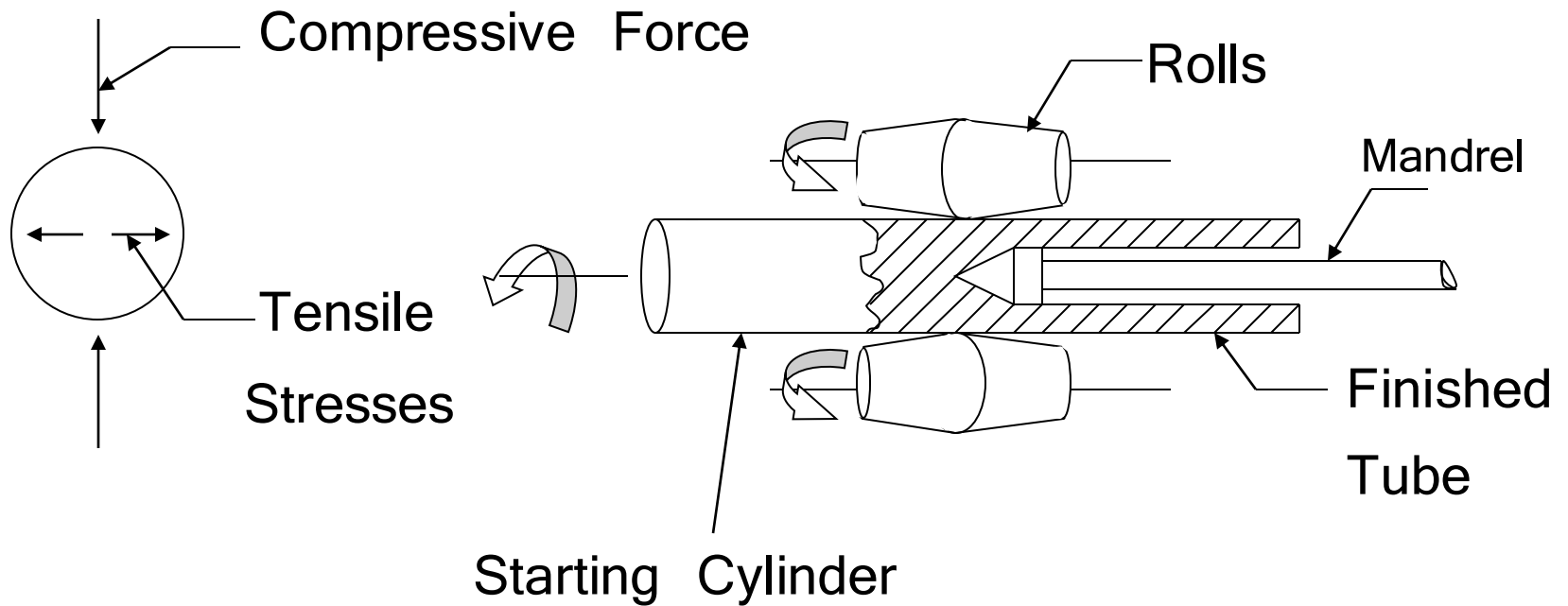


Figure 14.13 Two examples of the roll-forging operation, also known as *cross-rolling*. Tapered leaf springs and knives can be made by this process. *Source:* (a) J. Holub; (b) reprinted with permission of General Motors Corporation.

Roll piercing



Friction and lubrication

- Friction is undesirable;
 - Metal flow in the work is retarded
 - The forces to perform are increased
 - Rapid wear of tooling

sticking

Lubrication

- Mineral oils
- Graphite
- Glass
- Graphite in water or mineral oil is a common for hot forging

Sheet Metal Processes

Raw material: sheets of metal, rectangular, large

Raw material Processing: Rolling (anisotropic properties)

Processes:

Shearing

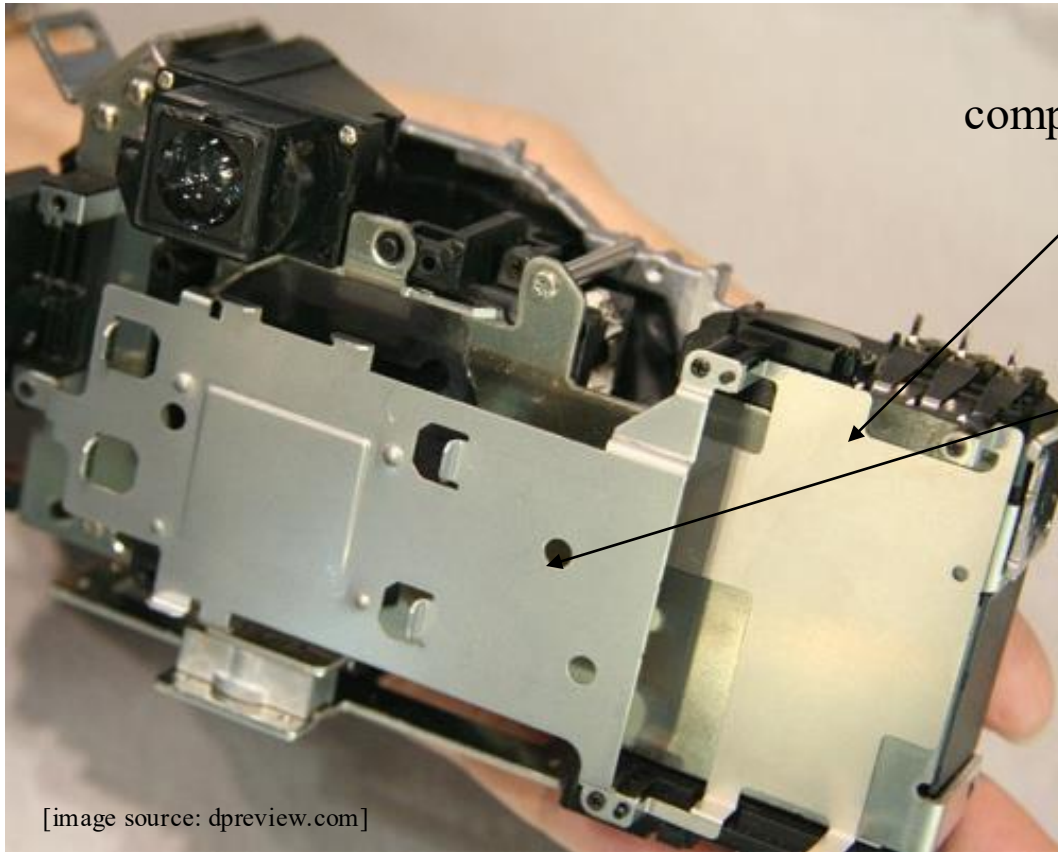
Punching

Bending

Deep drawing

Bending

Body of Olympus E-300 camera



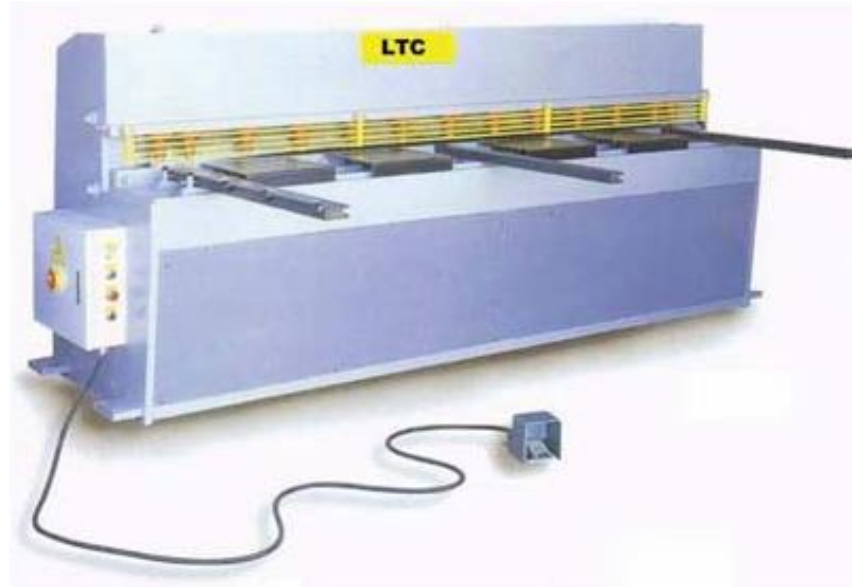
component with multiple bending operations

component with punching,
bending, drawing operations

[image source: dpreview.com]

Shearing

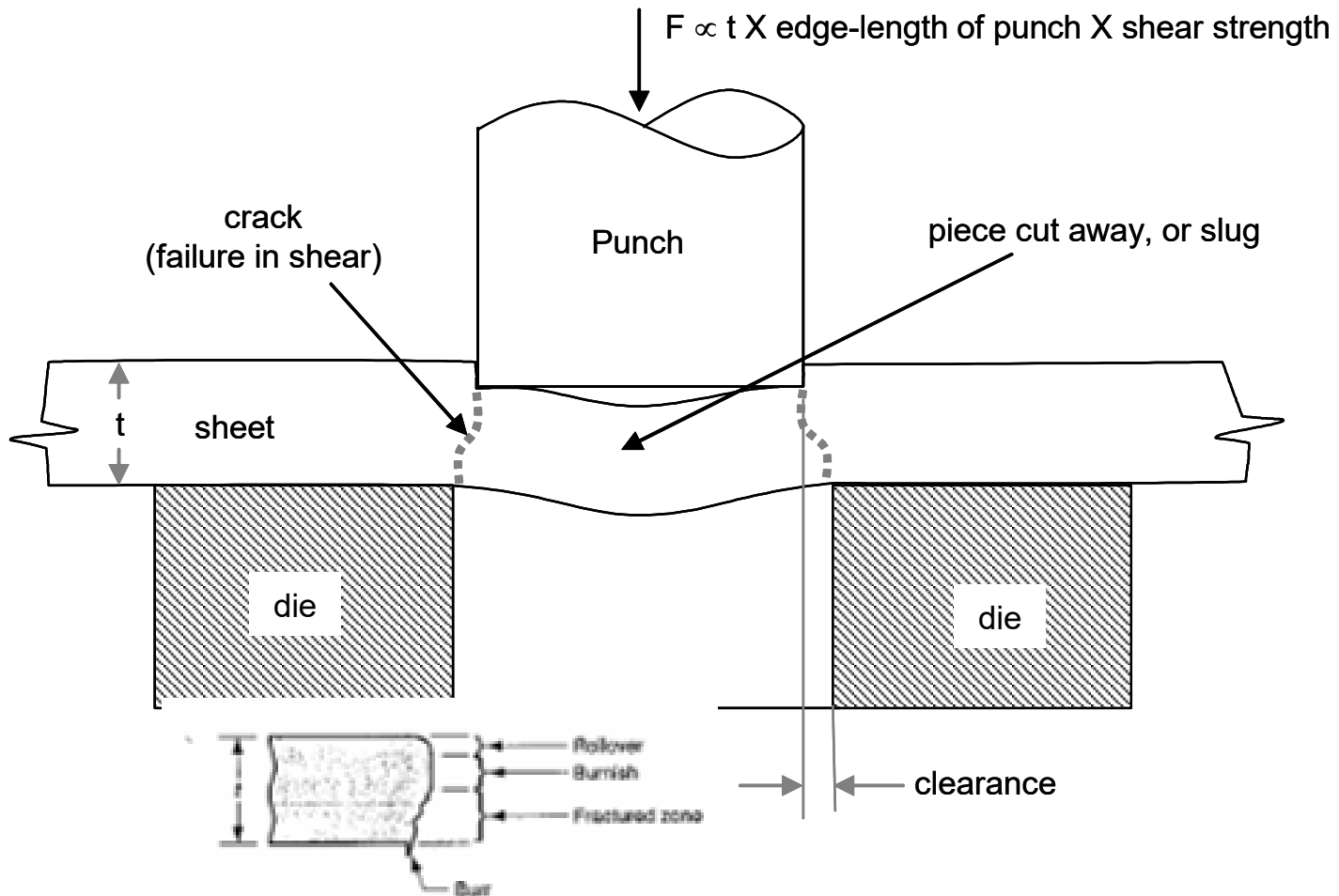
A large scissors action, cutting the sheet along a straight line



Main use: to cut large sheet into smaller sizes for making parts.

Punching

Cutting tool is a round/rectangular punch,
that goes through a hole, or die of same shape

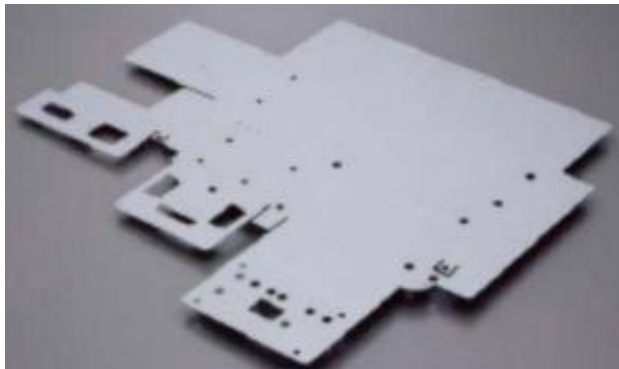


Punching

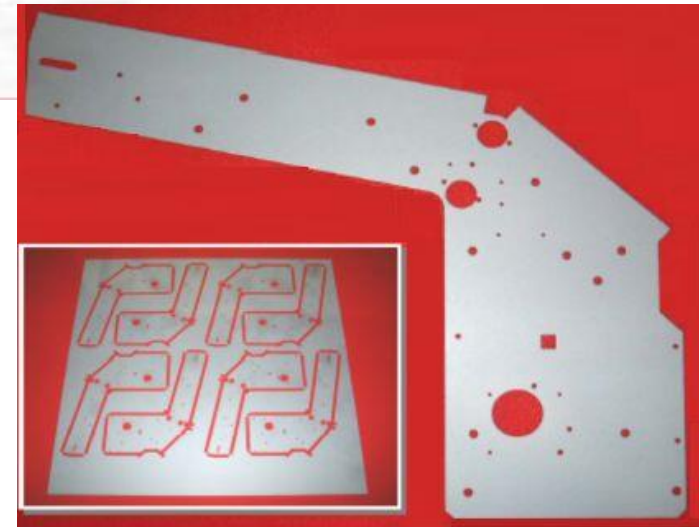
Main uses: cutting holes in sheets; cutting sheet to required shape



nesting of parts

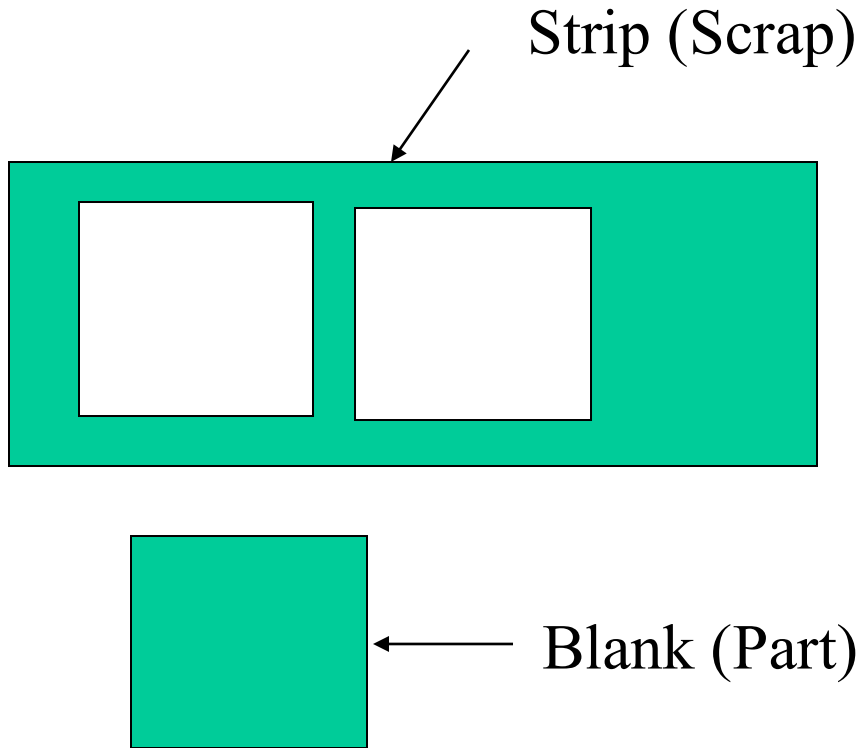


typical punched part

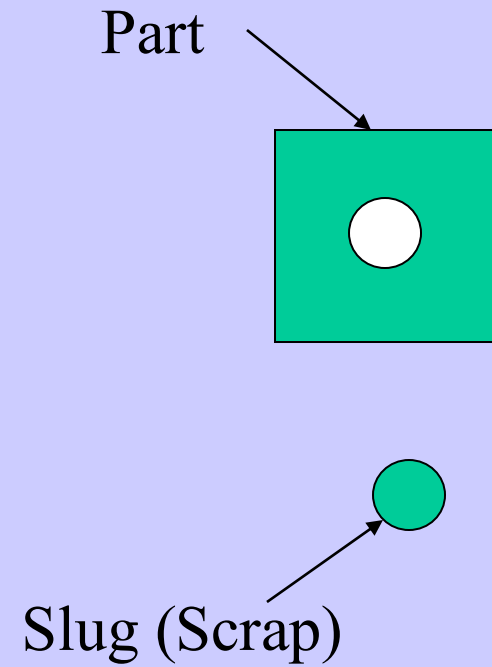


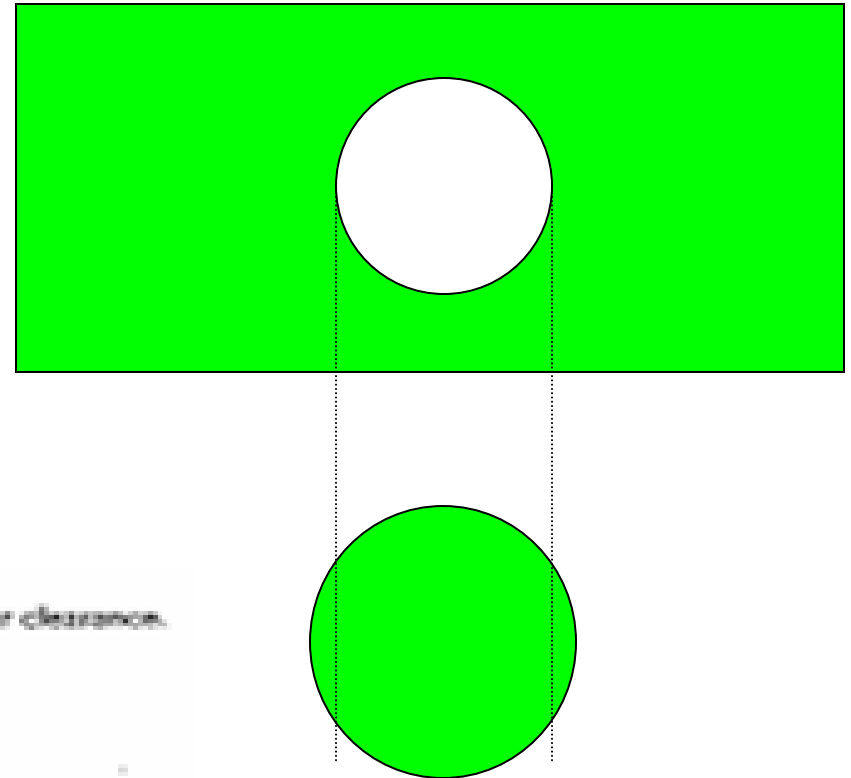
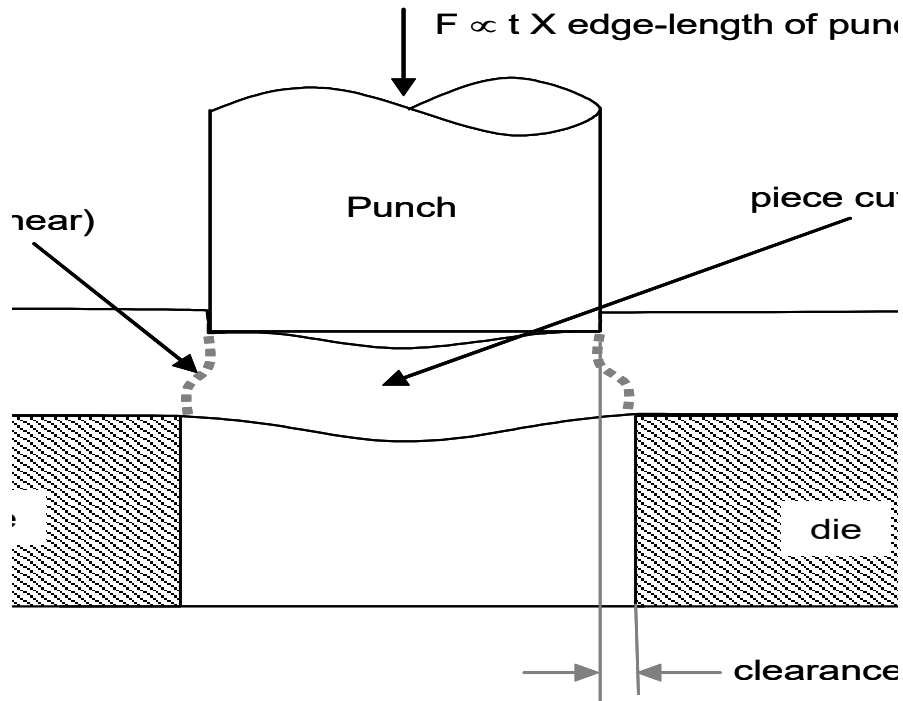
Exercise: how to determine optimal nesting?

Blanking



Piecing



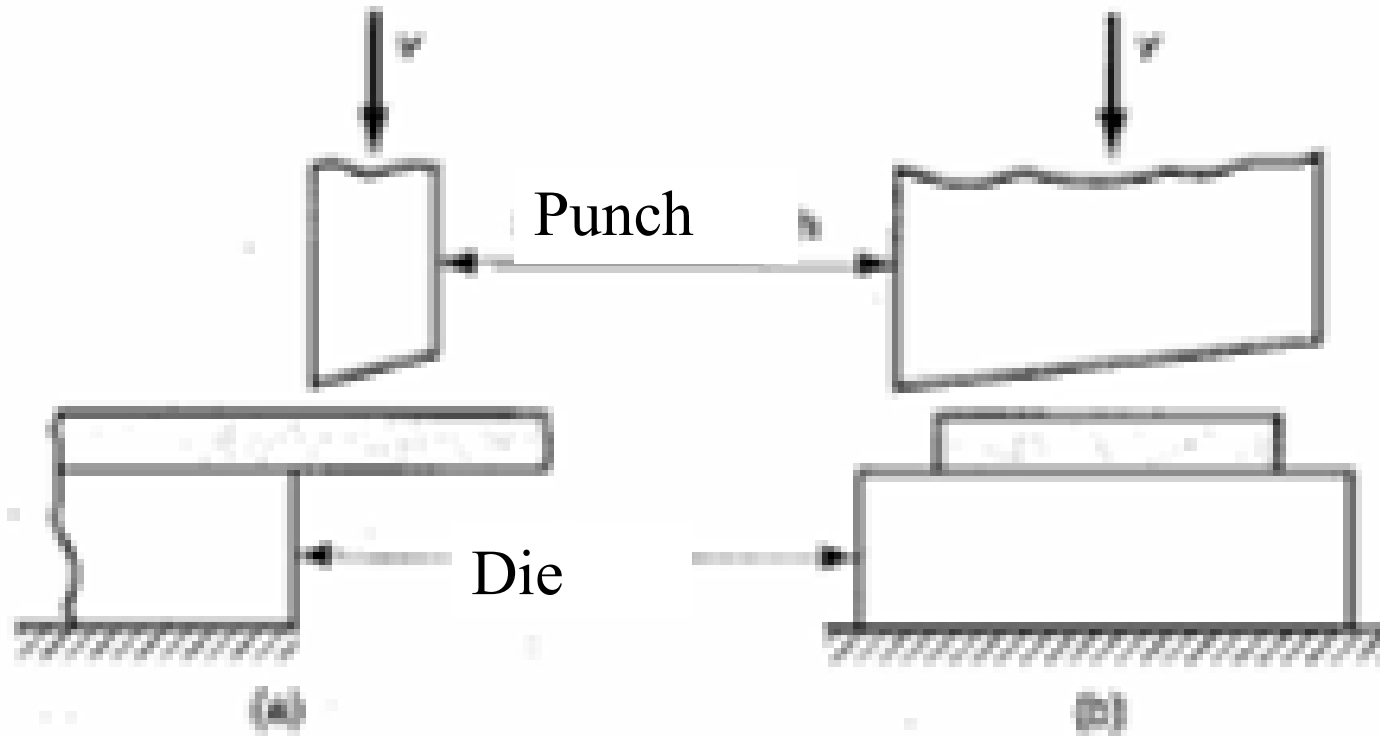


Straight portion (for resharpening)

FIGURE 22.7 Angular clearance.

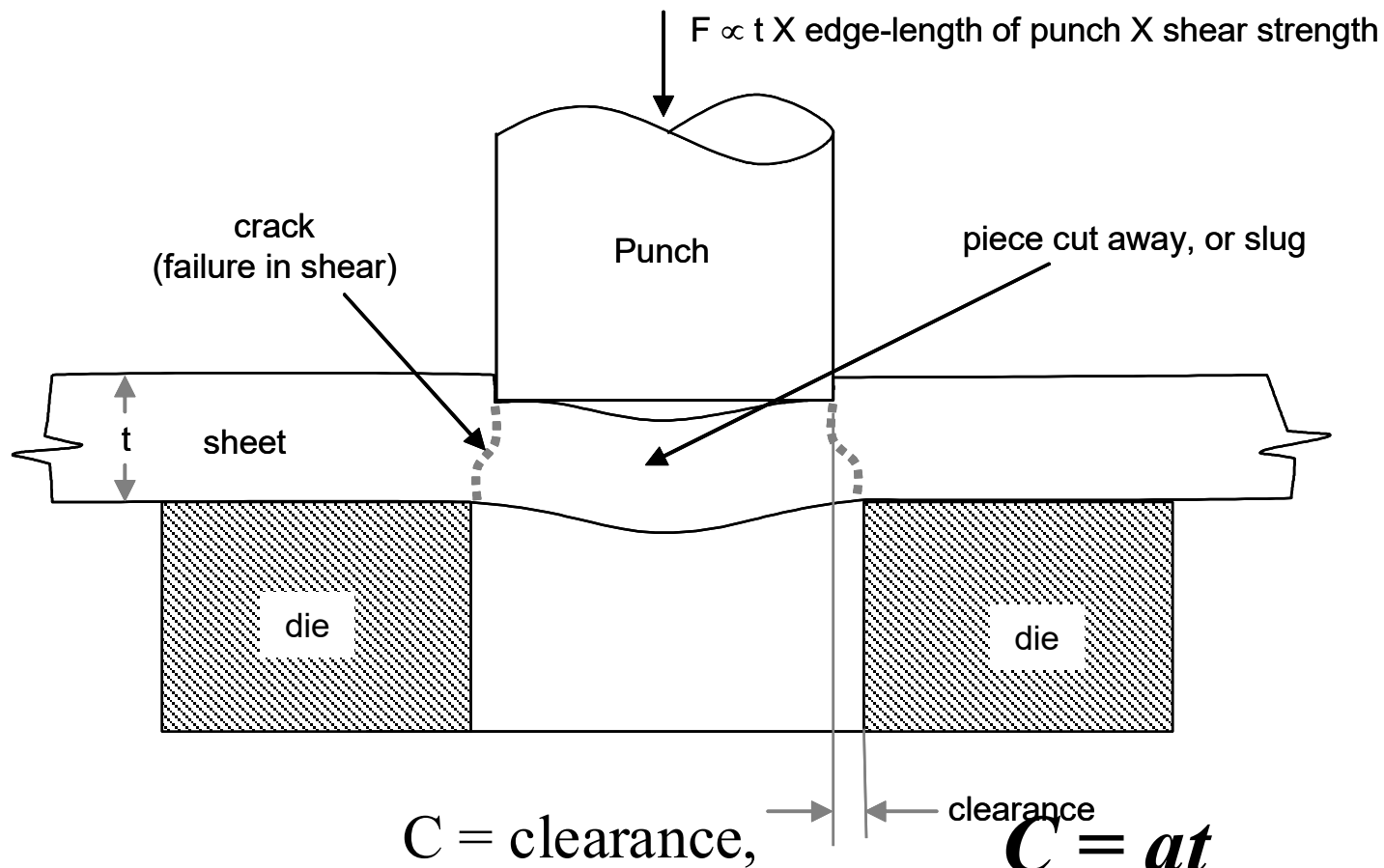


Shearing Operation



Side View

Front View



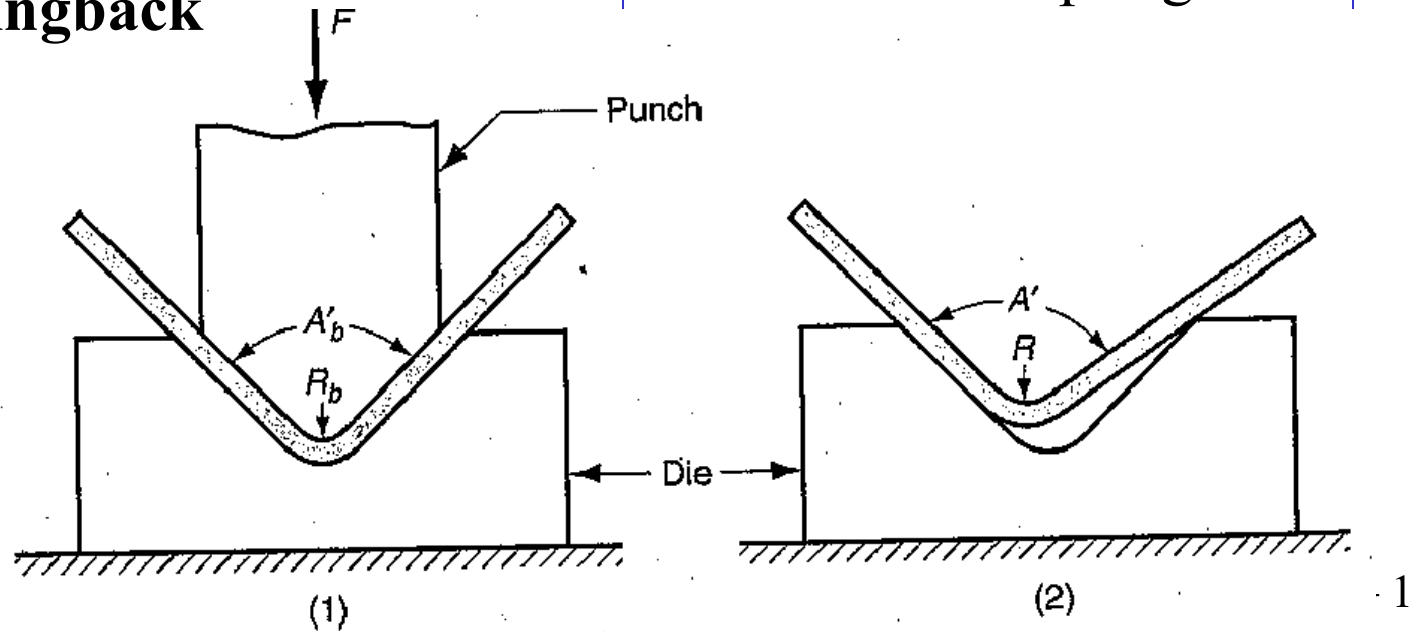
C = clearance,

a = allowance

t = Thickness (plate)

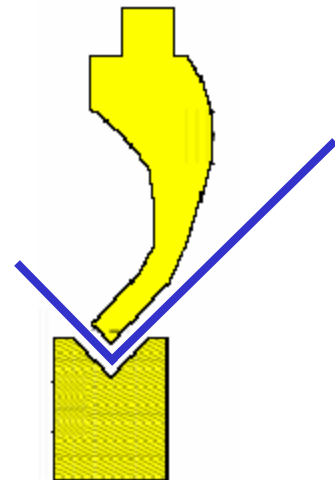
Bending: springback

How to handle springback:



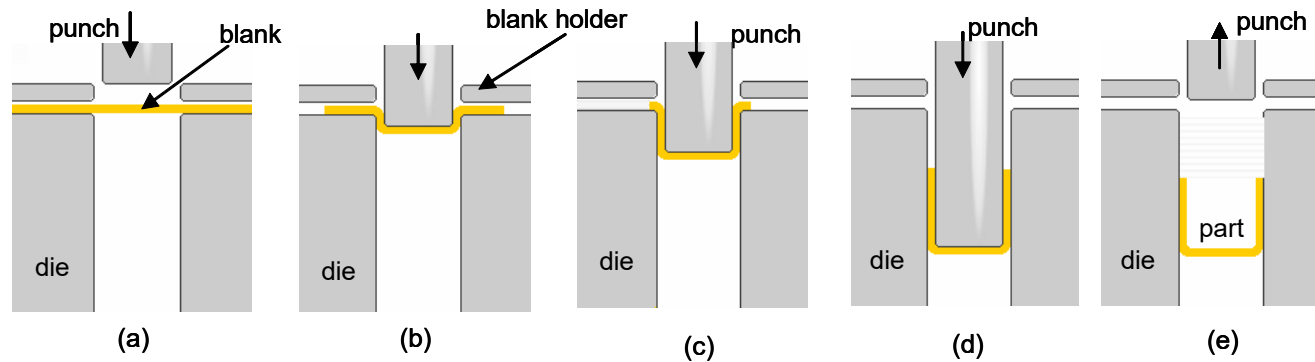
- (a) Compensation: the metal is bent by a larger angle
- (b) Coining the bend:
at end of bend cycle, tool exerts large force, dwells

coining: press down hard, wait, release



Deep Drawing

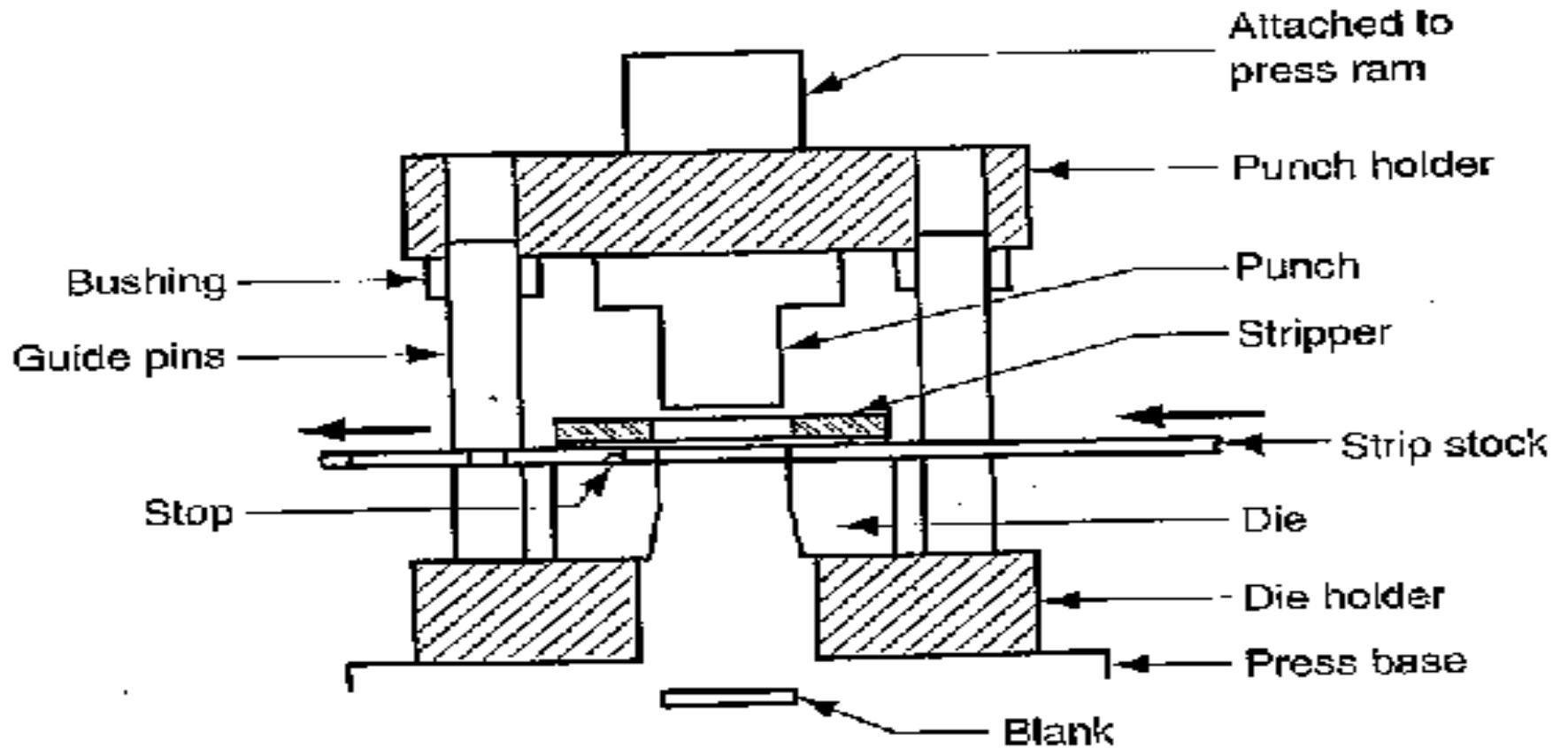
Tooling: similar to punching operation,
Mechanics: similar to bending operation



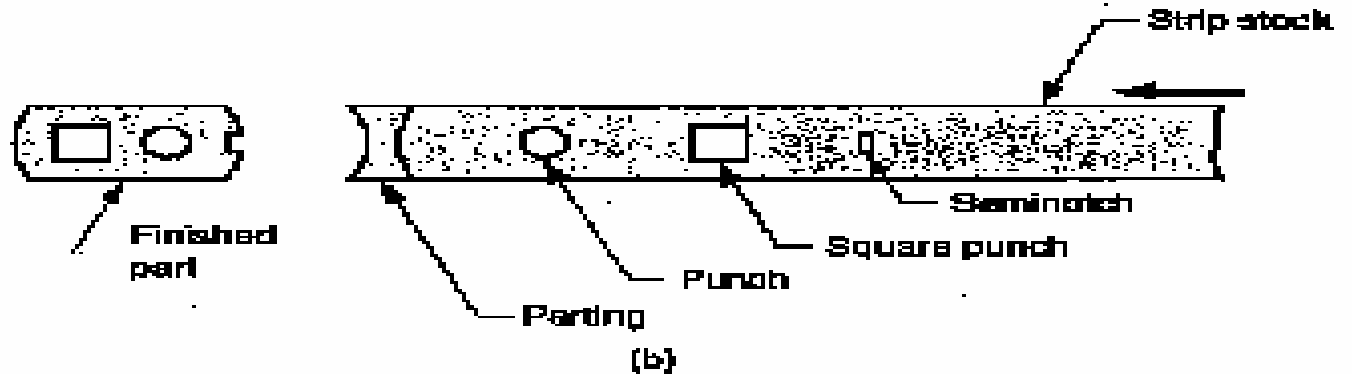
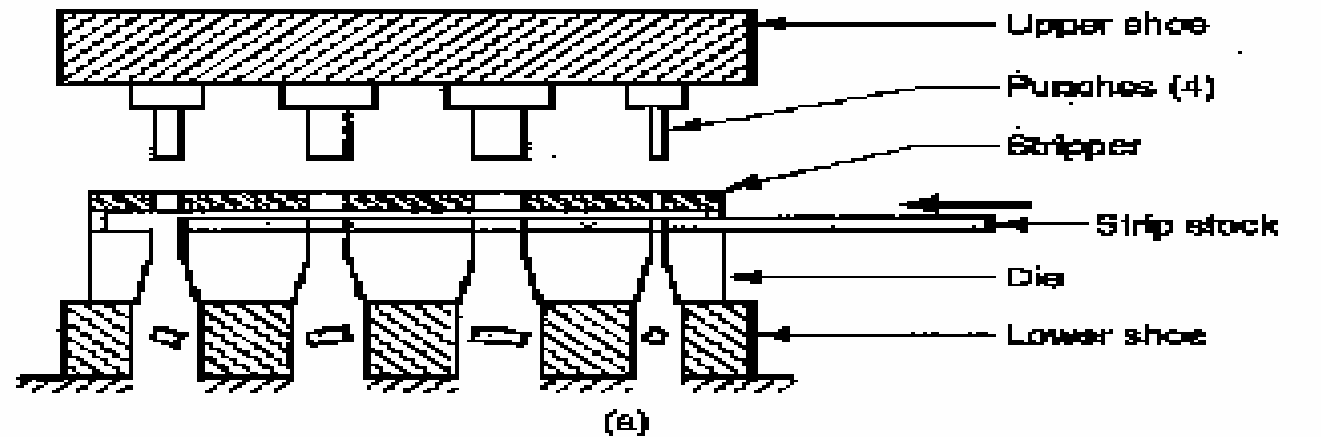
Examples of deep drawn parts

Common applications: cooking pots, containers, ...

Punch Die



Progressive Die



<https://youtu.be/NuFpzJLMnFs?t=88>

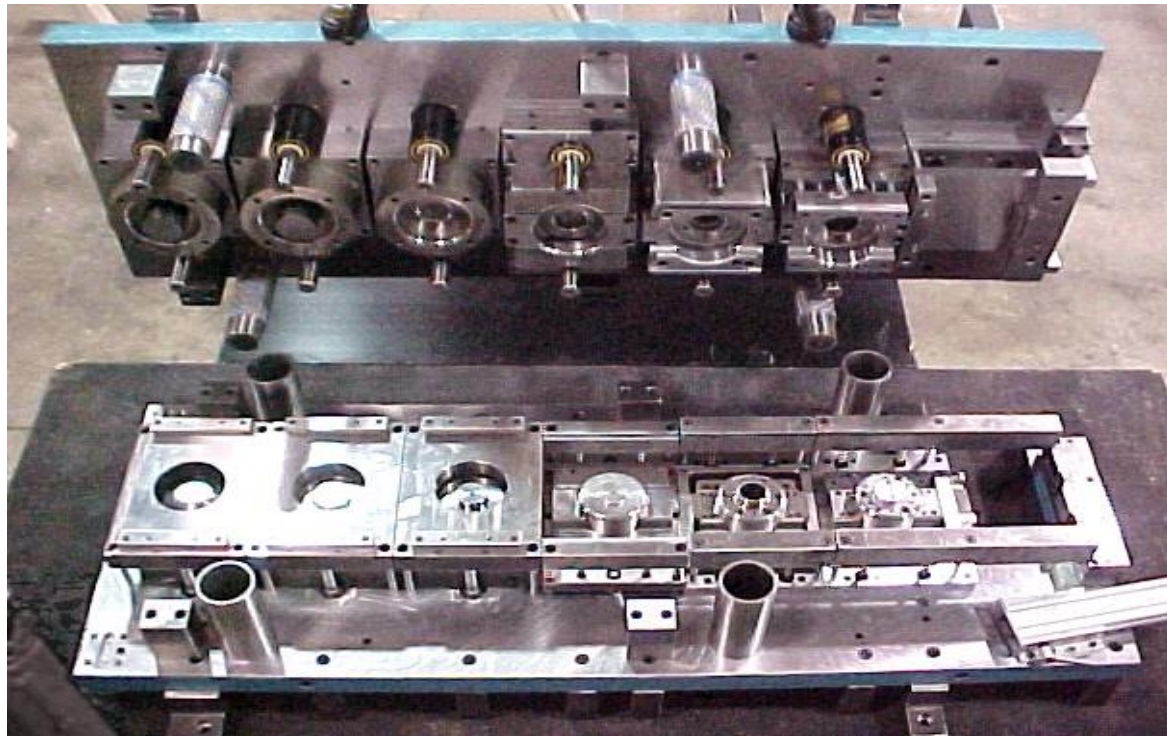
Progressive Die Stamping

Fabricating complex parts in earlier days was an inefficient and costly affair. Either a manufacturer could repeatedly change the tool in a single press or use multiple presses each performing single action to produce one part. This resulted in long load times and a high labor cost.

With progressive die stamping, a manufacturer can eliminate all these hassles and produce better quality parts faster and in a cost-effective manner.



Progressive Die



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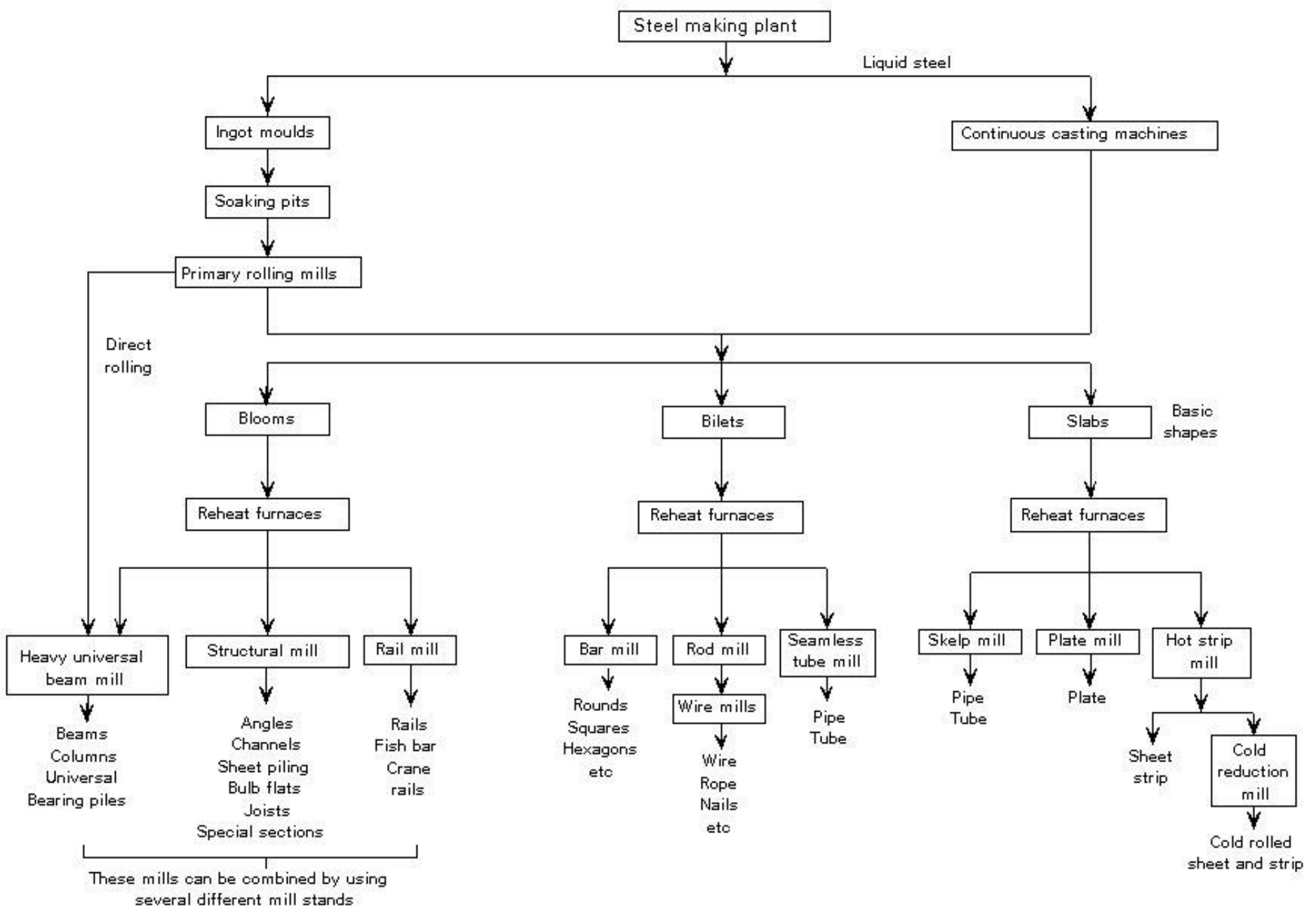


Figure 9 Principal product routes



Manufacturing Processes

Chapters 5:

Bulk Deformation Processes in Metal Working

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Introduction

- Bulk deformation processes in metal working include:
 - Rolling.
 - Other deformation processes related to rolling.
 - Forging.
 - Other deformation processes related to forging.
 - Extrusion.
 - Wire and Bar Drawing.



Introduction

- Bulk deformation processes accomplish significant shape change in metal parts whose initial form is bulk rather than sheet.
- The starting forms include (1) cylindrical bars and billets, (2) rectangular billets and slabs, and (3) similar elementary geometries.
- The bulk deformation processes **refine the starting shapes**, sometimes improving mechanical properties, and **always** adding commercial value.
- Deformation processes work by **stressing** the metal sufficiently to cause it to plastically flow into the desired shape.



Introduction

- Bulk deformation processes are performed as (1) cold, (2) warm, and (3) hot working operations.
- Cold and warm working is appropriate when the shape change is less severe, and there is a need to improve mechanical properties and achieve good finish on the part.
- Hot working is generally required when massive deformation of large workparts is involved.



Introduction

- The commercial and technological importance of bulk deformation processes derives from the following:
 - When performed as hot working operations, they can achieve significant change in the shape of the workpart.
 - When performed as cold working operations, they can be used not only to shape the product, but also to increase its strength through strain hardening.
 - These processes produce little or no waste as a byproduct of the operation. Some bulk deformation operations are near net shape or net shape processes; they achieve final product geometry with little or no subsequent machining.



Rolling

- **Rolling:** is a deformation process in which the thickness of the work is reduced by compressive forces exerted by two opposing rolls.
- The rolls rotate to pull and simultaneously squeeze the workpart between them.

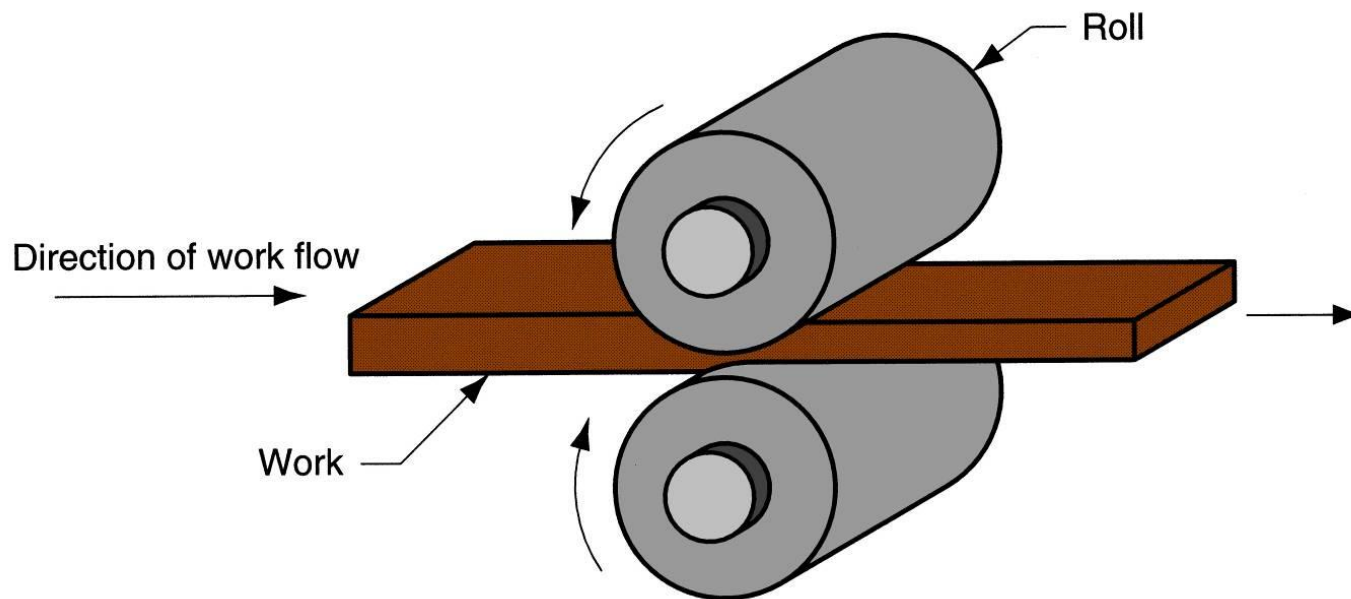


Figure 19.1 The rolling process (specifically, flat rolling).



Rolling

- According to the part geometry, the rolling processes can be divided into:
 - **Flat rolling**: used to reduce the thickness of a rectangular cross section.
 - **Shape rolling**: related to flat rolling, in which a square cross section is formed into a shape such as an I-beam.



Rolling

- Rolling can be carried out at high or low (ambient) temperatures.
 - **Hot rolling**: most rolling is carried out by hot working, due to the large amount of deformation required.
 - Hot-rolled metal is generally free of residual stresses, and its properties are isotropic (similar properties in different directions).
 - Disadvantages of hot rolling are that the product cannot be held to close tolerances, and the surface has a characteristic oxide scale.



Rolling

- Rolling can be carried out at high or low (ambient) temperatures.
 - ***Cold rolling***: less common than hot rolling.
 - Cold rolling strengthens the metal and permits a tighter tolerance on thickness.
 - the surface of the cold-rolled sheet is absent of scale and generally superior to the corresponding hot-rolled product.



Rolling

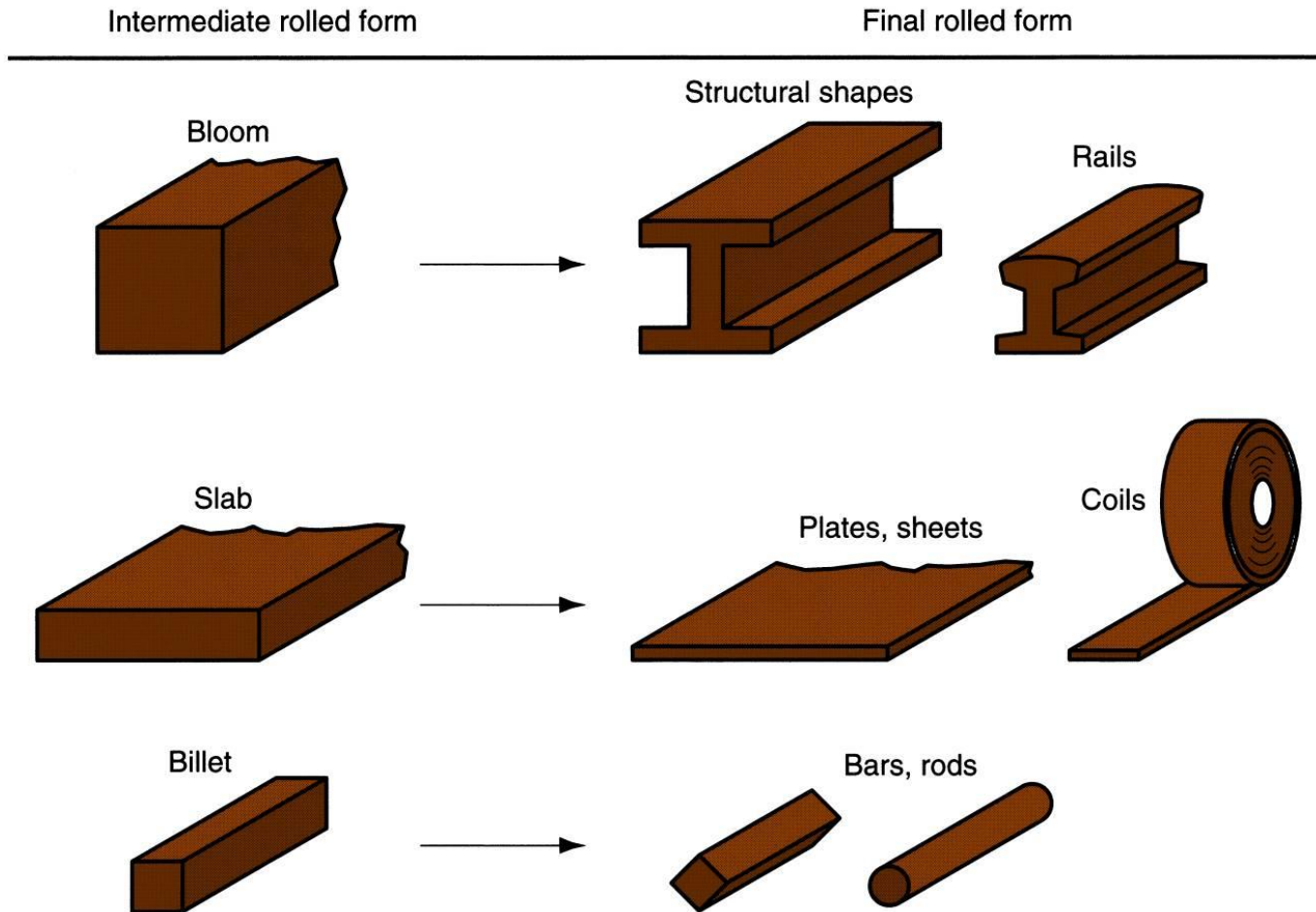


Figure 19.2 Some of the steel products made in a rolling mill.



Rolling

Flat Rolling and Its Analysis

- Flat rolling involves the rolling of workparts of rectangular cross section in which the width is greater than the thickness; e.g. slabs, strips, sheets and plates.
- **Draft** is amount of thickness reduction and described as:

$$d = t_0 - t_f$$

where d = draft, mm; t_0 = starting thickness, mm; and t_f = final thickness, mm.

- Draft is sometimes expressed as a fraction of the starting stock thickness, called the **Reduction** (r):

$$r = \frac{d}{t_0}$$



Rolling

Flat Rolling and Its Analysis

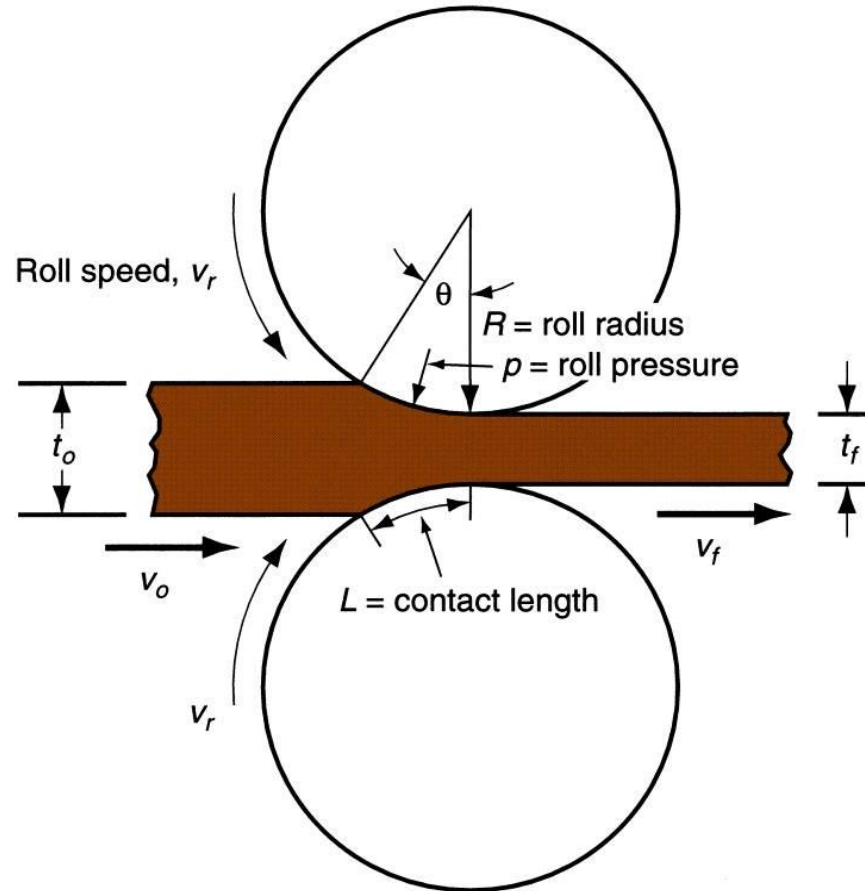


Figure 19.3 Side view of flat rolling, indicating before and after thicknesses, work velocities, angle of contact with rolls, and other features.



Rolling

Flat Rolling and Its Analysis

- **Spreading:** the increase in width due to rolling, described as:

$$t_o w_o L_o = t_f w_f L_f$$

where w_o and w_f are the before and after work widths, mm; and L_o and L_f are the before and after work lengths, mm.

- Similarly, before and after volume rates of material flow must be the same, so the before and after velocities can be related:

$$t_o w_o v_o = t_f w_f v_f$$

where v_o and v_f are the entering and exiting velocities of the work.



Rolling

Flat Rolling and Its Analysis

- True strain is expressed by:

$$\varepsilon = \ln \frac{t_o}{t_f}$$

- The true strain can be used to determine the average flow stress $\overline{Y}_f$ (MPa) applied to the work material in flat rolling:

$$\overline{Y}_f = \frac{K\varepsilon^n}{1+n}$$

The average flow stress is used to compute estimates of force and power in rolling.



Rolling

Flat Rolling and Its Analysis

- There is a limit to the maximum possible draft that can be accomplished in flat rolling with a given coefficient of friction, defined by:

$$d_{\max} = \mu^2 R$$

where $d_{\max}$ = maximum draft, mm; μ = coefficient of friction; and R = roll radius, mm.

- Rolling force (F , N) can be expressed as:

$$F = \overline{Y_f} wL$$

- Contact length (L , mm) is described as:

$$L = \sqrt{R(t_o - t_f)}$$

- The torque (T) and the power required to drive each roll (P , J/s) are:

$$T = 0.5FL$$

and

$$P = 2\pi NFL$$

where P = power, J/s or W; N = rotational speed, 1/s; F = rolling force, N; and L = contact length, m.



Rolling Shape Rolling

- In shape rolling, the work is deformed into a contoured cross section.
- Products include construction shapes such as I-beams, L-beams, and U-channels; rails for railroad tracks; and round and square bars and rods.
- The process is accomplished by passing the work through rolls that have the reverse of the desired shape.
- Most of the principles that apply in flat rolling are also applicable to shape rolling.
- Shaping rolls are more complicated; and the work, usually starting as a square shape, requires a gradual transformation through several rolls in order to achieve the final cross section.



Rolling Rolling Mills

- Rolling mill configurations:
 - ***Two-high***: consists of two opposing rolls, and the configuration can be either reversing or nonreversing.

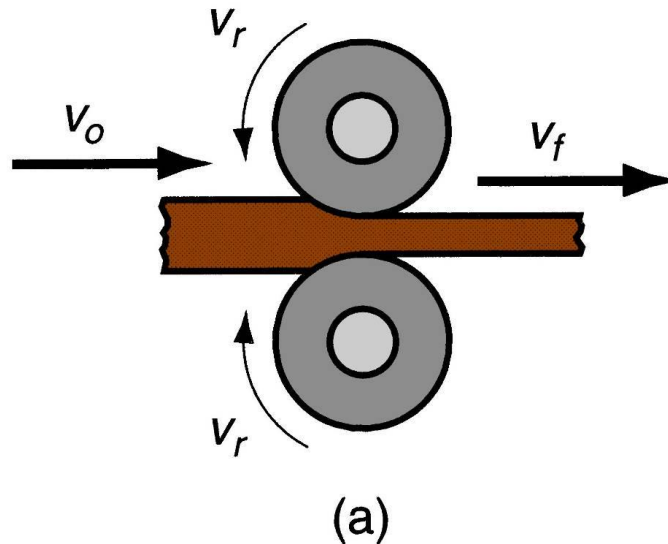


Figure 19.4 Various configurations of rolling mills: (a) two-high rolling mill.



Rolling Rolling Mills

- Rolling mill configurations:
 - **Three-high**: three rolls in a vertical column, and the direction of rotation of each roll remains unchanged.

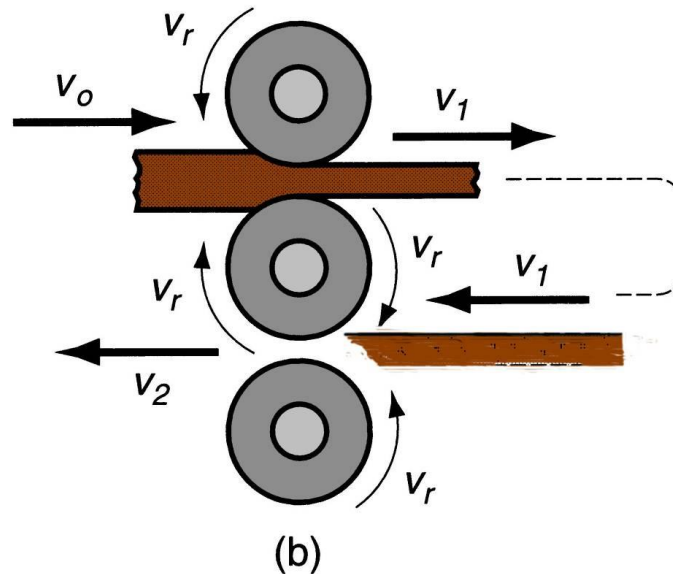


Figure 19.4 Various configurations of rolling mills: (b) three-high rolling mill.



Rolling Rolling Mills

- Rolling mill configurations:
 - **Four-high**: uses two smaller-diameter rolls to contact the work and two backing rolls behind them.

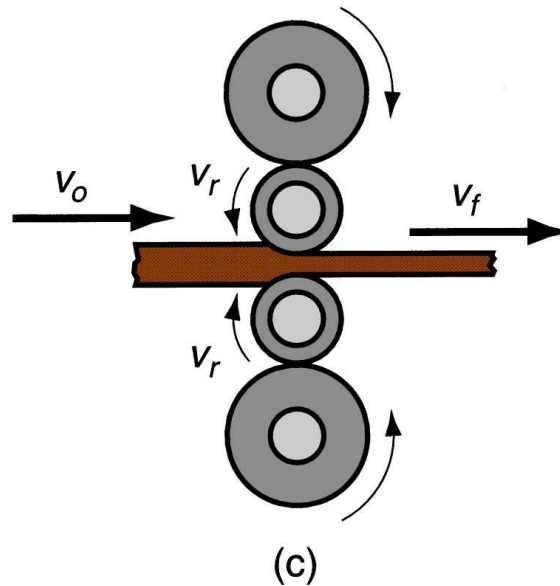


Figure 19.4 Various configurations of rolling mills: (c) four-high rolling mill.



Rolling Rolling Mills

- Rolling mill configurations:
 - **Cluster mill:** roll configuration that allows smaller working rolls against the work (smaller than in four-high mills).

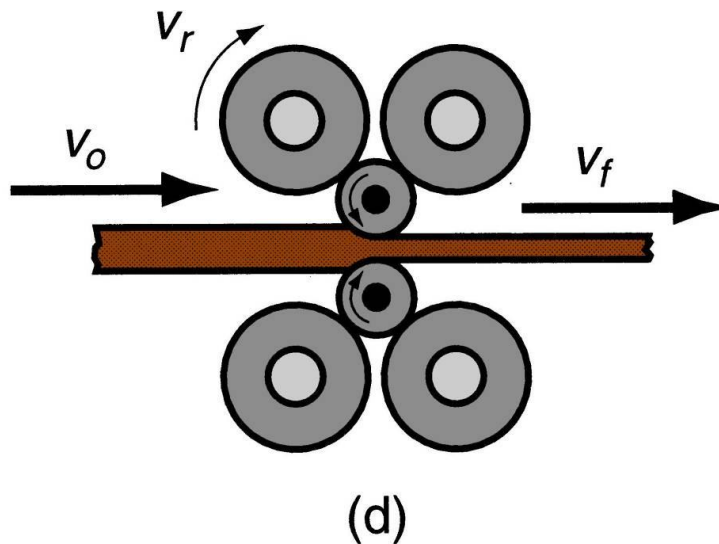


Figure 19.4 Various configurations of rolling mills: (d) cluster mill.



Rolling Rolling Mills

- Rolling mill configurations:
 - ***Tandem rolling mill*** : consists of a series of rolling stands, aimed at higher throughput rates.

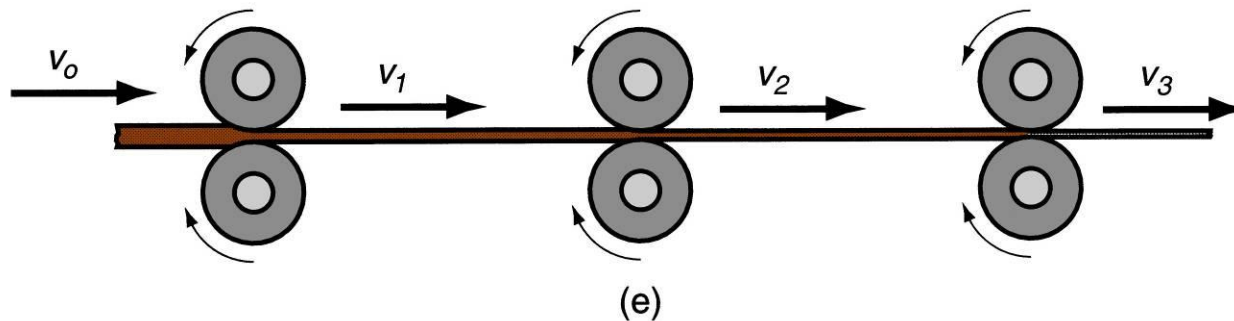


Figure 19.4 Various configurations of rolling mills: (e) tandem rolling mill.



Other Deformation Processes Related to Rolling

- ***Thread Rolling:***

- Used to form threads on cylindrical parts by rolling them between two dies.
- The most important commercial process for mass producing external threaded components.
- Performed by cold working in thread rolling machines. These are equipped with special dies that determine the size and form of the thread.
- Advantages of thread rolling over thread cutting and rolling include:
 - Higher production rates.
 - Better material utilization.
 - Smoother surface.
 - Stronger threads and better fatigue resistance due to work hardening.



Other Deformation Processes Related to Rolling

- ***Thread Rolling:***

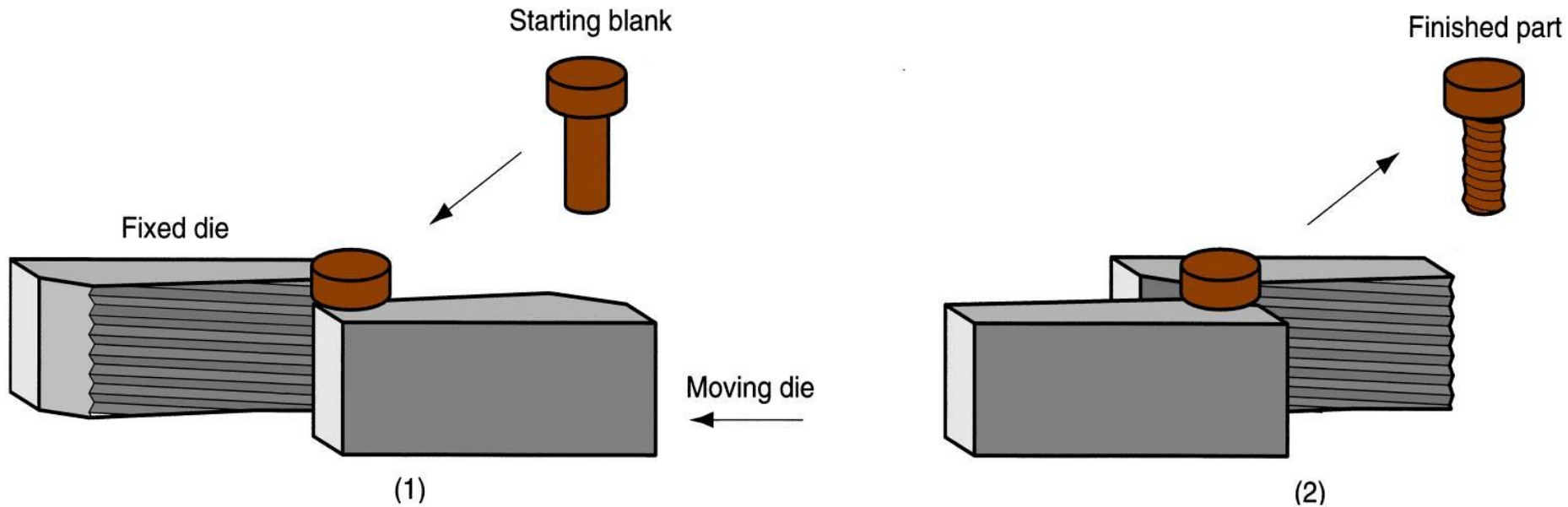


Figure 19.5 Thread rolling with flat dies: (1) start, and (2) end of cycle.



Other Deformation Processes Related to Rolling

- **Ring Rolling:** a deformation process in which a thick-walled ring of smaller diameter is rolled into a thin-walled ring of larger diameter.
 - As the thick-walled ring is compressed, the deformed material elongates, causing the diameter of the ring to be enlarged.

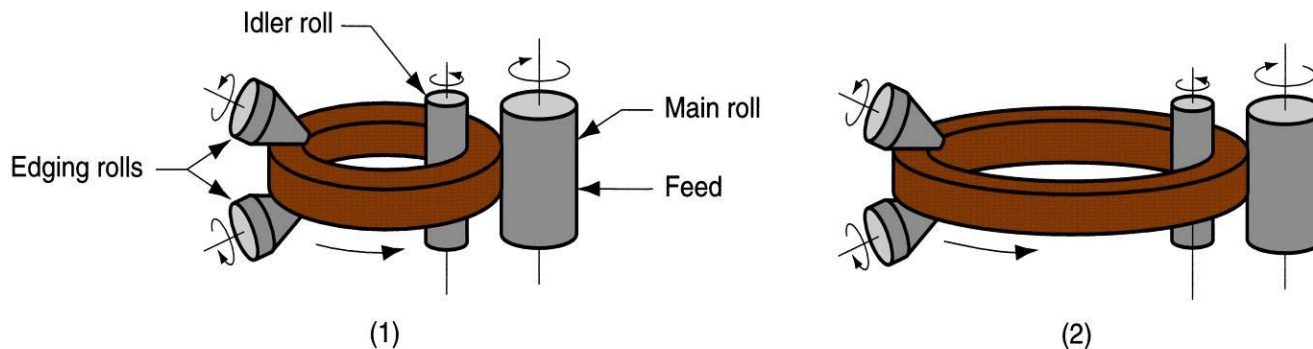


Figure 19.6 Ring rolling used to reduce the wall thickness and increase the diameter of a ring: (1) start, and (2) completion of process.



Other Deformation Processes Related to Rolling

- ***Ring Rolling:***

- Usually performed as a hot-working process for large rings and as a cold-working process for smaller rings.
- Applications include ball and roller bearing races, steel tires for railroad wheels, and rings for pipes, pressure vessels, and rotating machinery.
- Advantages over processes producing similar products include: (1) raw material savings, (2) ideal grain orientation for the application, and (3) strengthening through cold working.



Other Deformation Processes Related to Rolling

- ***Roll Piercing***: a specialized hot working process for making seamless thick-walled tubes.
 - Based on the principle that when a solid cylindrical part is compressed on its circumference, high tensile stresses are developed at its center. If compression is high enough, an internal crack is formed.
 - Compressive stresses on a solid cylindrical billet are applied by two rolls, whose axes are oriented at slight angles (6°) from the axis of the billet, so that their rotation tends to pull the billet through the rolls. A mandrel is used to control the size and finish of the hole created by the action.



Other Deformation Processes Related to Rolling

- ***Roll Piercing:***

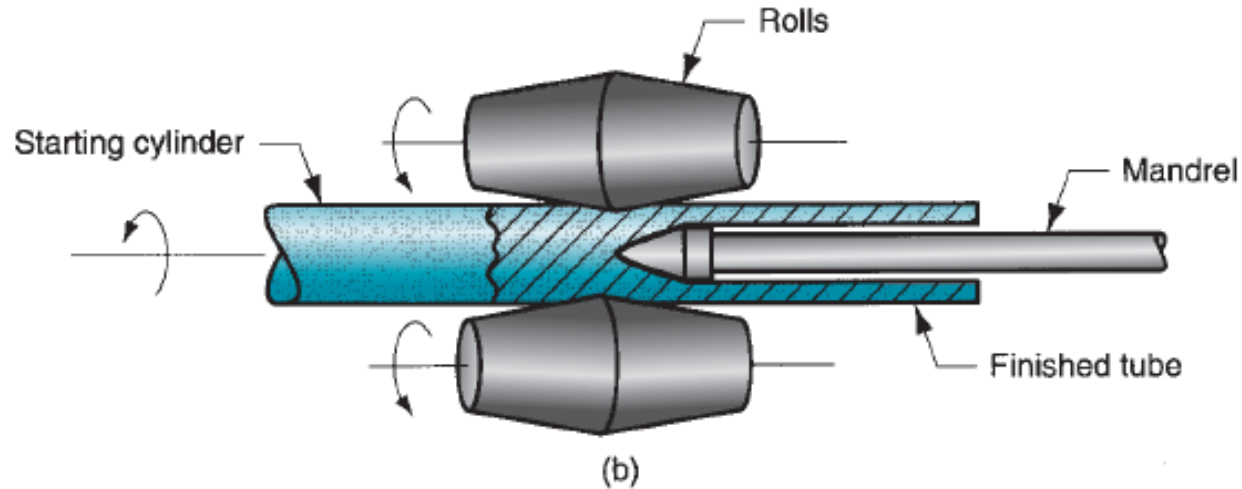
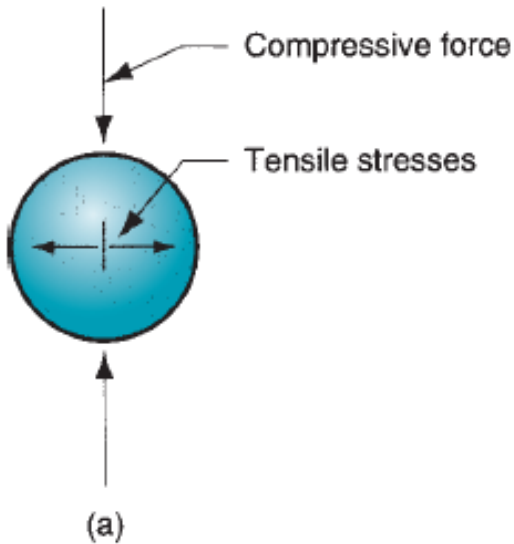


Figure 19.7 Roll piercing: (a) formation of internal stresses and cavity by compression of cylindrical part; and (b) setup of Mannesmann roll mill for producing seamless tubing.



Forging

- **Forging**: a deformation process in which the work is compressed between two dies, using either impact or gradual pressure to form the part.
 - Dates back to perhaps 5000 BCE.
 - Today, forging is an important industrial process used to make a variety of high-strength components for automotive, aerospace, and other applications.
 - These components include engine crankshafts and connecting rods, gears, aircraft structural components, and jet engine turbine parts.
 - In addition, steel and other basic metals industries use forging to establish the basic form of large components that are subsequently machined to final shape and dimensions.



Forging

- Forging can be classified in many ways, one is working temperature.
 - **Hot or warm forging**: done when significant deformation is demanded by the process and when strength reduction and increase of ductility is required.
 - **Cold forging**: its advantage is the increased strength that results from strain hardening of the component.
- The other way is by the way the forging is carried out:
 - **Forging hammer**: a forging machine that applies an impact load.
 - **Forging press**: a forging machine that applies gradual load.



Forging

- Forging can be also classified according to the degree to which the flow of the work metal is constrained by the dies.
 - ***Open-die forging***: the work is compressed between two flat dies, thus allowing the metal to flow without constraint in a lateral direction relative to the die surfaces.
 - ***Impression-die forging***: the die surfaces contain a shape or impression that is imparted to the work during compression, thus constraining metal flow to a significant degree. Here, flash will form.
 - ***Flashless forging***: the work is completely constrained within the die and no excess flash is produced.



Forging

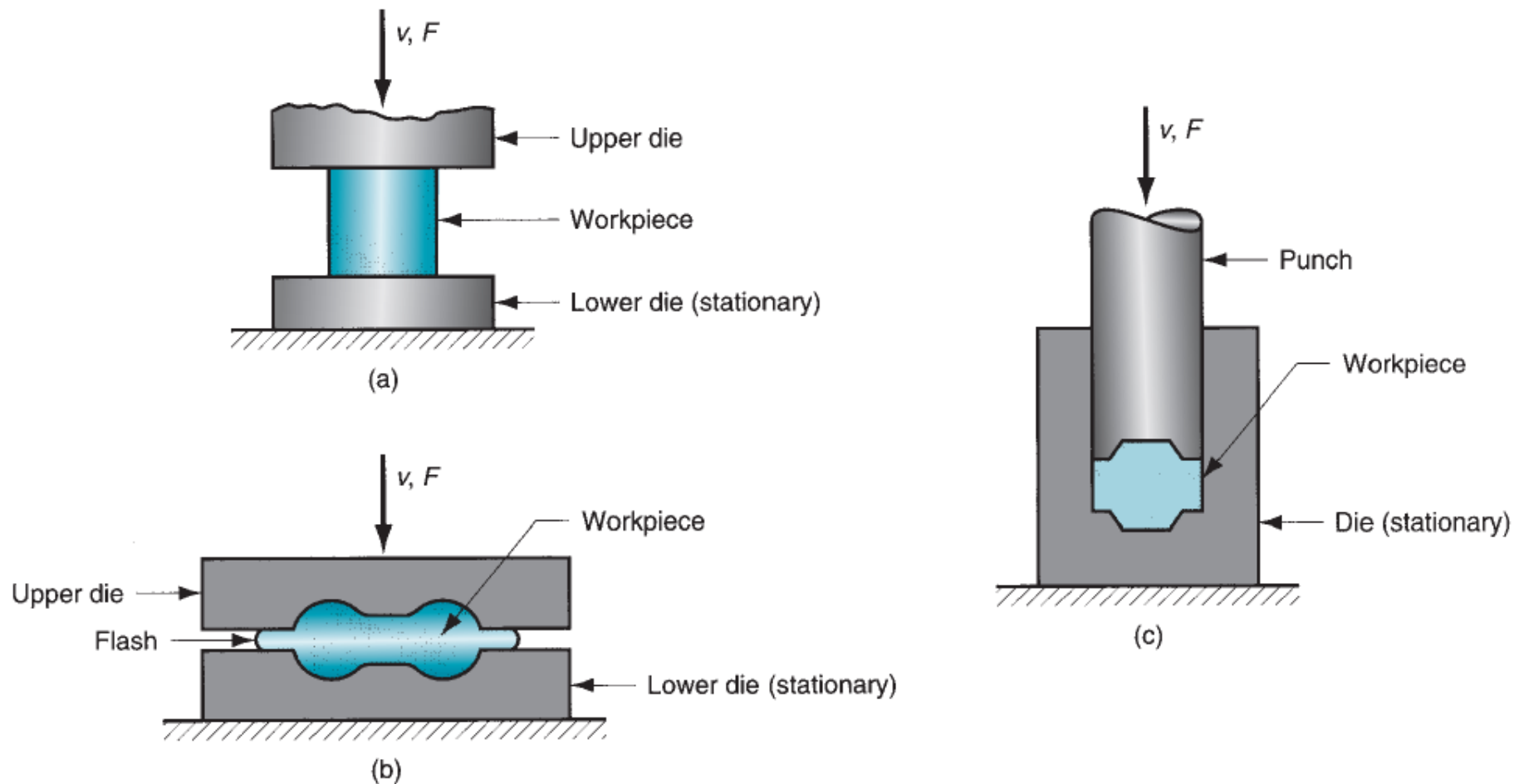


Figure 19.8 Three types of forging operation: (a) open-die forging, (b) impression-die forging, and (c) flashless forging.



Forging

Open-Die Forging

- Known as ***upsetting*** or ***upset forging***.
- Involves compression of a workpart of cylindrical cross section between two flat dies, much in the manner of a compression test.
- It reduces the height of the work and increases the diameter.



Forging

Open-Die Forging

- Analysis of Open-Die Forging:
 - If carried out under ideal conditions of no friction between work and die surfaces, then homogeneous deformation occurs, and the flow of the material is uniform throughout its height.

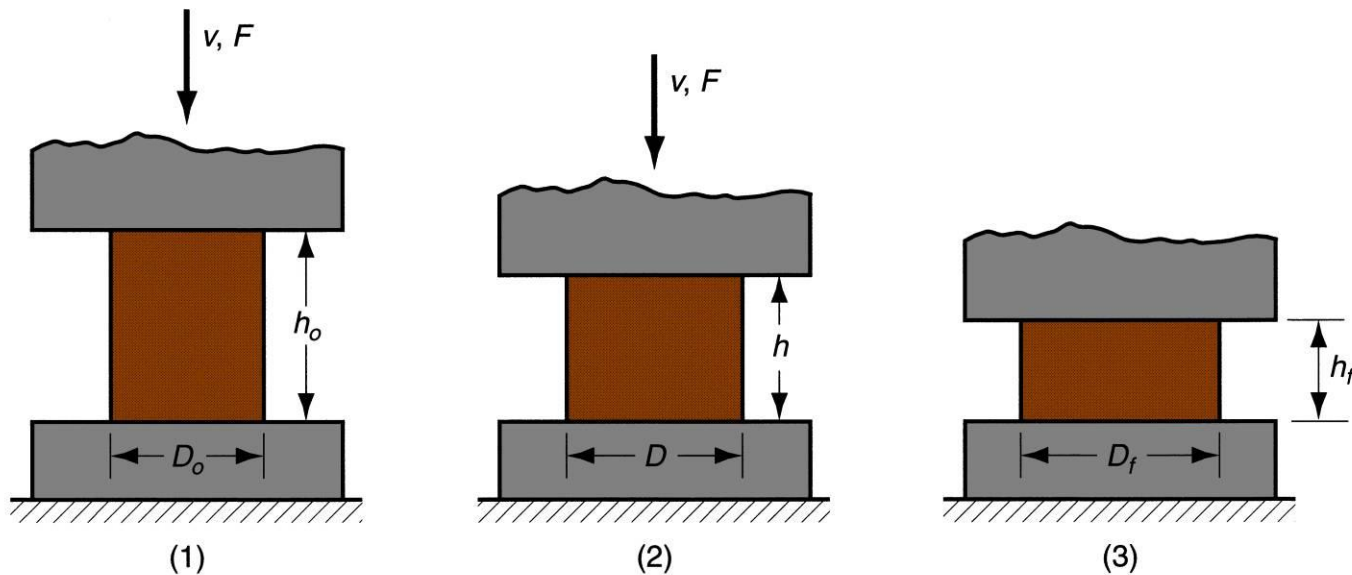


Figure 19.9 Homogeneous deformation of a cylindrical workpart under ideal conditions in an open-die forging operation: (1) start of process with workpiece at its original length and diameter, (2) partial compression, and (3) final size. 33



Forging

Open-Die Forging

- Analysis of Open-Die Forging:
 - Under these ideal conditions, the true strain experienced by the work during the process can be determined by:

$$\varepsilon = \ln \frac{h_o}{h}$$

- The force to perform upsetting at any height is given by:

$$F = Y_f A$$

where F = force, N; A = cross-sectional area, mm²; and Y_f = flow stress, MPa.



Forging

Open-Die Forging

- Analysis of Open-Die Forging:
 - If carried out under conditions where friction between work and die surfaces is accounted for, a barreling effect is created.

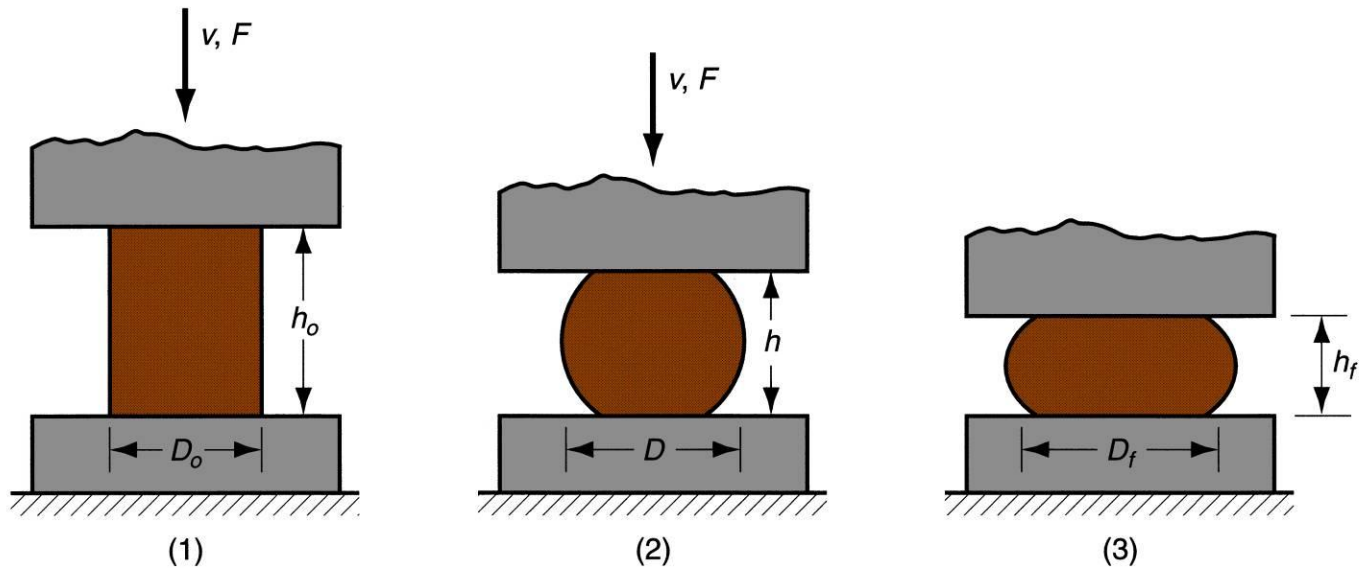


Figure 19.10 Actual deformation of a cylindrical workpart in open-die forging, showing pronounced barreling: (1) start of process, (2) partial deformation, and (3) final shape.



Forging

Open-Die Forging

- Analysis of Open-Die Forging:
 - Friction causes the actual upsetting force to be greater than what is predicted the previous equation:

$$F = K_f Y_f A$$

where K_f is the forging shape factor, defined as:

$$K_f = 1 + \frac{0.4\mu D}{h}$$

where μ = coefficient of friction; D = workpart diameter or other dimension representing contact length with die surface, mm; and h = workpart height, mm.



Forging

Open-Die Forging

- In practice, open-die forging can be classified into:
 - **Fullering**: a forging operation performed to reduce the cross section and redistribute the metal in a workpart in preparation for subsequent shape forging (dies have convex surfaces).
 - **Edging**: similar to fullering, except that the dies have concave surfaces.
 - **Cogging**: consists of a sequence of forging compressions along the length of a workpiece to reduce cross section and increase length.



Forging

Open-Die Forging

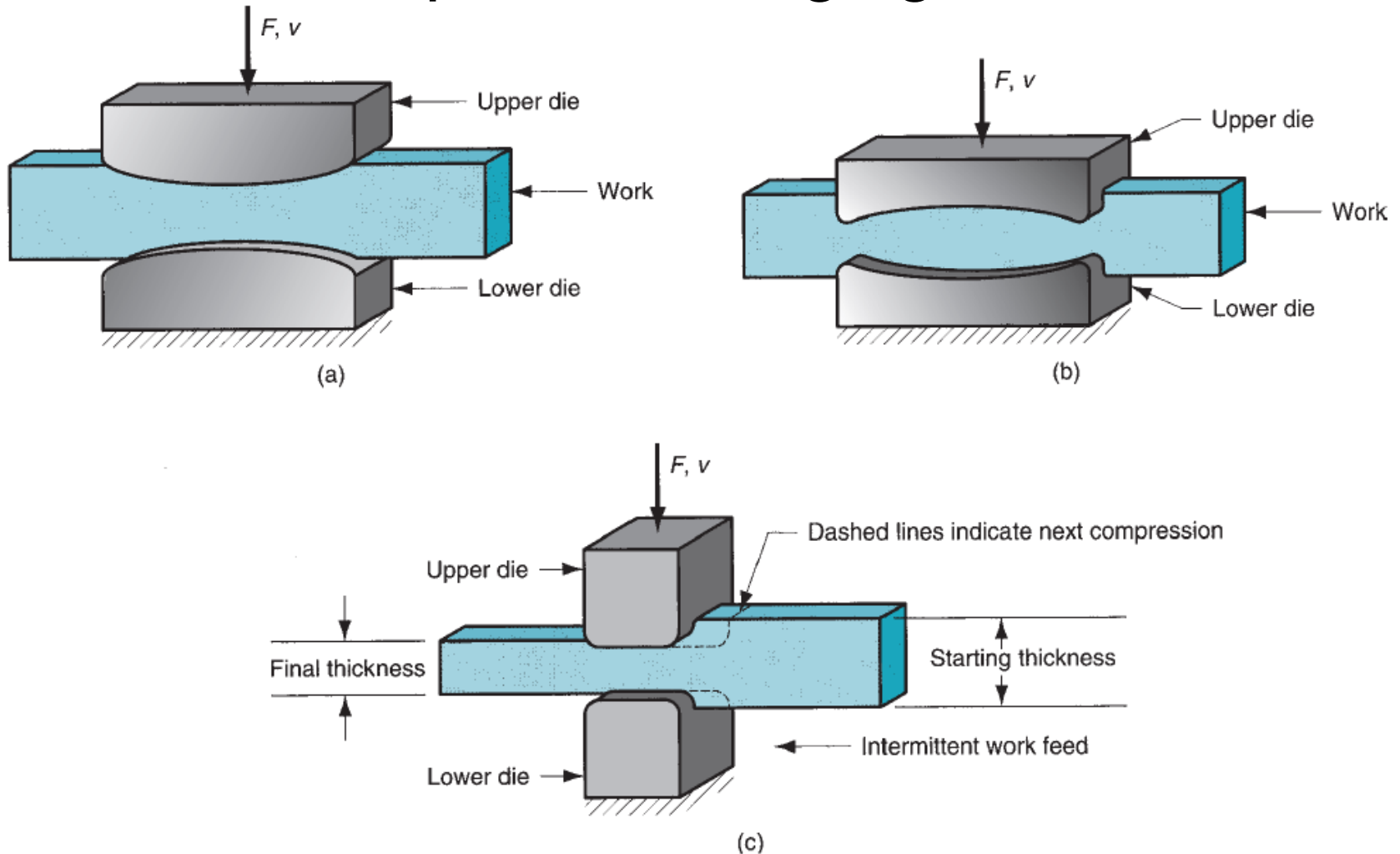


Figure 19.11 Several open-die forging operations: (a) fullering, (b) edging, and (c) cogging.



Forging

Impression-Die Forging

- ***Impression-die forging*** (sometimes called ***closed-die forging***): performed with dies that contain the inverse of the desired shape of the part.
 - As the die closes to its final position, flash is formed by metal that flows beyond the die cavity and into the small gap between the die plates.
 - Although this flash must be finally cut away, it serves an important function during impression-die forging.
 - As the flash begins to form, friction resists continued flow of metal into the gap, thus constraining the bulk of the work material to remain in the die cavity.
 - In hot forging, metal flow is further restricted because the thin flash cools quickly against the die plates, thereby increasing its resistance to deformation.
 - Accordingly, compression pressure is increased, thus forcing the material to fill the whole cavity.



Forging

Impression-Die Forging

- Sequence in impression-die forging:

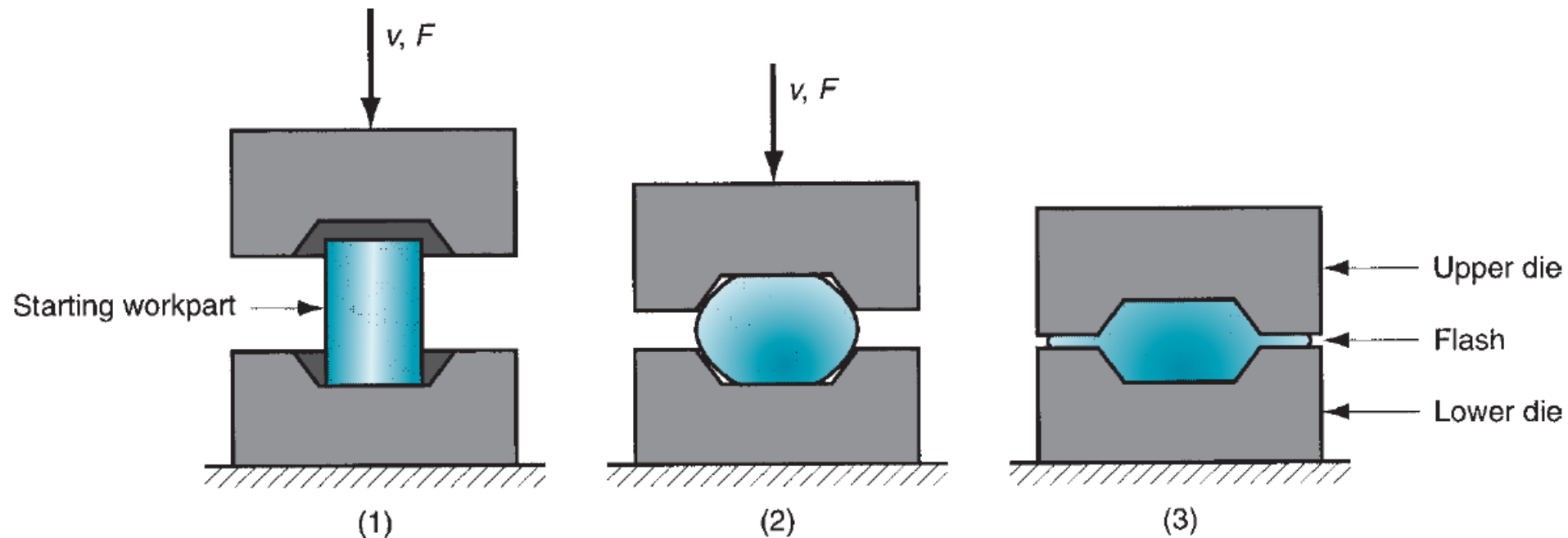


Figure 19.12 Sequence in impression-die forging: (1) just prior to initial contact with raw workpiece, (2) partial compression, and (3) final die closure, causing flash to form in gap between die plates.



Forging

Impression-Die Forging

- Advantages of impression-die forging compared to machining from solid stock include: higher production rates, less waste of metal, greater strength and favorable grain orientation in the metal.
- Limitations include: the incapability of close tolerances and machining is often required to achieve accuracies and features needed.



Forging

Flashless Forging

- ***Flashless Forging***: the raw workpiece is completely contained within the die cavity during compression, and no flash is formed.
- Several requirements:
 - The work volume must equal the space in the die cavity within a very close tolerance.
 - If the starting blank is too large, excessive pressures may cause damage to the die or press. If the blank is too small, the cavity will not be filled.
 - Simple geometries required.
 - Best for soft metals, such as aluminum and copper and their alloys.
 - Sometimes classified as ***Precision Forging***.



Forging

Flashless Forging

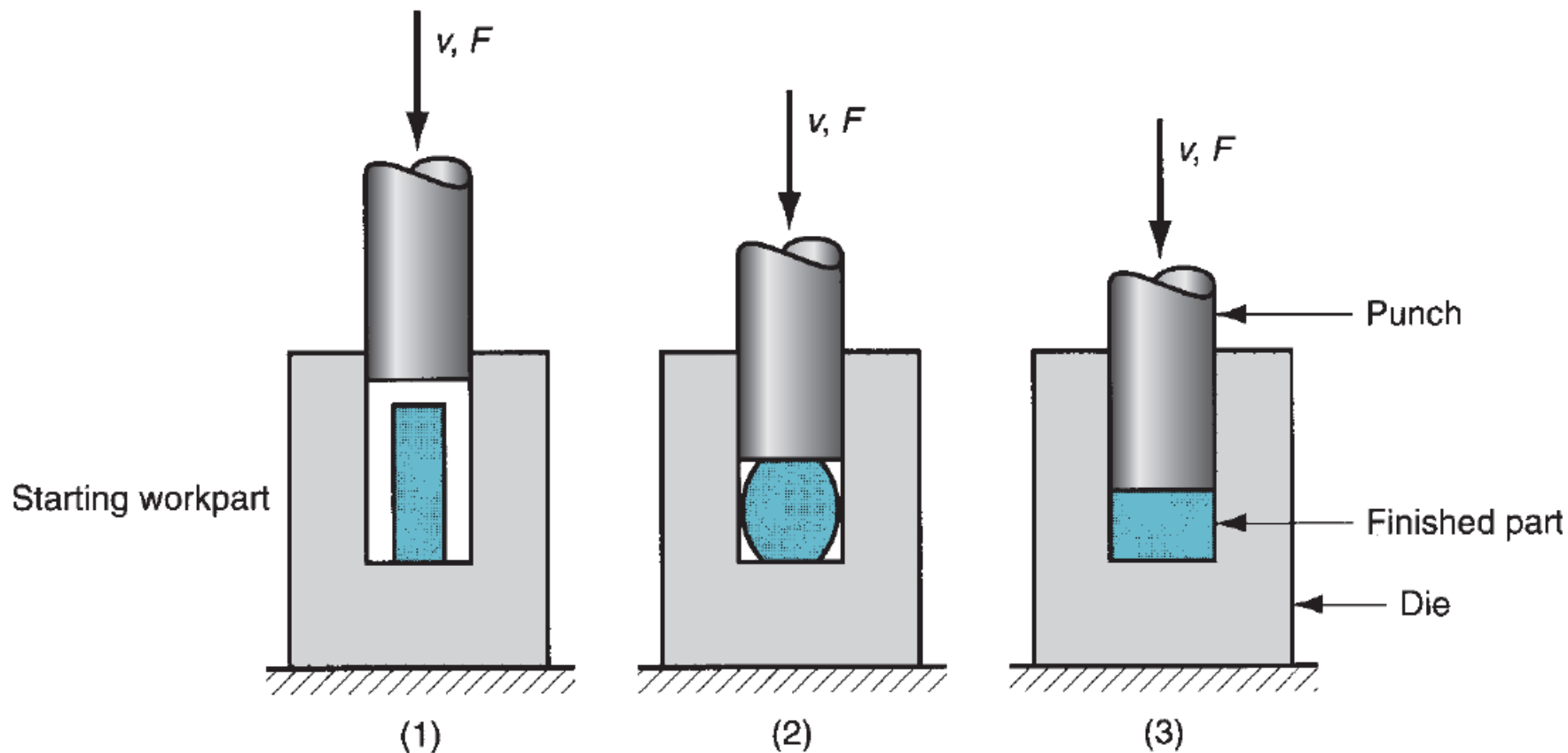


Figure 19.13 Flashless forging: (1) just before initial contact with workpiece, (2) partial compression, and (3) final punch and die closure.



Forging

Flashless Forging

- **Coining:** is a type of flashless forging, in which fine details in the die are impressed into the top and bottom surfaces of the workpart. There is little flow of metal in coining.

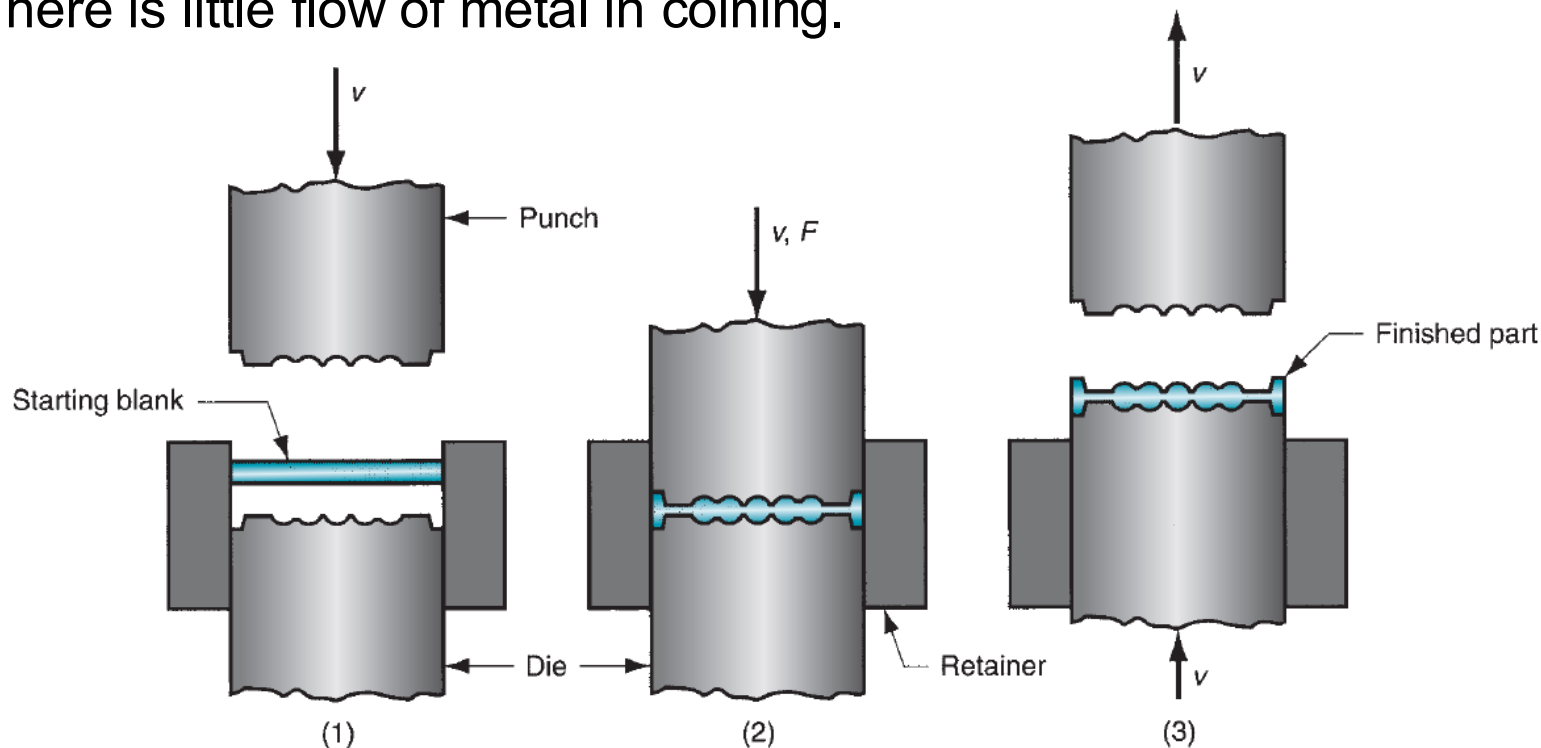


Figure 19.14 Coining operation: (1) start of cycle, (2) compression stroke, and (3) ejection of finished part.



Forging

Forging Hammers and Presses

- Equipment used in forging consists of forging machines, classified as hammers or presses, and forging dies.
- In addition, auxiliary equipment is needed, such as furnaces to heat the work, mechanical devices to load and unload the work, and trimming stations to cut away the flash in impression-die forging.



Forging

Forging Hammers and Presses

(1) **Forging Hammers**: operate by applying an impact loading against the work. They deliver impact energy to the workpiece.

- Used for impression-die forging.
- The upper portion of the forging die is attached to the ram, and the lower portion is attached to the anvil.
- The work is placed on the lower die, and the ram is lifted and then dropped.
- When the upper die strikes the work, the impact energy causes the part to assume the form of the die cavity.
- Several blows of the hammer are often required to achieve the desired change in shape.



Forging

Forging Hammers and Presses

- Forging hammers are classified into:
 - (1) **Gravity drop hammers:** achieve their energy by the falling weight of a heavy ram, and the force of the blow is determined by the height of the drop and the weight of the ram.
 - (2) **Power drop hammers:** accelerate the ram by pressurized air or steam.
- Disadvantage: a large amount of the impact energy is transmitted through the anvil and into the floor of the building.



Forging

Forging Hammers and Presses

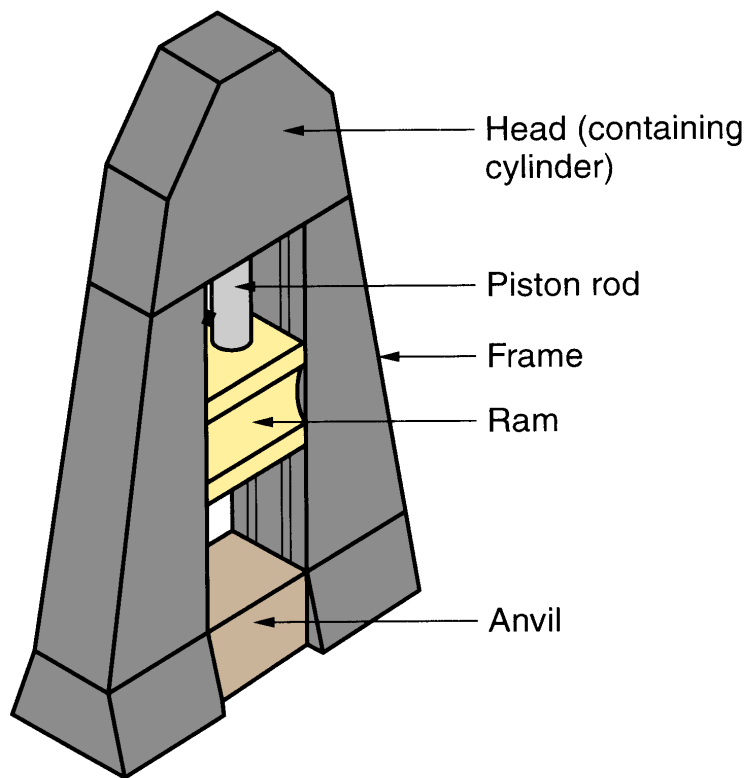


Figure 19.15 Diagram showing details of a drop hammer for impression-die forging.



Forging

Forging Hammers and Presses

(2) ***Forging Presses***: apply gradual pressure, rather than sudden impact, to accomplish the forging operation.

- Include mechanical presses, hydraulic presses, and screw presses.
- Mechanical presses convert the rotating motion of a drive motor into the translation motion of the ram.
- Hydraulic presses use a hydraulically driven piston to drive the ram.
- Screw presses apply force by a screw mechanism that drives the vertical ram.



Other Deformation Processes Related to Forging

- **Upsetting** and **Heading**: a deformation operation in which a cylindrical workpart is increased in diameter and reduced in length.
- Used in the fastener industry to form heads on nails, bolts, etc (in these applications, it is referred to as heading).
- More parts produced by upsetting than any other forging operation.
- Performed cold, hot or warm on special upset forging machines, called headers or formers.
- Long wire is fed into the machines, the end of the stock is upset forged, and then the piece is cut to length to make the desired hardware item.



Other Deformation Processes Related to Forging

- **Upsetting** and **Heading**: a deformation operation in which a cylindrical workpart is increased in diameter and reduced in length.

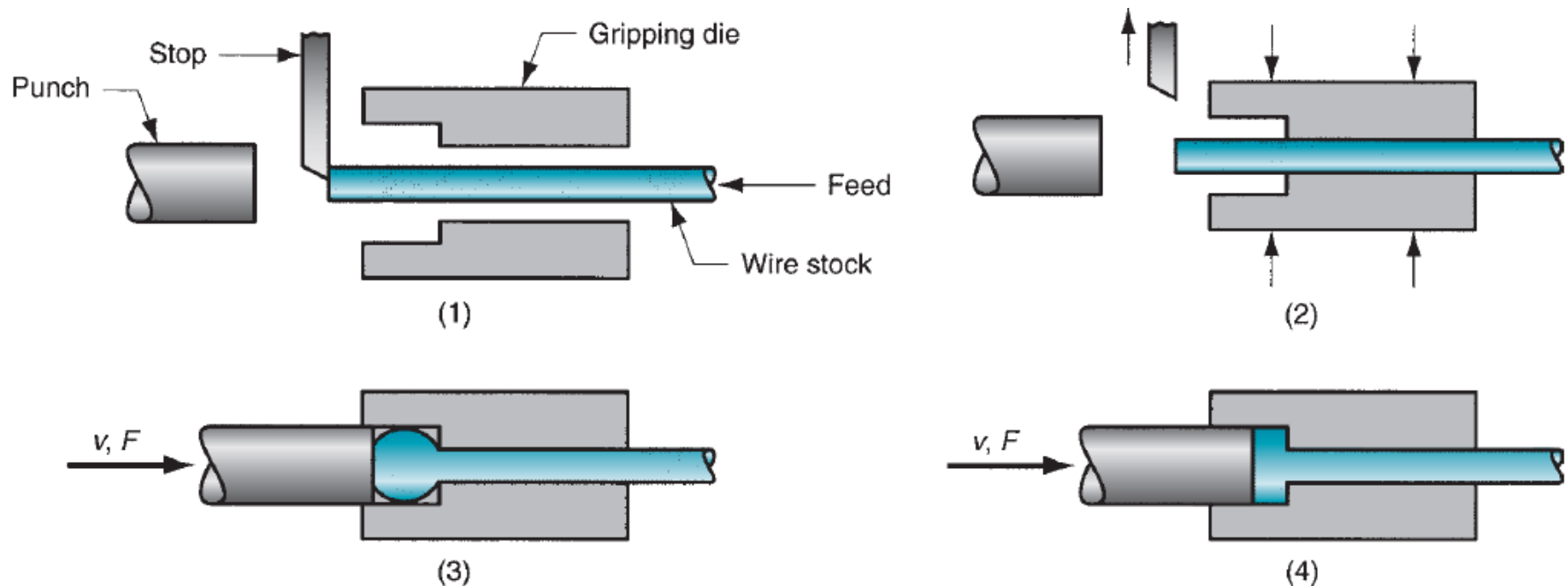


Figure 19.16 An upset forging operation to form a head on a bolt. (1) wire stock is fed to the stop; (2) gripping dies close on the stock and the stop is retracted; (3) punch moves forward; and (4) bottoms to form the head.



Other Deformation Processes Related to Forging

- **Upsetting** and **Heading**: a deformation operation in which a cylindrical workpart is increased in diameter and reduced in length.

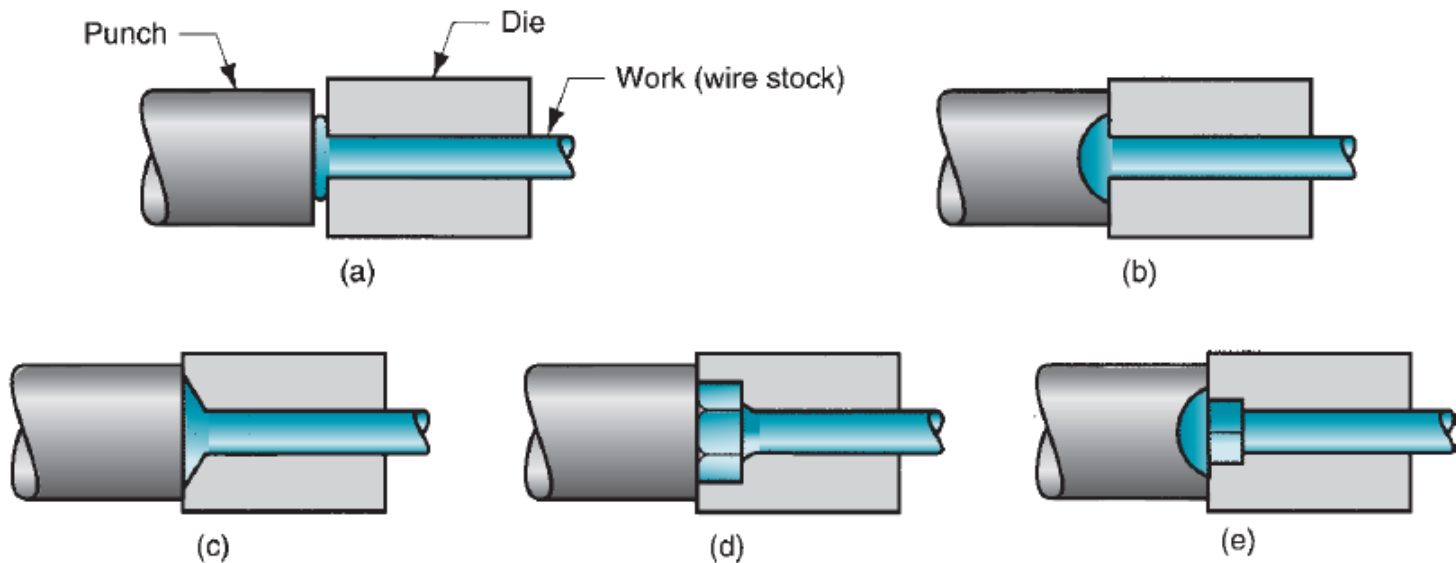


Figure 19.17 Examples of heading (upset forging) operations: (a) heading a nail using open dies, (b) round head formed by punch, (c) and (d) heads formed by die, and (e) carriage bolt head formed by punch and die.



Other Deformation Processes Related to Forging

- **Swaging** and **Radial Forging**: forging processes used to reduce the diameter of a tube or solid rod.
- The **swaging** process is accomplished by means of rotating dies that hammer a workpiece radially inward to taper it as the piece is fed into the dies.
- **Radial forging** is similar to swaging in its action against the work and is used to create similar part shapes. The difference is that in radial forging the dies do not rotate around the workpiece; instead, the work is rotated as it feeds into the hammering dies.



Other Deformation Processes Related to Forging

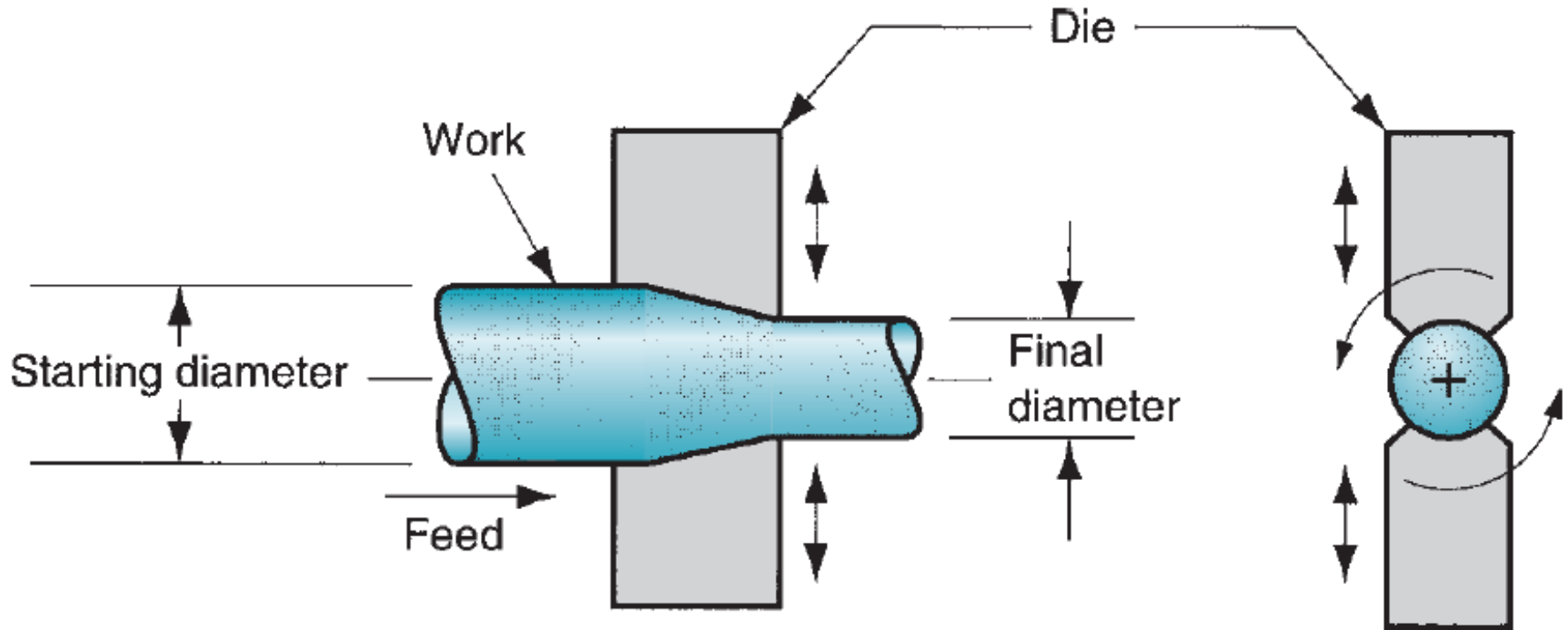


Figure 19.18 Swaging process to reduce solid rod stock; the dies rotate as they hammer the work. In radial forging, the workpiece rotates while the dies remain in a fixed orientation as they hammer the work.



Other Deformation Processes Related to Forging

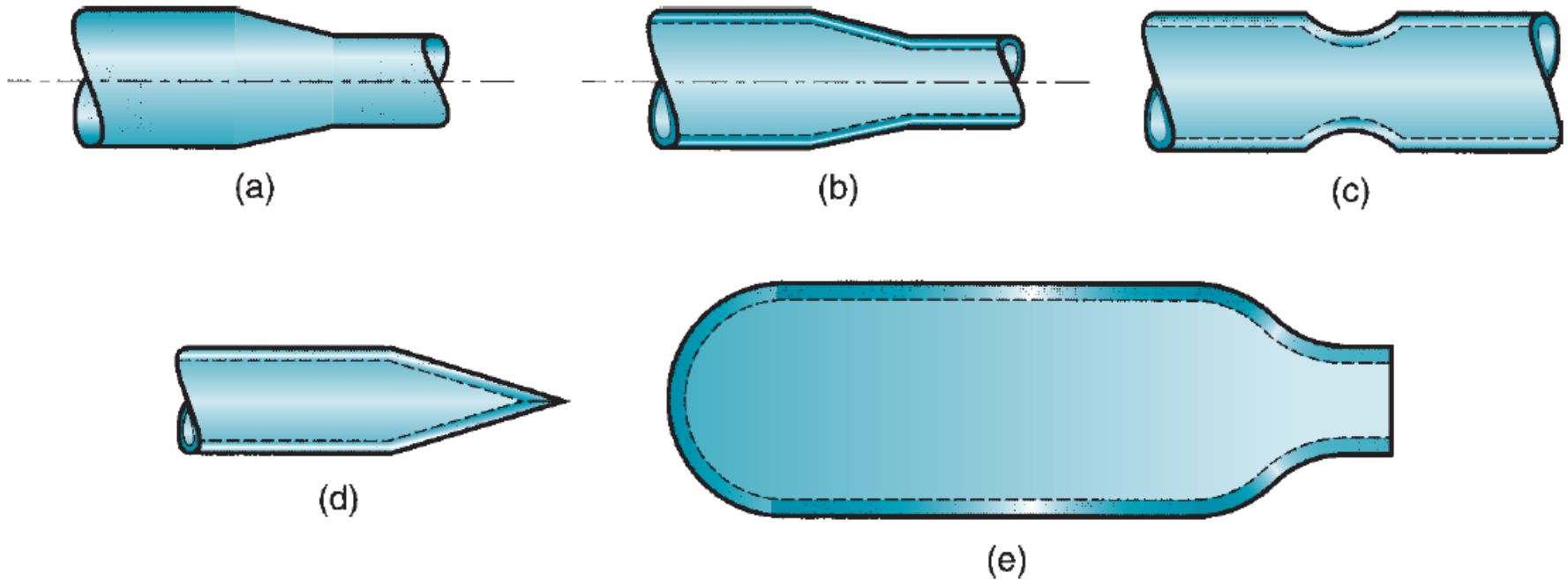


Figure 19.19 Examples of parts made by swaging: (a) reduction of solid stock, (b) tapering a tube, (c) swaging to form a groove on a tube, (d) pointing of a tube, and (e) swaging of neck on a gas cylinder.



Other Deformation Processes Related to Forging

- ***Trimming:*** an operation used to remove flash on the workpart in impression-die forging.
- In most cases, trimming is accomplished by shearing.
- Trimming is usually done while the work is still hot.
- In cases where the work might be damaged by the cutting process, trimming may be done by alternative methods, such as grinding or sawing.



Other Deformation Processes Related to Forging

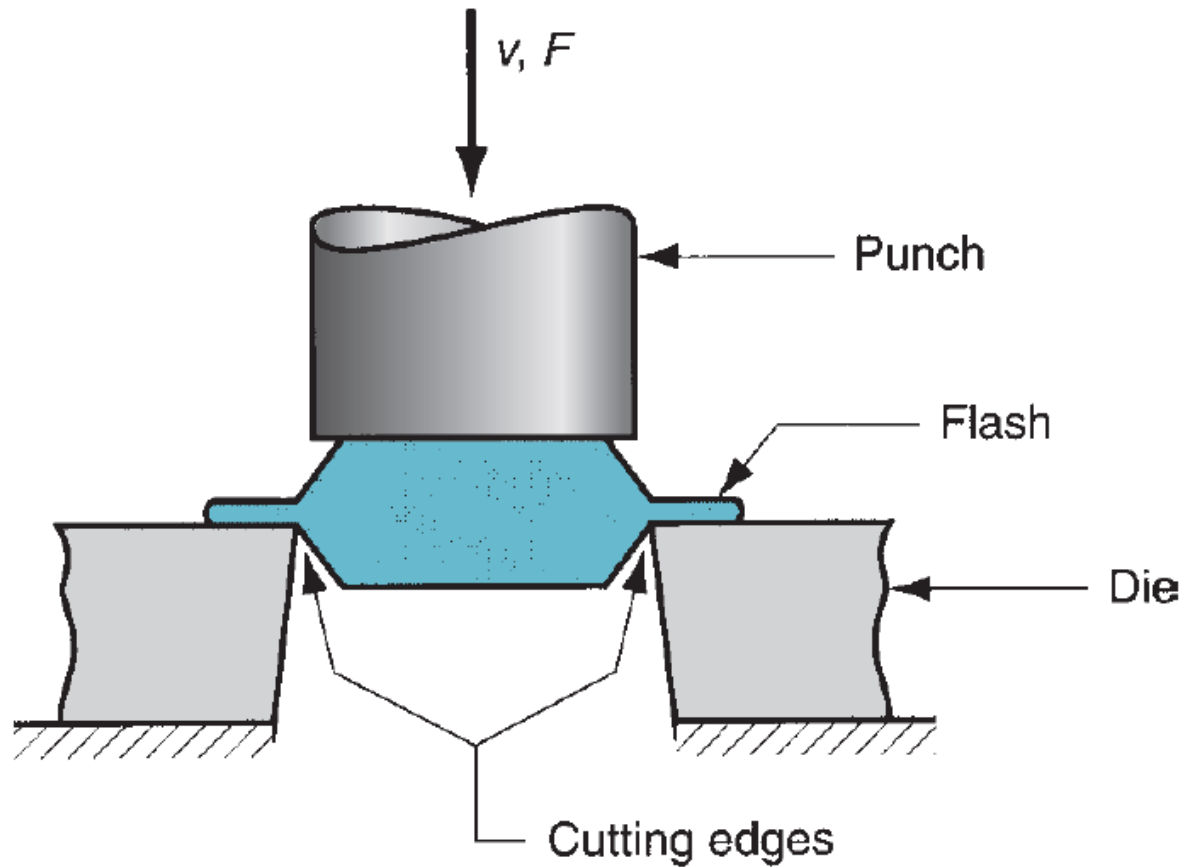


Figure 19.20 Trimming operation (shearing process) to remove the flash after impression-die forging.



Extrusion

- **Extrusion:** a compression process in which the work metal is forced to flow through a die opening to produce a desired cross-sectional shape.
- Imagine squeezing toothpaste out of toothpaste tube.
- Advantages include:
 - A variety of shapes are possible (especially in hot extrusion).
 - Microstructure and strength are enhanced in cold and warm extrusion.
 - Close tolerances are possible, especially in cold extrusion.
 - in some extrusion operations, little or no wasted material is created.



Extrusion

Types of Extrusion

- Extrusion can be classified in various ways:
 - By physical configuration: ***Direct Extrusion*** and ***Indirect Extrusion***.
 - By working temperature: ***Cold***, ***Warm***, or ***Hot Extrusion***.
 - Finally, it is performed as either a ***Continuous*** or ***Discrete process***.



Extrusion

Types of Extrusion

- **Direct** versus **Indirect Extrusion**: (1) Direct extrusion (also called **forward extrusion**) is illustrated in the Figure below.

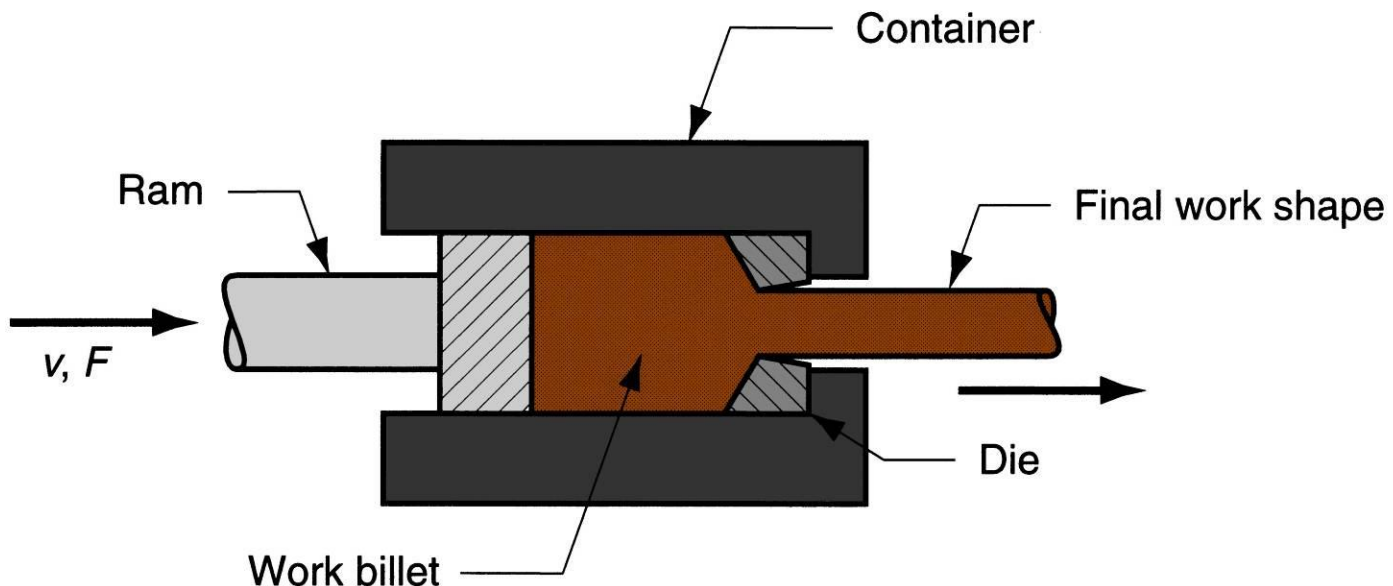


Figure 19.21 Direct extrusion.



Extrusion

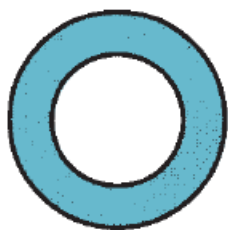
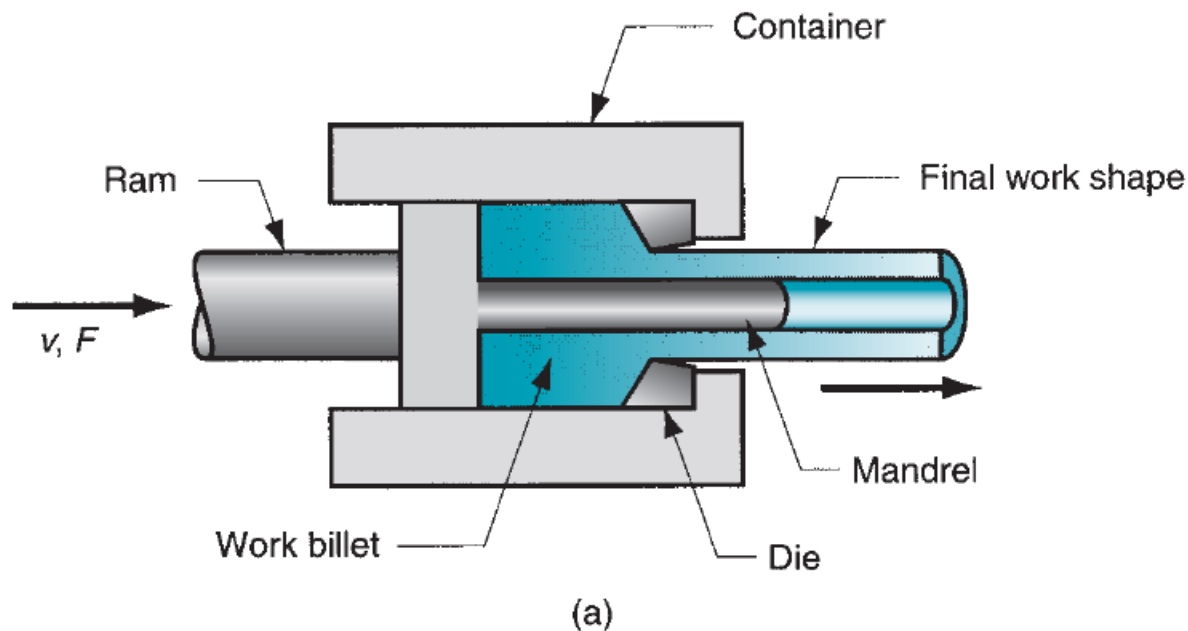
Direct Extrusion

- A metal billet is loaded into a container, and a ram compresses the material, forcing it to flow through one or more openings in a die at the opposite end of the container.
- As the ram approaches the die, a small portion of the billet remains that cannot be forced through the die opening.
- This extra portion, called the **butt**, is separated from the product by cutting it just beyond the exit of the die.
- Friction between container's walls and workpiece is one big problem in extrusion (so higher forces are needed to accomplish the process).
- The problem is aggravated in hot extrusion due to formation of oxide layer.

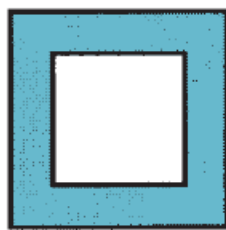


Extrusion

Direct Extrusion



(b)



(c)



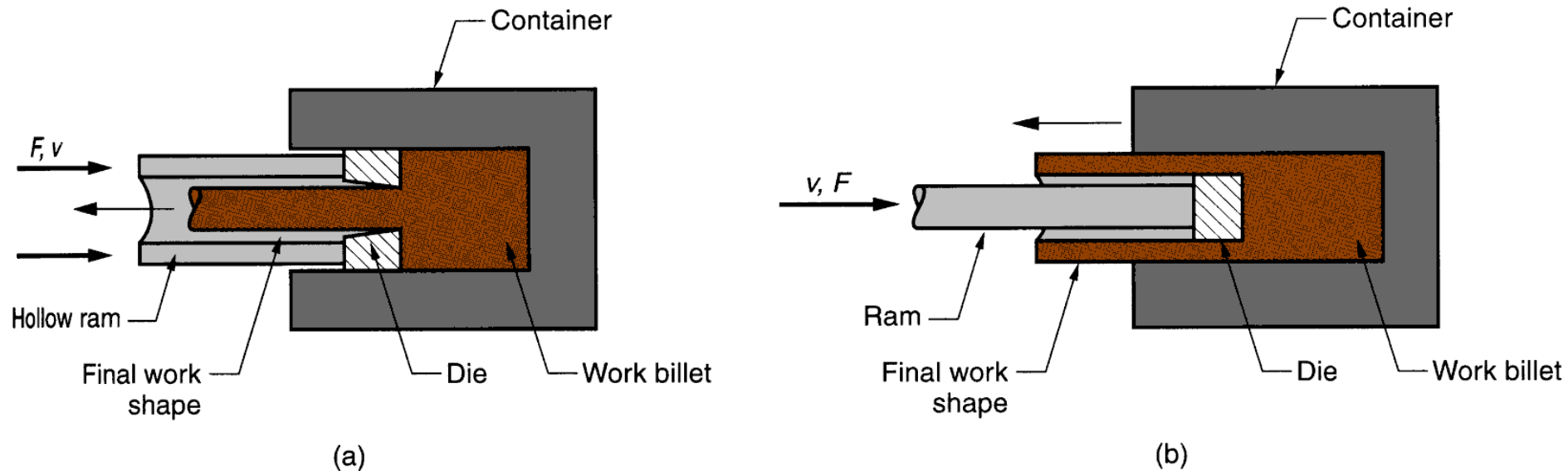
Figure 19.22 (a) Direct extrusion to produce a hollow or semi-hollow cross section; (b) hollow and (c) semi-hollow cross sections.



Extrusion

Types of Extrusion

- **Direct** versus **Indirect Extrusion**: (2) Indirect extrusion (also called **backward extrusion**) is illustrated in the Figure below.



(a)

(b)

Figure 19.23 Indirect extrusion to produce (a) a solid cross section and (b) a hollow cross section.



Extrusion

Indirect Extrusion

- The die is mounted to the ram rather than at the opposite end of the container.
- As the ram penetrates into the work, the metal is forced to flow through the clearance in a direction opposite to the motion of the ram.
- Since the billet is not forced to move relative to the container, there is no friction at the container walls, and the ram force is therefore lower than in direct extrusion.
- Limitations of indirect extrusion are imposed by the lower rigidity of the hollow ram and the difficulty in supporting the extruded product as it exits the die.



Extrusion

Types of Extrusion

- ***Hot*** versus ***Cold Extrusion***:
 - Extrusion can be performed either hot or cold, depending on work metal and amount of strain to which it is subjected during deformation.
 - Hot extruded metals include: Al, Cu, Mg, Zn, Sn, and their alloys (sometimes extruded cold as well).
 - Steel alloys are usually extruded hot, although the softer, more ductile grades are sometimes cold extruded (e.g. low C-steels).
 - Al is probably the most ideal metal for extrusion (hot and cold).
 - Products include: doors and window frames.



Extrusion

Types of Extrusion

- ***Hot Extrusion:***
 - Involves prior heating of the billet to a temperature above its recrystallization temperature.
 - This reduces strength and increases ductility.
 - Additional advantages include reduction of ram force, and increased ram speed.



Extrusion

Types of Extrusion

- ***Cold Extrusion:***
 - Used to produce discrete parts, in finished (or near finished) form.
 - Impact Extrusion: indicates high-speed cold extrusion.
 - Advantages: increased strength due to strain hardening, close tolerances, improved surface finish, absence of oxide layers, and high production rates.



Extrusion

Types of Extrusion

- ***Continuous*** versus ***Discrete Extrusion***:
 - ***Continuous Extrusion***: producing very long sections in one cycle, but these operations are limited by the size of the starting billet that can be loaded into the extrusion container. In nearly all cases, the long section is cut into smaller lengths in a subsequent sawing or shearing operation.
 - ***Discrete Extrusion***: a single part is produced in each extrusion cycle. Impact extrusion is an example of the discrete processing case.



Extrusion

Analysis of Extrusion

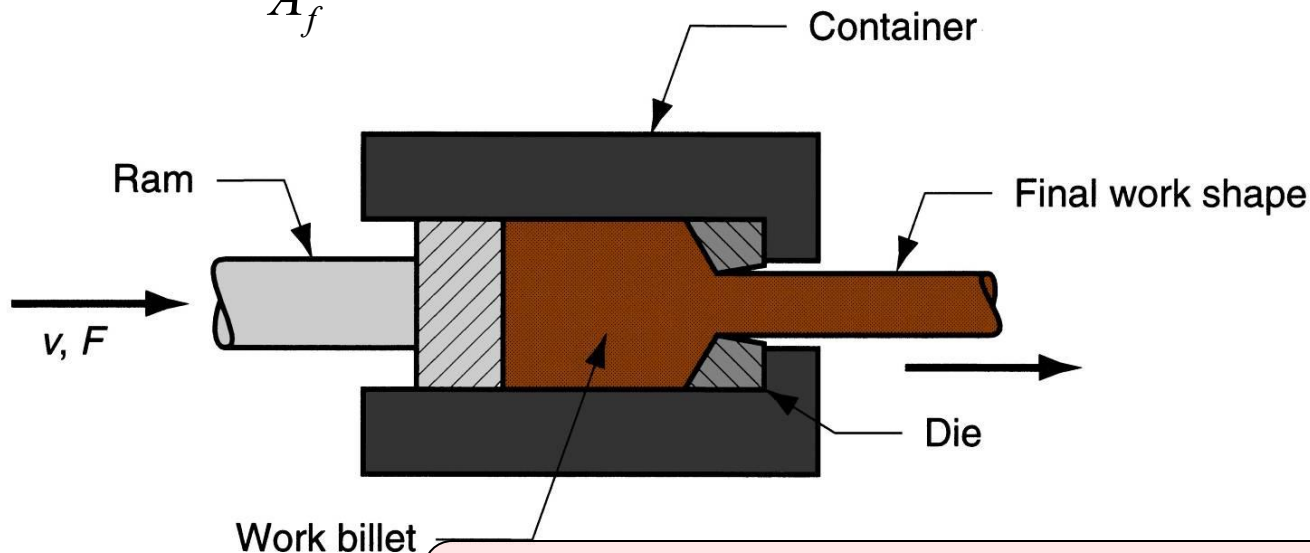
- Consider the figure below:

Extrusion ratio: $r_x = \frac{A_o}{A_f}$

True strain: $\varepsilon = \ln \frac{A_o}{A_f}$

Idea (no friction) case, pressure p : $p = \overline{Y}_f \ln r_x$

Average flow stress (MPa): $\overline{Y}_f = \frac{K \varepsilon^n}{1 + n}$



NOTE: This is ideal case (no friction considered).
The workpiece has round cross section



Extrusion

Analysis of Extrusion

- If friction is considered:

Extrusion strain: $\varepsilon_x = a + b \ln r_x$ where a & b are constants for a given die angle: $a = 0.8$ & $b = 1.2$ to 1.5 .

For indirect extrusion: $p = \overline{Y}_f \varepsilon_x$

For direct extrusion, friction is higher, so: $p = \overline{Y}_f \left(\varepsilon_x + \frac{2L}{D_o} \right)$

Ram forces in indirect or direct extrusion, F (N): $F = pA_o$

Power required P (J/s): $P = Fv$ v is velocity in m/s

NOTE: friction considered
and cross section is round.



Extrusion

Extrusion Dies and Presses

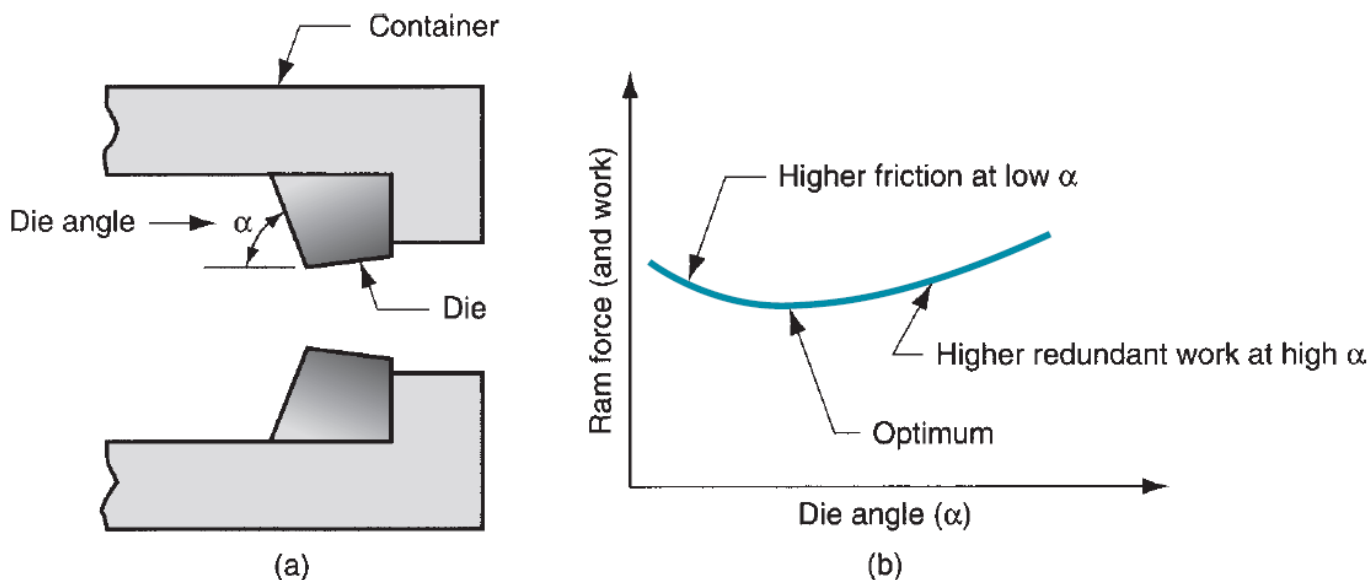


Figure 19.24 (a) Definition of die angle in direct extrusion; (b) effect of die angle on ram force.

Low die angles (α): high friction so high ram force.

High die angles (α): more turbulence, so increased ram force.

An optimum die angle exists.



Extrusion

Extrusion Dies and Presses

- The effect of the die orifice shape can be assessed by the die shape factor, can be expressed as follows:

$$K_x = 0.98 + 0.02 \left(\frac{C_x}{C_c} \right)^{2.25}$$

where K_x = die shape factor in extrusion; C_x = perimeter of the extruded cross section, mm; and C_c = perimeter of a circle of the same area as the extruded shape, mm.

K_x for circular shape = 1

K_x for hollow, thin-walled sections is higher.

For indirect extrusion: $p = K_x \bar{Y}_f \epsilon_x$

For direct extrusion: $p = K_x \bar{Y}_f \left(\epsilon_x + \frac{2L}{D_o} \right)$



For shapes other than round.



Extrusion

Extrusion Dies and Presses

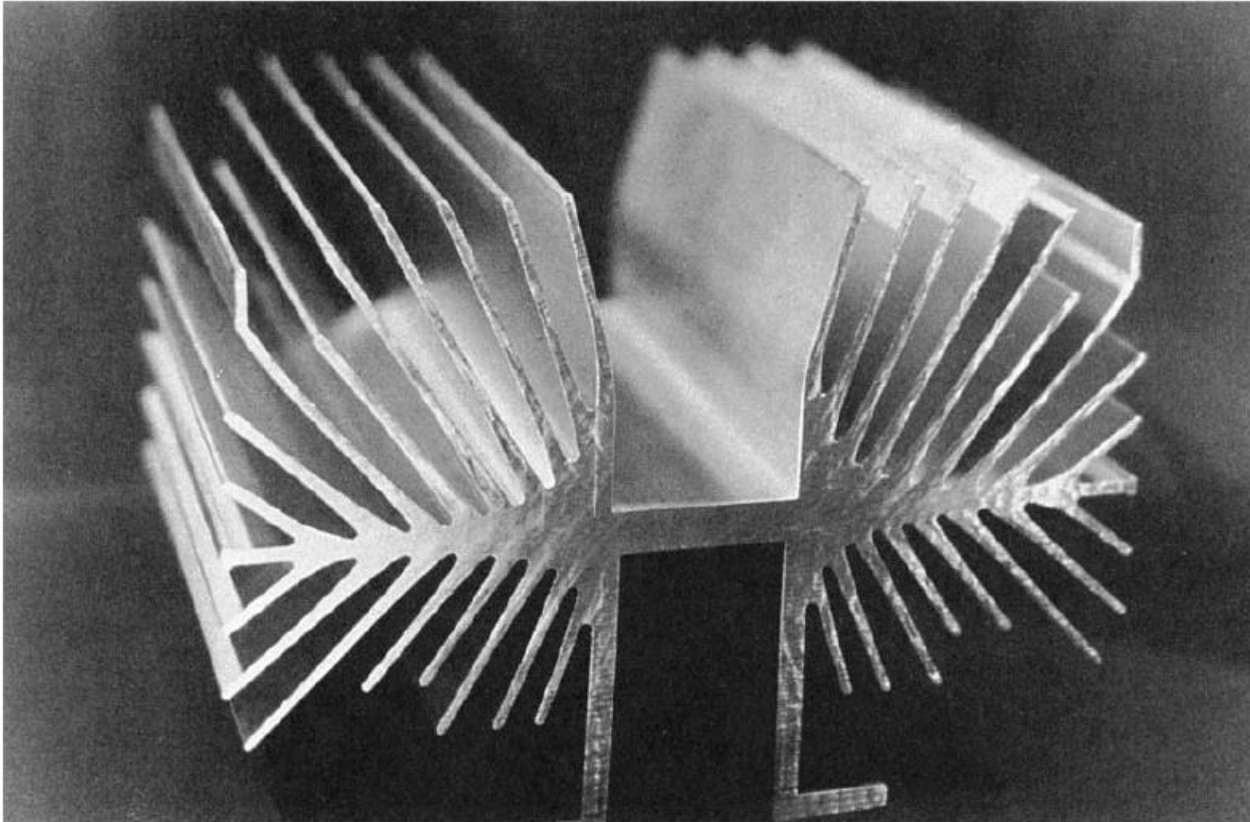


Figure 19.25 A complex extruded cross section for a heat sink. (Photo courtesy of Aluminum Company of America, Pittsburg, Pennsylvania).



Extrusion

Extrusion Dies and Presses

- ***Extrusion presses***: either horizontal or vertical, depending on orientation of the work axis.
- Usually hydraulically driven.
- This drive is especially suited to semi-continuous production of long sections, as in direct extrusion.
- Mechanical drives are often used for cold extrusion of individual parts, such as in impact extrusion.



Extrusion

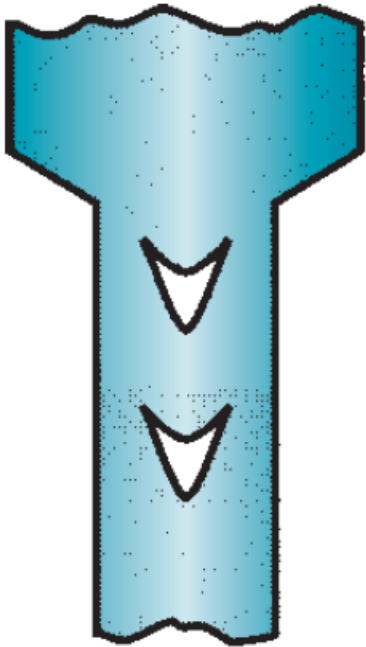
Defects in Extrusion

- **Centerburst:** an internal crack that develops as a result of tensile stresses along the centerline of the workpart during extrusion. Conditions that promote centerburst are high die angles, low extrusion ratios, and impurities.
- **Piping:** a defect associated with direct extrusion. It is the formation of a sink hole in the end of the billet. The use of a dummy block whose diameter is slightly less than that of the billet helps to avoid piping.
- **Surface cracking:** results from high workpart temperatures that cause cracks to develop at the surface. They often occur when extrusion speed is too high, leading to high strain rates and associated heat generation.

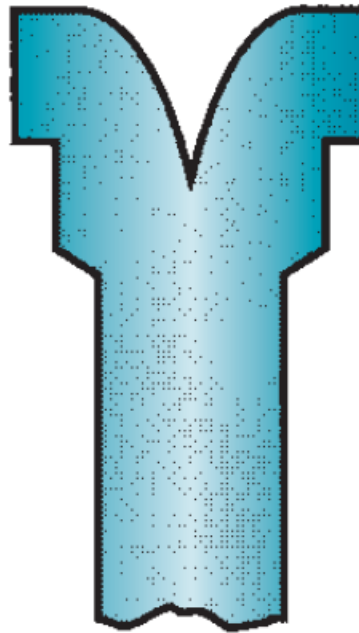


Extrusion

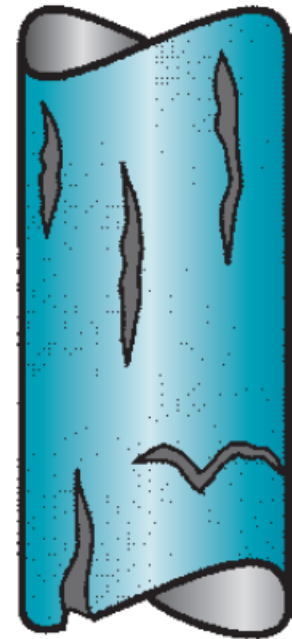
Defects in Extrusion



(a)



(b)



(c)

Figure 19.26 Some common defects in extrusion: (a) centerburst, (b) piping, and (c) surface cracking.



Wire and Bar Drawing

- **Drawing:** is an operation in which the cross section of a bar, rod, or wire is reduced by pulling it through a die opening.
- The difference between drawing and extrusion: the work is pulled through the die in drawing, whereas it is pushed through the die in extrusion.

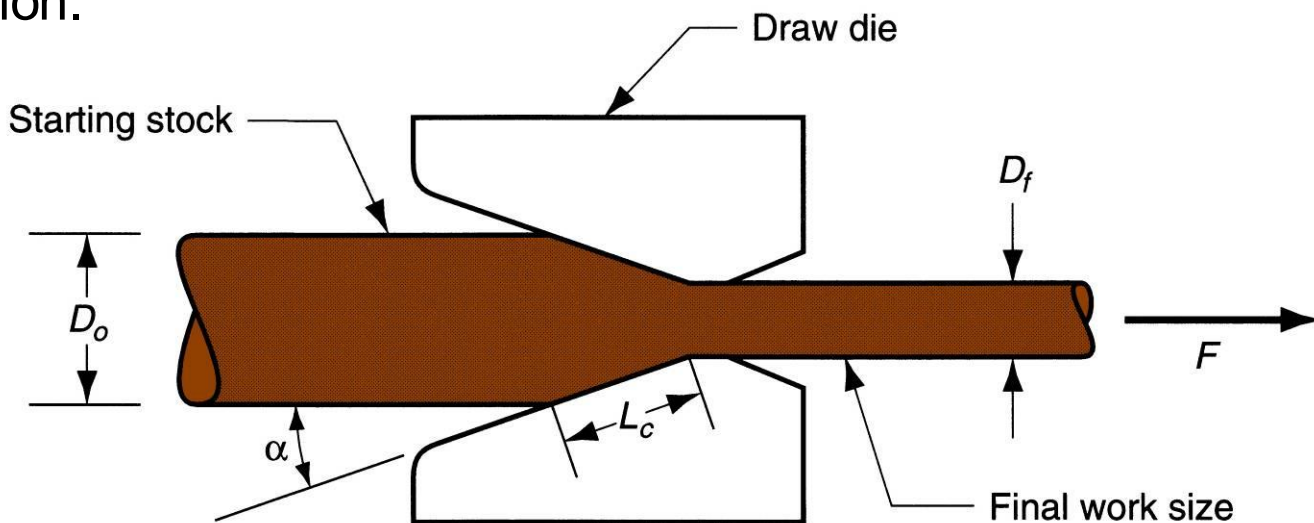


Figure 19.27 Drawing of bar, rod, or wire.



Wire and Bar Drawing

- **Bar drawing**: the term used for large diameter bars.
- **Wire drawing**: applies to small diameter bars (wire sizes down to 0.03 mm are possible in wire drawing).
- Two stress components are present in drawing; **tensile stresses** due to the pulling action and **compressive stresses** because the metal is squeezed down as it passes through the die opening.
- Change in size of work (given by area reduction):
$$r = \frac{A_o - A_f}{A_o}$$
- Draft: difference between original and final diameter:
$$d = D_o - D_f$$

Note: A is in (mm²) and D is in (mm).



Wire and Bar Drawing

Analysis of Drawing

- ***Mechanics of Drawing***: assume no friction.

True strain: $\varepsilon = \ln \frac{A_o}{A_f} = \ln \frac{1}{1-r}$

Stress: $\sigma = \overline{Y}_f \varepsilon = \overline{Y}_f \ln \frac{A_o}{A_f}$ where $\overline{Y}_f = \frac{K \varepsilon^n}{1+n}$



Wire and Bar Drawing

Analysis of Drawing

- **Mechanics of Drawing:** assuming friction, consider Figure 19.27 .

$$\sigma_d = \overline{Y}_f \left(1 + \frac{\mu}{\tan \alpha} \right) \phi \ln \frac{A_o}{A_f} \quad \text{where } \sigma_d = \text{draw stress, MPa; } \mu = \text{die-work coefficient of friction; } \alpha = \text{die angle; and } \phi \text{ is a factor that accounts for inhomogeneous deformation.}$$

$$\phi = 0.88 + 0.12 \frac{D}{L_c} \quad \text{where } D = \text{average diameter of work during drawing, mm; and } L_c = \text{contact length of the work with the draw die.}$$

$$D = \frac{D_o + D_f}{2} \quad \text{and} \quad L_c = \frac{D_o - D_f}{2 \sin \alpha}$$

Accordingly, $F = A_f \sigma_d = A_f \overline{Y}_f \left(1 + \frac{\mu}{\tan \alpha} \right) \phi \ln \frac{A_o}{A_f}$ where F = drawing force, N.



Wire and Bar Drawing

Analysis of Drawing

- ***Maximum Reduction per Pass***: why entire reduction is not taken in one pass?
 - As the reduction increases, draw stress increases.
 - If the reduction is large enough, draw stress will exceed the yield strength of the exiting metal.
 - When that happens, the drawn wire will simply elongate instead of new material being squeezed through the die opening.
 - For wire drawing to be successful, maximum draw stress must be less than the yield strength of the exiting metal.



Wire and Bar Drawing

Analysis of Drawing

- **Maximum Reduction per Pass:** assuming perfectly plastic material; then ($n = 0$ hence $\bar{Y}_f = Y$), and no friction:

$$\sigma_d = \bar{Y}_f \ln \frac{A_o}{A_f} = Y \ln \frac{A_o}{A_f} = Y \ln \frac{1}{1-r} = Y$$

- This means that $\ln (A_o/A_f) = \ln (1/(1-r)) = 1$. Hence, $A_o/A_f = 1/(1-r)$ must equal the natural logarithm base e . that is, the maximum possible strain is 1.0:

$$\varepsilon_{\max} = 1.0$$

- The maximum possible area ratio is: $\frac{A_o}{A_f} = e = 2.7183$
- The maximum possible reduction is: $r_{\max} = \frac{e-1}{e} = 0.632$



Wire and Bar Drawing

Drawing Practice

- Drawing is usually performed as a cold working operation.
- Most frequently used to produce round cross sections, but other shapes are also drawn.
- Drawn products include:
 - Electrical wire and cable; wire stock for fences, coat hangers, and shopping carts.
 - Rod stock to produce nails, screws, rivets, springs, and other hardware items.
 - Bar drawing is used to produce metal bars for machining, forging, and other processes.



Wire and Bar Drawing Drawing Practice

- Advantages include:
 - Close dimensional control.
 - Good surface finish.
 - Improved mechanical properties such as strength and hardness.
 - Adaptability to mass production.



Wire and Bar Drawing

Drawing Practice

- ***Drawing Equipment:*** (Bar Drawing)
 - Draw bench: consists of an entry table, die stand, carriage, and exit rack.
 - The carriage is used to pull the stock through the draw die.
 - Powered by hydraulic cylinders or motor-driven chains.

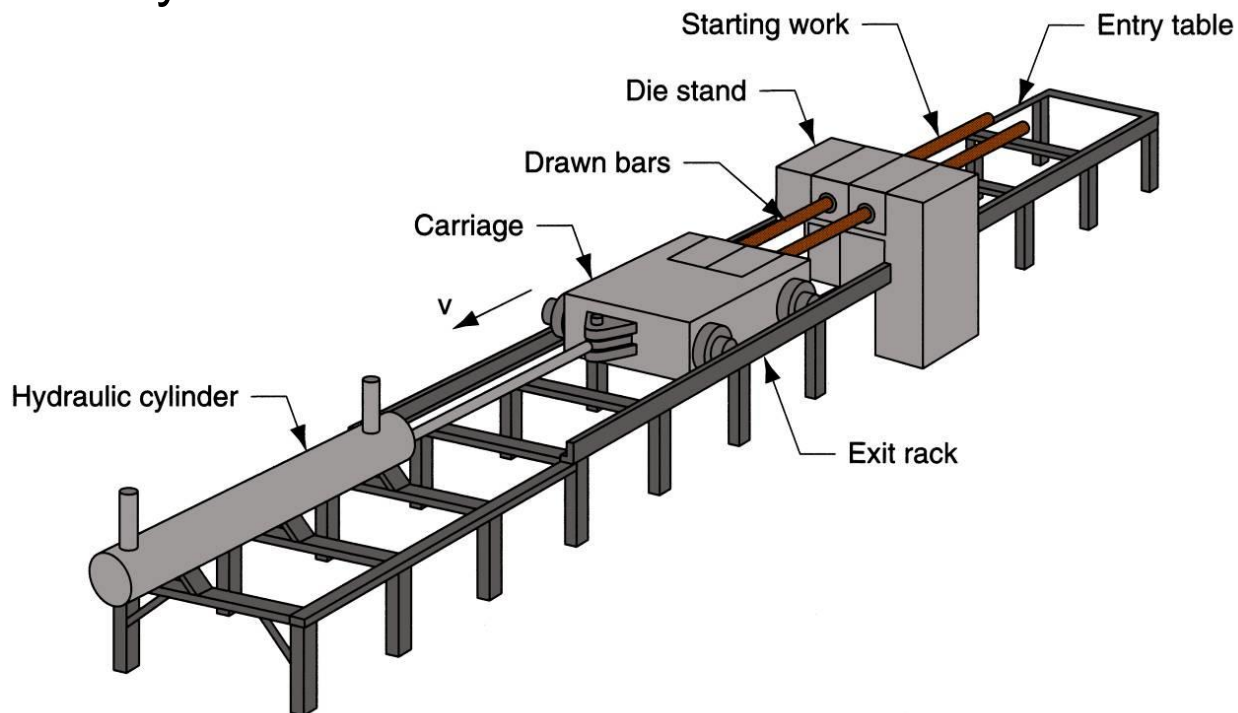


Figure 19.28 Hydraulically operated draw bench for drawing metal bars.



Wire and Bar Drawing

Drawing Practice

- ***Drawing Equipment:*** (Wire Drawing)
 - Done on continuous drawing machines that consist of multiple draw dies, separated by accumulating drums between the dies.
 - Each drum, called a ***capstan***, is motor driven to provide the proper pull force to draw the wire stock through the upstream die.
 - It also maintains a modest tension on the wire as it proceeds to the next draw die in the series.
 - Each die provides a certain amount of reduction in the wire, so that the desired total reduction is achieved by the series.



Wire and Bar Drawing

Drawing Practice

- ***Drawing Equipment:*** (Wire Drawing)

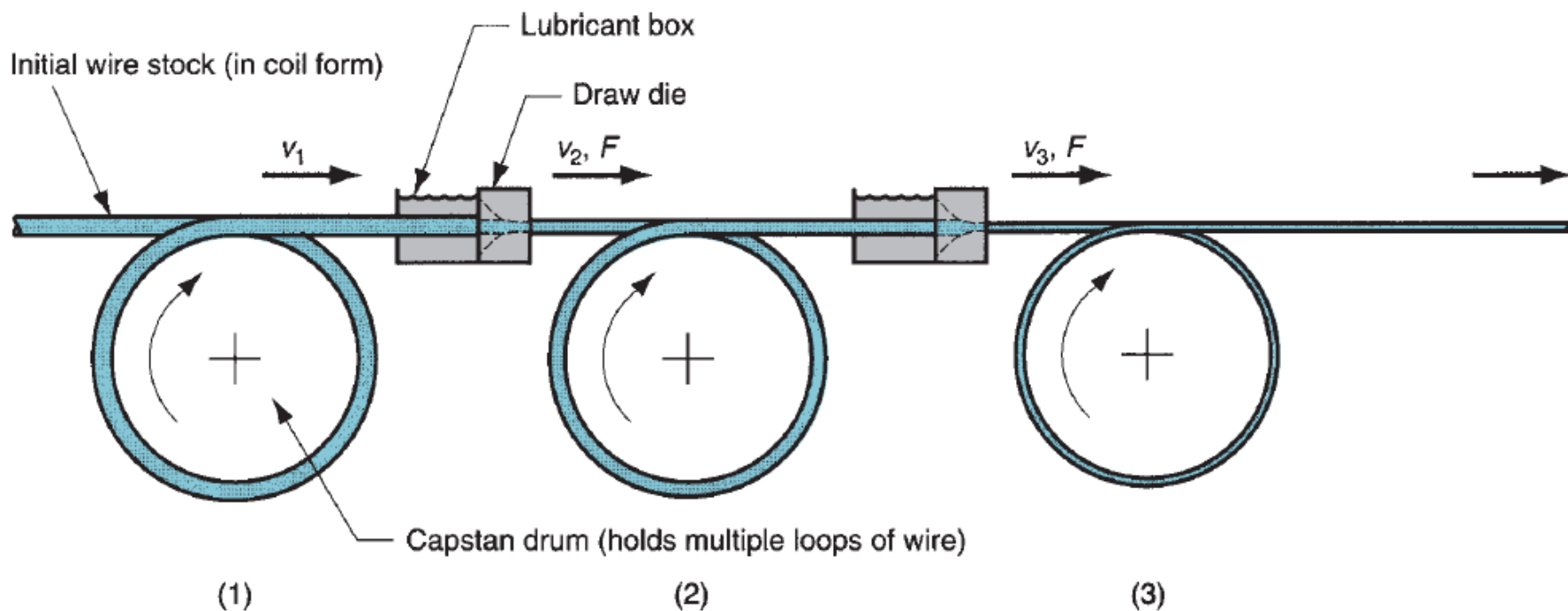


Figure 19.29 Continuous drawing of wire.



Wire and Bar Drawing

Drawing Practice

- **Drawing Dies** are made of tool steel, cemented carbides or diamond and they consist of 4 regions:
 - (1) **Entry Region**: usually a bell-shaped mouth that does not contact the work. Its purpose is to funnel the lubricant into the die and prevent scoring of work and die surfaces.
 - (2) The **Approach Region**: is where the drawing process occurs. It is cone-shaped with an angle (half-angle) normally ranging from about 6 to 20°.
 - (3) The **Bearing Surface (Land)**: determines the size of the final drawn stock.
 - (4) The **Back Relief**: is the exit zone. It is provided with a back relief angle (half-angle) of about 30°.



Wire and Bar Drawing

Drawing Practice

- ***Drawing Dies:***

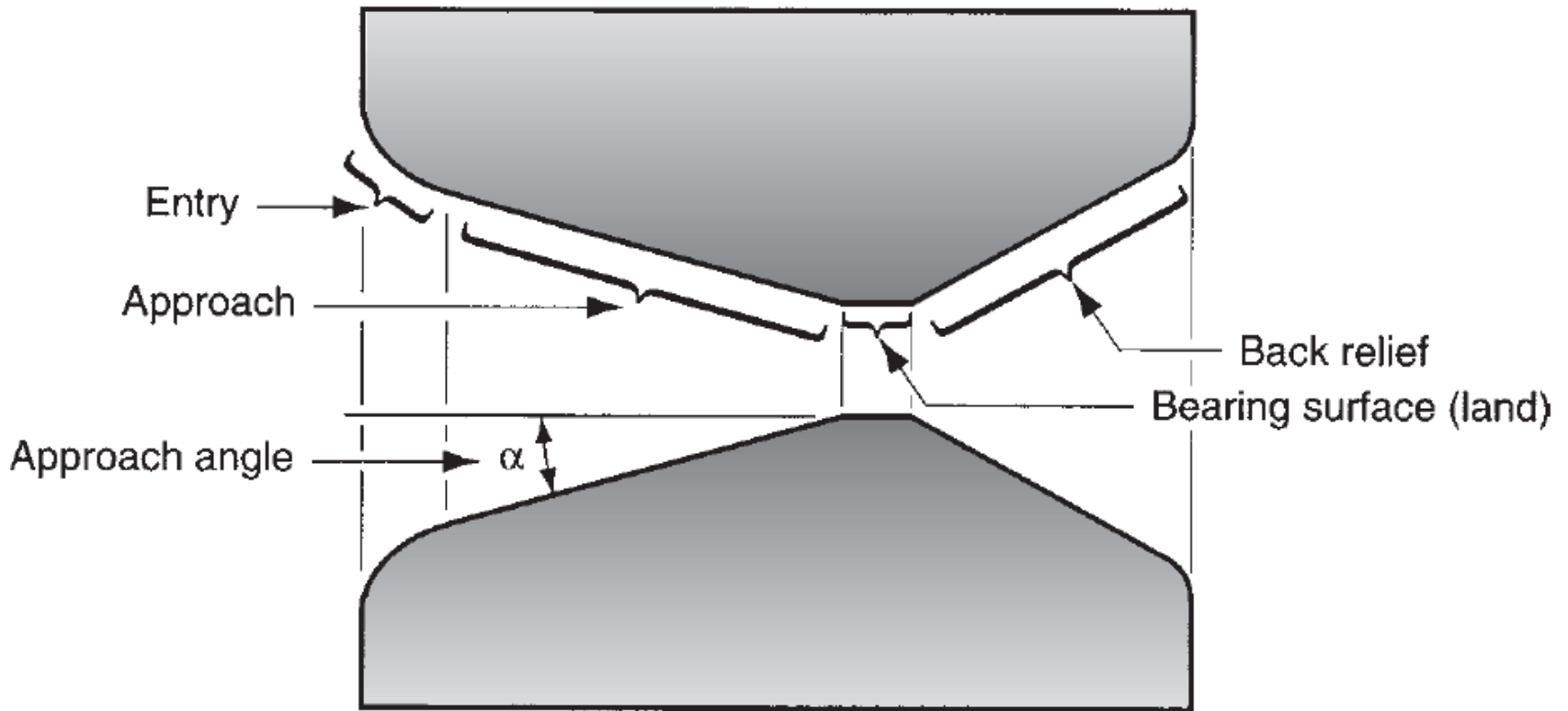


Figure 19.30 Draw die for drawing of round rod or wire.



Wire and Bar Drawing

Drawing Practice

- ***Preparation of work***: involves three steps: (1) annealing, (2) cleaning, and (3) pointing.

(1) ***Annealing***: done to increase the ductility of the stock.

(2) ***Cleaning***: required to prevent damage of the work surface and draw die.

(3) ***Pointing***: involves the reduction in diameter of the starting end of the stock so that it can be inserted through the draw die to start the process. This is usually accomplished by swaging, rolling, or turning.